



International
Institute of Information
Technology Bangalore

Powered By

upGrad

EXECUTIVE PROGRAM IN GENERATIVE AI FOR LEADERS

Duration: 20 weeks



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The Power of **Generative AI in Businesses**



The Gen AI revolution is transforming industries with its advanced capabilities, driving innovation and efficiency.

For business leaders, embracing Gen AI is essential to unlock new business models, enhance customer experiences, and streamline operations, gaining a significant edge over competitors.

This program empowers business leaders to harness the full potential of Gen AI. Leaders will learn to identify and implement Gen AI solutions that align with their strategic goals, ensuring their organizations remain at the forefront of innovation.

The program combines practical, hands-on learning with expert mentorship, ensuring participants can immediately apply their newfound knowledge. personalised workshops provide mentoring tailored to their specific problem domains.

Adopting Gen AI is not just an opportunity but a pressing priority to stay ahead of the curve, and that's exactly what we aim to deliver with this program.

About **IIIT-Bangalore**



IIIT-Bangalore is a premier institute dedicated to undergraduate and postgraduate programs specialising in Computer Science and Engineering (CSE), Electronics and Communication Engineering (ECE), Data Science and Artificial Intelligence (DSAI), Digital Humanities and Societal Systems (DHSS) and broader fields of IT and research. Situated in the heart of Electronic City, Bangalore, it holds a prominent position in the academic landscape. IIIT-Bangalore is graded A+ by National Assessment and Accreditation Council (NAAC).

IIIT-Bangalore contains state-of-the-art infrastructure, eminently qualified faculty, a vibrant alumni community, cutting-edge research facilities, and close industry collaborations.

The institute's specially designed courses make the students' cognizant of the current technologies. Experiential learning and practices followed in the institute equip them with the tools and knowledge to solve contemporary real problems. IIIT-Bangalore has consistently achieved excellent placement records every year since its establishment in 1998, thanks to the unwavering support of the industry and the expanding pool of highly skilled alumni.

For more details, log onto: www.iiitb.ac.in

Why upGrad

upGrad is a leading global learning and workforce development company. We're on a single-minded mission of powering career success for every member of the global workforce as their trusted lifelong learning partner.

Established in 2015, we have over 10 million learners who have upskilled from a range of online and offline programs from top universities in India and the world; and in domains from ranging Data Science and Analytics, Business Administration, Marketing, Tech & Software Development, & Law.



Key Highlights

Program Overview



Designed for Business and Functional Leaders

Coding or programming experience is not required to achieve the desired learning outcomes in this program.



Exclusive One-Day Immersion at Microsoft

Access to one-day immersion at Microsoft & an optional immersion at IIIT-Bangalore.



Capstone Project: Domain-Driven AI Implementation

Develop a personalized Gen-AI implementation roadmap tailored to your specific domain.



Program Tools

Explore popular Gen AI tools like ChatGPT, DALL-E 2, Pinecone, and many more!



Triple Certificate from IIIT-Bangalore, Microsoft & eCornell

Get certified by IIIT-Bangalore, Microsoft and eCornell on successful completion of the program.

*Cost of immersion is not included in the program fees and has to be borne by the learner

Program Features

1

AI Leadership
Challenges

+

2

AI Insights:
Personalised Reports

+

3

AI-Powered
Thought Leadership

AI Leadership Challenges

- Tackle high-stakes scenarios, from building Gen-AI solution pitches in groups to handling simulated AI crisis management simulations.
- Get feedback from the mentors to master the Gen AI leadership scenarios.

AI Insights: Personalised Reports

- Receive timely, curated reports and case studies on emerging AI trends, hand-picked by our AI experts and tailored to your specific domain and industry.
- Get actionable insights and key takeaways, empowering them to make data-driven strategic decisions.

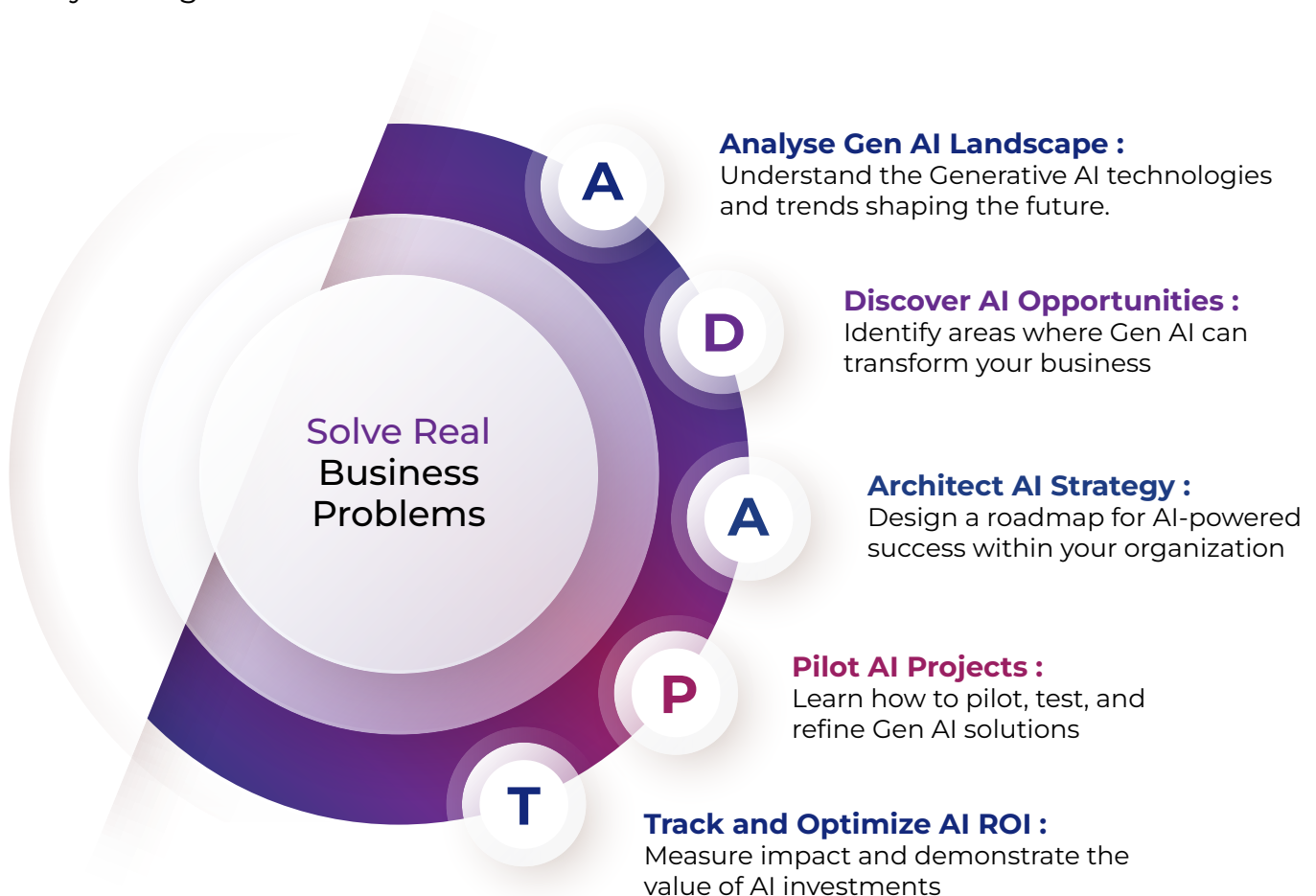
AI-Powered Thought Leadership

- Receive guidance on building a powerful professional brand as an AI-savvy leader.
- Get access to sessions on crafting compelling AI content, explore opportunities to publish their insights, and potentially teach within upGrad courses to expand your reach and establish yourself as a thought leader.

A.D.A.P.T Framework

A.D.A.P.T Now:

- Path to Gen AI Leadership Excellence
- Learn Gen-AI skills through A.D.A.P.T framework to lead AI-powered transformation in your organisation



Learn from Expert Faculty



Shailesh Kumar

Chief Data Scientist



Former Vice President



Former Chief Scientist & Co-Founder



Former Principal Development Lead



At Jio, Shailesh leads a top-tier team advancing AI in sectors like education, healthcare, and security. With 20+ years of experience, his expertise spans machine learning, quantum AI, and IoT. A renowned speaker, he collaborates with government agencies and top universities, pushing AI's boundaries.



Mukesh Jain

CTO, VP & Global Head People Analytics



Former CTO



Former Principal Manager



M-Tech in Software Systems



Mukesh Jain, CTO and VP at Capgemini, leads AI-driven innovation in people analytics. With 28 years of experience, he has led digital transformation at Microsoft, Jio, NICE, and VFS Global. An expert in technology and analytics, he's an international speaker and author of two books.



Hindol Basu

Cofounder
& CEO



Former
Cofounder
& CEO



Former
Partner



Former Senior
Manager



Hindol, CEO and Co-Founder of Leobrix and Actify Labs, has led analytics at Tata Industries and FICO. He specializes in risk analytics and data-driven decision-making. At Citi bank, he developed advanced risk management solutions and led process innovation as VP of risk modeling.



Dr. Chandrashekar Ramanathan

Dean,
Academics



Former Sr.
Solutions Architect



Ph.D



Dr. Chandrashekar Ramanathan, a professor at IIIT-Bangalore since 2007, specializes in data science, software engineering, and large-scale applications. With over 25 years of experience, he was named a top data science academician in India by Analytics India magazine in 2018 and 2019.



Dr. V. Sridhar

Professor



Former
Research Fellow



Former
Professor



Former Assistant
Professor



Dr. V. Sridhar, a professor at IIIT-Bangalore's CITAPP, specializes in telecommunications technology, management, and regulation. He has co-authored two books, published numerous articles in leading journals, and written about 300 articles on Telecom and IT in major business newspapers and magazines.



Rajesh Sabapathy

Health-tech
start-up



Former
Manager



Former Software
Engineer



MSc,
Economics



Rajesh, an AI/ML leader, has driven innovation in healthcare using deep learning and big data. With expertise in Python, R, and cloud platforms, he has delivered significant business value and collaborated across various industries. Formerly Distinguished Engineer at Fortune 5 Healthcare Giant

Our Class Profile



Microsoft-Certified Advantage –

This program doesn't just prepare you for the world of Data Science & AIML—it gives you the Microsoft edge.

Learners earn industry backed certification from upGrad in association with Microsoft by completing specially designed modules integrated into the program, boosting both credibility and career readiness.



Microsoft Learn content modules Certification Modules:

- Introduction to Generative AI Concepts
- Introduction to GitHub Copilot
- Design & Manage Analytics Solutions using Power BI
- Designing & Implementing a Data Science Solution on Azure

Microsoft AI Immersion | Bangalore

Go beyond theory. In this exclusive, hands-on immersion led by Microsoft experts, you'll:

- Master advanced AI-powered collaboration & automation tools
- Build muscle memory with real-world Microsoft applications
- Gain insider strategies directly from the creators of the technology

Why It Matters:

-  Global Recognition – Microsoft badges showcase your skills worldwide
-  Career Differentiator – Employers value certified Microsoft professionals
-  Practical Edge – Learn tools you'll use daily in analytics, AI, and cloud

With industry backed certification from upGrad in association with Microsoft certification + IITB's academic rigour, you graduate with **dual credibility**: academic excellence and industry validation.



Your Personalised Gen AI Learning Journey



Module 1: Understanding Gen AI Fundamentals

15+ hours of live learning and case studies covering AI fundamentals like LLMs and transformers.



Module 2: Applications of Gen AI in Business

Engage in workshops focused on industry-specific business problems with 7+ hours of live learning and case studies.



Module 3: Crafting Gen AI Strategy

Develop long-term Gen AI initiatives with 5+ hours of live sessions, case studies.



Module 4: GenAI Project Planning

Develop hands-on skills in managing Gen AI projects through a workshop and 9+ hours of live learning and case studies.



Module 5: Future of Gen AI

Understand emerging trends in Gen AI and develop a forward-thinking approach through a roundtable on "The Future of Work with AI".



Module 6: Capstone

Complete an industry-mentored capstone project in a chosen domain with live reviews and feedback from mentors.



Explore the Curriculum

Module 01 | (Week 1-4)

Understanding AI and Gen AI Landscape



Outcomes

- A thorough understanding of the foundational principles of AI and Generative AI.
- Familiarity with key AI technologies, including Large Language Models (LLMs) and Transformers.
- Acquire basic skills in navigating and understanding the technological aspects of Gen AI applications.

Topics

Understanding AI and Gen AI Landscape

- History of AI
- AI Paradigms
- Generative AI Landscape
- Leveraging Data for AI

Generative AI in Business

- Generative Models: Principles and Applications
- Impact of Generative AI on Business Functions: Opportunities & Automation
- Tools for Business Leaders: Applications and Contributions

Explore the Curriculum

Module 01 | (Week 1-4)

Topics

Foundations of Gen AI

- Prerequisites for Gen AI for technical leaders
- Prerequisites for Gen AI for non-technical leaders
- Basics of ML, NLP, and language models

Introduction to Large Language Models (LLMs)

- Overview and Evolution of LLM and LLM Architecture
- Understanding Attention Mechanism
- Introduction to Reinforcement Learning
- Brief History of the GPT Model Series

Advanced Prompt Engineering Workshop

- Basic Prompting
- Prompt Engineering using Few Shot and Chain of Thought Reasoning Techniques
- Application of Prompt Engineering in Industry Tools: ChatGPT, DALL-E 2, MidJourney, Stable Diffusion, Runway
- Building chatbots using RAG and Finetuning

Leveraging Gen AI for Improving Productivity

- Chrome extension integration (ChatGPT for Sheets, etc.)
- Automation with agent chains
- Generative AI for data analysis and conversation insights

Tech Stack for Generative AI Applications - I

- LLMs (Closed vs OpenSource)
- Orchestrators for creating systems (LangChain and LlamaIndex)
- VectorDBs for storage (ChromaDB, Pinecone)
- Multimodal LLMs (Text, Images, Audio, Video)
- Choosing between Generative AI and Traditional ML
- Integrating Gen AI Components
- Future of Generative AI

Explore the Curriculum

Module 02 | (Weeks 5-7)

Applications of Gen AI in Business



Outcomes

- Learn to identify and implement practical applications of Gen AI in various business contexts.
- Understand how Gen AI can drive business innovation and improve operational efficiencies.
- Develop the ability to analyze and solve business problems using Gen AI solutions.

Topics

Applications of Gen AI in Various Industries

- Industry-Specific AI Solutions
- Gen AI Applications
- AI in Diverse Sectors
- Gen AI Business Use Cases

Frameworks for Identifying Generative AI Opportunities

- AI Opportunity Identification
- Generative AI Frameworks
- Opportunity Assessment
- AI Innovation

Explore the Curriculum

Module 02 | (Weeks 5-7)

Topics

Design Thinking in AI Solution Development

- Design Thinking
- AI Solution Development
- User-Centric Design
- Innovation in AI

Prototyping and Validating Gen AI Concepts for Business

- Develop prototypes of Gen AI concepts
- Validate prototypes for feasibility and effectiveness
- Refine concepts based on validation results

Workshop: Exploring Generative AI Potential in Your Own Business



Learn to understand where all Gen AI can be implemented and processes can be automated for your business. Apply basic frameworks and design thinking to leverage Gen AI to your business.

Explore the Curriculum

Module 03 | (Weeks 8-9)

Crafting Gen AI Strategy



Outcomes

- Master the skills to develop strategic frameworks for integrating Gen AI into business operations.
- Learn to plan, deploy, and scale Gen AI initiatives effectively within an organisation.
- Enhance strategic decision-making skills with a focus on long-term integration of Gen AI technologies.

Topics

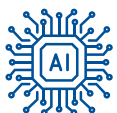
Strategic Implications of AI in Business

- Business Strategy
- Automation
- Decision-making
- Competitive Advantage
- Cost Organisation

Building AI-Driven Business Models

- AI Opportunity Identification
- Generative AI Frameworks
- Opportunity Assessment
- AI Innovation

Workshop: Developing an AI Strategy for Your Business



Build an AI strategy for your business and understand implications of AI integration, data driven decisions and automation.

Explore the Curriculum

Module 04 | (Weeks 10-13)

Gen AI Project Planning



Outcomes

- Develop hands-on skills in managing the lifecycle of Gen AI projects, from inception to deployment.
- Gain practical experience in troubleshooting and overcoming common challenges in Gen AI project management.
- Acquire project management skills specific to AI-driven initiatives, enhancing efficiency and effectiveness.

Topics

AI Project Lifecycle & Management

- Planning and Scope Definition
- Development and Implementation
- Monitoring and Iteration

Planning and Executing AI Projects

- Project scoping and goal setting
- Data collection, preprocessing, and feature engineering
- Model selection, training, and evaluation

Explore the Curriculum

Module 04 | (Weeks 10-13)

Topics

Scaling and Maintaining AI Projects

- Scale infrastructure as needed
- Monitor model performance

Measuring Gen AI Impact & ROI on Projects

- Define project goals and metrics
- Track KPIs to assess impact and ROI
- Analyse results for improvements

Workshop: Executing AI Strategy for Your Business



Learn to execute AI strategy for your business by applying Gen AI project lifecycle and management principles, measure ROI and Gen AI impact on the business.

Build an AI strategy for your business and understand implications of AI integration, data driven decisions and automation.

Explore the Curriculum

Module 05 | (Weeks 14-15)

Future of Gen AI



Outcomes

- Understand emerging trends and innovations in the field of Gen AI.
- Prepare to adapt and respond to future advancements in AI technologies.
- Cultivate a forward-thinking approach to utilizing upcoming Gen AI capabilities in various sectors.

Topics

Applications of Gen AI in Various Industries

- Emerging AI Trends Identification
- Entrepreneurial Opportunities
- Future Scope in Generative AI Landscape
- Ethical and Legal Considerations in AI Usage

Round Table Discussion

- The Future of Work with AI [Interactive]

Responsible AI: Ethics, Bias & Regulation

- Ethics and Bias
- Algorithmic Bias
- AI Responsibility

Explore the Curriculum

Module 05 | (Weeks 14-15)

Topics

Review and Reflect: Key Learnings and Next Steps

- Reflect and summarise the learnings of the program
- Implement feedback on the workflow created in interactive workshops

Capstone Project (Weeks 16-18): Domain-specific AI Implementation

eCornell Course

- Quality and Service Excellence
- Leading Collaborative Teams
- Strategic Decision Making
- Leading with Credibility
- Motivating People for High Performance
- Coaching Skills for Leaders

Note*- Completion of this optional course is a must in order to obtain a certification from eCornell.

Microsoft Course

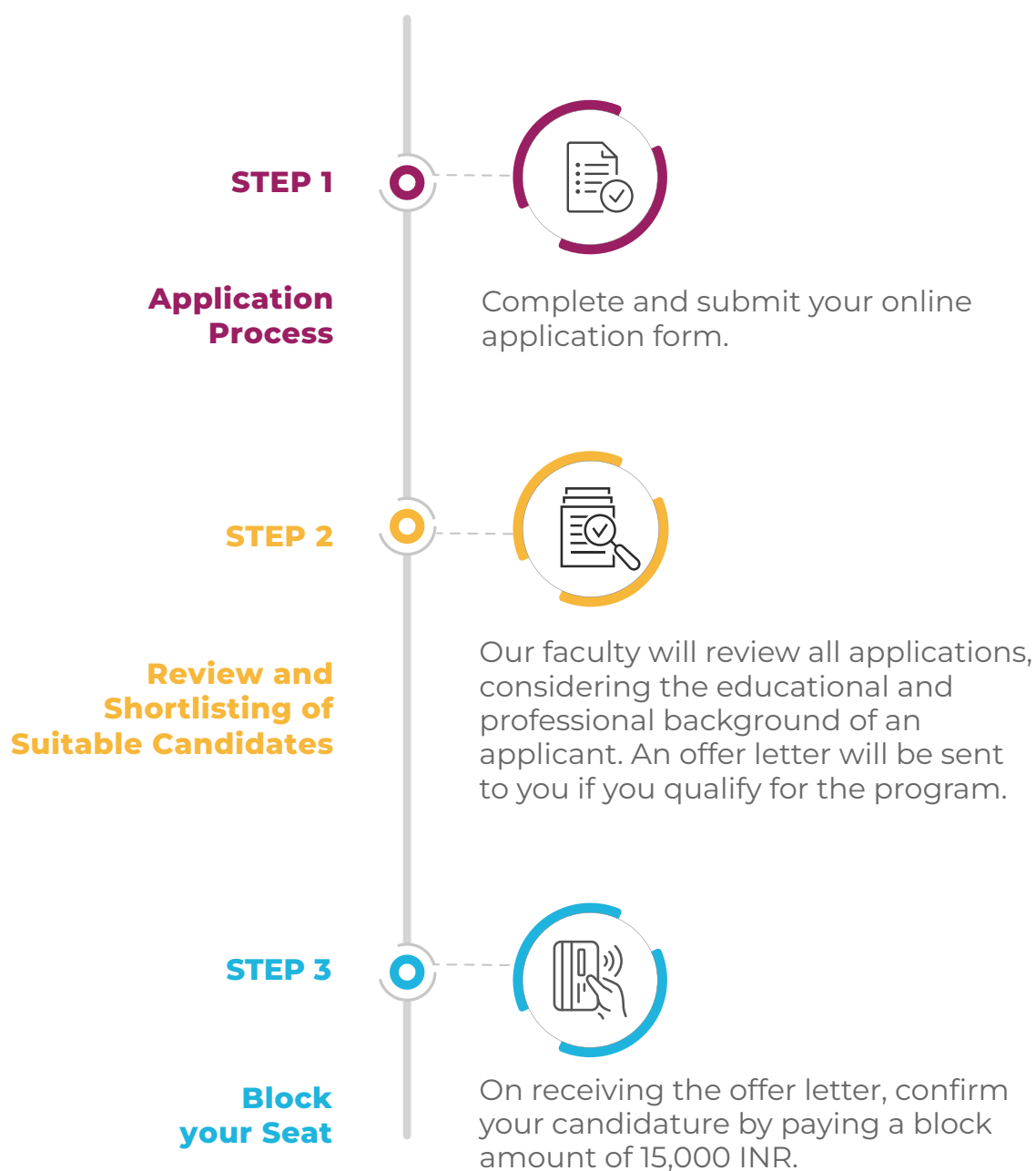
- Transform your business with Microsoft AI
- Introducing Microsoft 365 Copilot, Getting Started and Getting Ready
- Copilot for Microsoft 365 Interactive Experience for Executives
- What Can Copilot's Earliest Users Teach Us About Generative AI at Work?
- Craft effective prompts for Microsoft Copilot for Microsoft 365
- Data, Privacy, and Security for Microsoft Copilot for Microsoft 365
- Manage data security for Copilot for Microsoft 365
- Manage data compliance for Copilot for Microsoft 365
- Microsoft 365 reports in the Admin Center – Copilot for Microsoft 365 readiness and usage

Note*- Completion of this optional course is a must in order to obtain a certification from Microsoft.

Ready to Get Started? **Apply Now.**

Eligibility:

Bachelor's Degree | 4+years of work experience preferred.





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International
Institute of Information
Technology Bangalore

powered by

upGrad

Executive Diploma in

Machine Learning & Artificial Intelligence

With MLOps and
Generative AI Specialisations

Duration 12-15 months



Largest ML & AI program in India

with 9+ Years of Legacy &
10,000+ learners





powered by

upGrad

Glimpses From Convocation Ceremony





About IIITB

The International Institute of Information Technology Bangalore (IIITB)

Established in 1998, IIITB is a premier institute known for its interdisciplinary approach, integrating technology with social sciences. Supported by the Government of Karnataka and the IT industry, IIITB fosters strong academic-industry partnerships and attracts top talent from across India and abroad through its merit-based selection process.

The institute has graduated over 3,500 students, many of whom work at leading IT companies globally. With a focus on research and development in fields like Artificial Intelligence (AI) and Machine Learning (ML), IIITB is recognised as a leader in AI education.

Ranked 74th in the Engineering category of the National Institutional Ranking Framework (NIRF) in 2022, IIITB continues to excel in education and research, making it a preferred destination for aspiring technologists and future leaders.



AICTE



UGC

ABOUT upGrad

“upGrad is a leading global learning and workforce development company. We’re on a single-minded mission of powering career success for every member of the global workforce as their trusted lifelong learning partner. Established in 2015, we have over 10 million learners who have upskilled in a range of online and offline programs from top universities in India and the world.”



Ronnie Screwvala

Co-founder & Executive Chairman



GSV EDTECH 150
Top 150 Most Transformative
Growth Companies



**BRANDON HALL
GOLD AWARDS**
15+ awards across 2 years



PROMISING BRAND 2022
The Economic Times



**EDUCATION COMPANY
OF THE YEAR**
VC Circle, 2022



Entrepreneur India
**EDUCATION INNOVATION
AWARDS 2020**
Best Communication &
Teaching Platform



Current Industry Trends

15 LPA

Average salary
for Machine Learning
Engineers in India (2025)

Source: Glassdoor

36.08%

CAGR of global
Machine Learning
market till 2030

Source: Statista

50%

Increase in search
volume for 'Data Science
and Machine Learning'
since 2020

Source: Exploding Topics

74%

Annual increase in
growth of AI and
Machine Learning jobs
since 2020

Source: Careerist

\$503.4B

Size of global
machine learning market
by 2030

Source: Statista

Program Highlights

Here are the
top reasons why
you should consider
this program



Future-Ready Curriculum

Master In-Demand and Trending Competencies



Personalised Learning Experience

Learning Experience Tailored to Your Needs



Specialisations

Specialise in two in-demand Data Science specialisations



In-Demand Tools

80+ Industry Tools, Languages, Libraries



Outcome-Driven Learning Experience

Personalised Portfolio-Building Support and Career Preparation Sessions



Best-in-Industry Experts

Decorated IITB Faculty and Top Industry Practitioners



Golden Learning Ratio

Perfect Blend of Mathematics, Technology, and Business Understanding



Hands-on Learning

Solve 30+ Domain-Focused Assignments and Case Studies

Offline Graduation Function

On-Campus Graduation Ceremony for a Complete Program Experience

Program Impact Success Stories

Before upGrad

Tejasvy Gunturu

Application Development Associate

accenture



After upGrad

accenture

Application Development Analyst

Before upGrad

Vijeet Ved

System Engineer

tcs TATA CONSULTANCY SERVICES



After upGrad

tcs TATA CONSULTANCY SERVICES

IT Analyst

Before upGrad

Krish

Deputy Manager

MARUTI SUZUKI



After upGrad

SAMSUNG

Lead Engineer

Before upGrad

Vikas Kumar Gupta

Data Scientist

TATA POWER



After upGrad

TIBIL SOLUTIONS

Senior Data Scientist

Before upGrad

Sunil Acharya

Software Engineer

intel.



After upGrad

intel.

AI Engineer

Our Transition Statistics

INR 1.23 Cr Highest Salary

433% Highest Hike

50% Average Hike

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Program Completion Certificate



Earn valuable credentials with an Executive Diploma in Machine Learning and Artificial Intelligence-equivalent to a 1-year PG Diploma and accredited with NAAC A+ (2021). Join India's largest ML AI alumni network of over 10,000 professionals.

Earn Second Certificate

π Professional pack for dual specialisations



In just 3 more months learners can earn 1 more specialization across both Data Science & Machine learning specializations

Faculty



Dr. Debabrata Das
Director of IIITB



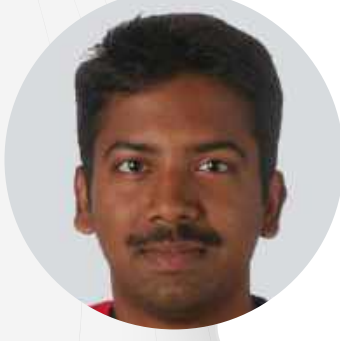
He has received his PhD from IIT-KGP. His main areas of research are IoT and Wireless Access Network.



Prof. G. Srinivasaraghavan
Professor, IIITB



Prof. Srinivasaraghavan has a PhD in Computer Science from IIT-K and 18 years of experience with Infosys Technologies and several other companies.



Dr. Dinesh Babu Jayagopi
Professor, IIITB



Dr. Dinesh is currently an Associate Professor at IIIT-B where he heads the Multimodal Perception Lab. His research interests are in Audio-Visual Signal Processing, Machine Learning, and Social Computing. He obtained his doctorate from Ecole Polytechnic Federale Lausanne (EPFL), Switzerland.



Chandrashekar Ramanathan
Professor & Dean (Academics)



Prof. Chandrashekar is a faculty member at IIIT-B since 2007 serving as professor, researcher and administrator. He has been working in the field of Computing for over 25 years in various capacities across industry and academia.



Tricha Anjali
Ex-Associate Dean



Prof. Anjali has a PhD from Georgia Institute of Technology as well as an integrated MTech (EE) from IIT Bombay.

Industry Experts



Abhishek Vijayvargia
Senior Data Scientist



Having worked with Microsoft as a Senior Data Scientist, he is an alumnus of IIT Kharagpur with 10+ years of experience in a Data Science domain



Ex-Senior Data Scientist



Anand
CEO



CEO, Gramener A gold medallist from IIM Bangalore, an alumnus of IIT Madras and London Business School, Anand is among the top 10 data scientists in India with 20 years of experience.



Faculty



Principal



Ex-Consultant



Manish Shukla
Head of Generative AI



Leading cutting-edge GenAI platform development at NatWest Group. Expertise in OpenAI products and MLOps for optimisation of operational efficiency and seamless project delivery with high user satisfaction.



Release Manager



Release Manager



Certified Scrum Master



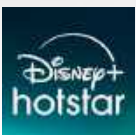
Deependra Singh
VP & Head of Data Science



Over 15 years of experience in leading analytics practices, data science, deep learning, and AI product development. Successfully led teams at Jungle Games, American Express Digital Business, and National Insurance Company, pioneering key projects like the analytics engine for the GOI PMJAY policy. Respected speaker at top educational institutes like IMT Hyderabad, BIT Mesra, and NMIMS.



Sajan Kedia
Sr. Engineering manager



Senior Engineering Manager, Hotstar Sajan has extensive experience in the field of ML, Big Data, Data Science, and AI.



Sr. Engineering manager



Machine Learning Engineer



Machine Learning Research Engineer



Mirza Rahim Baig
Startup Mentor



Analytics Lead, Zalando Mirza is a veteran professional with 10+ years of experience in application of data science, machine learning in e-commerce and healthcare.



Team Lead - Product



Marketing Analytics



Ex-Analytics Lead

Assignments and Case Studies from 12+ In-Demand Business Domains



Retail & Ecommerce

ETL Pipelining with Spark



Media & Entertainment

Data Analysis using SQL



Transportation

EDA
using Python



Education

Model Selection
using Sklearn



Civil Engineering

Classification using
CNNs



HR

Semantic Classification
using Word2Vec



Manufacturing

Regularisation using
Sklearn



Healthcare

Classification using
Sklearn



Law

RAG using
LangChain



InfoSec

Feature Engineering
using Sklearn



FMCG

Big Data Analysis
using Spark



BFSI

Sequence Data Prediction
using RNN

Learn by Doing

Your Program Journey

Phase 0

Math and Programming Bootcamp (12 weeks)

Phase I

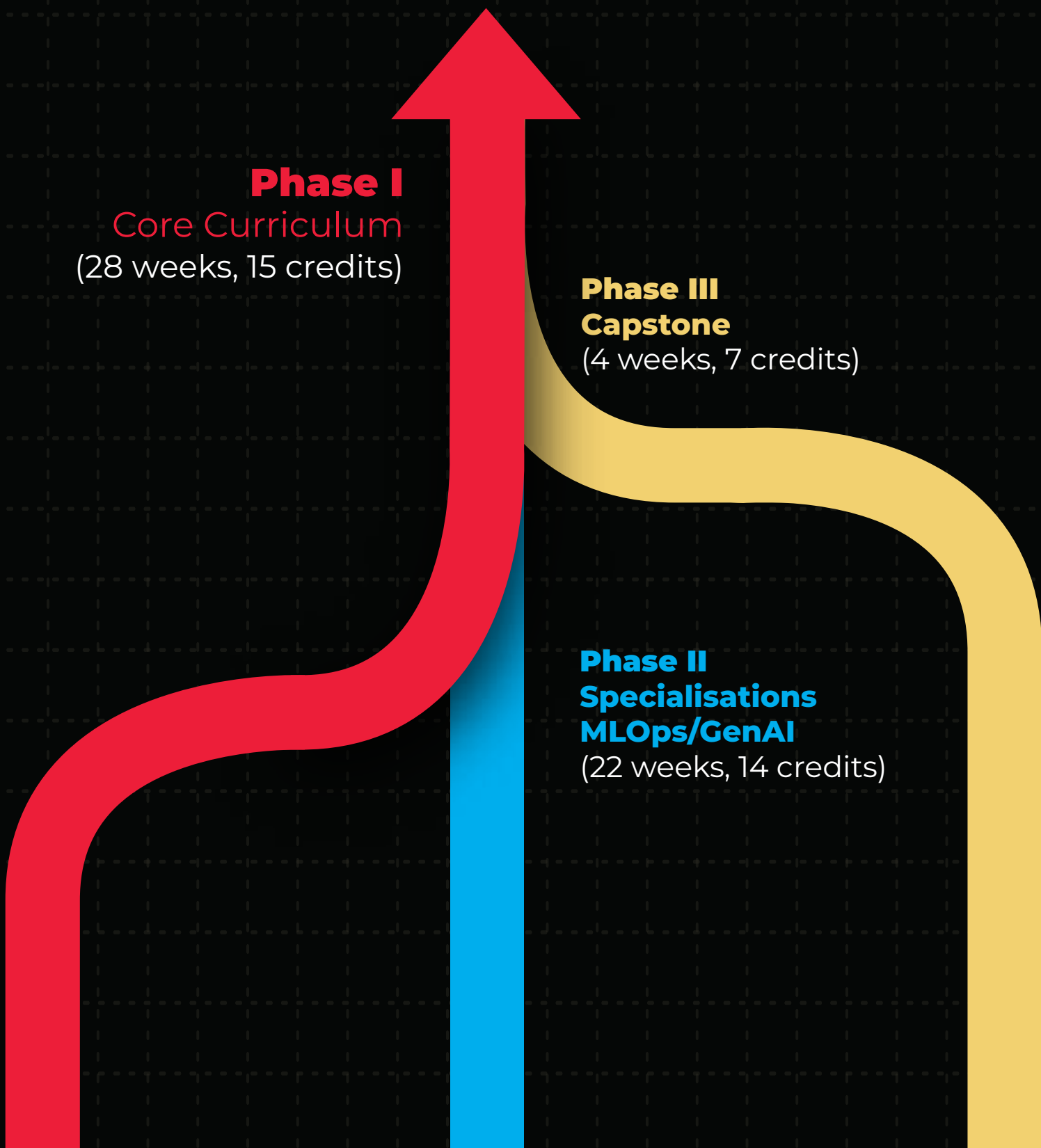
Core Curriculum
(28 weeks, 15 credits)

Phase III Capstone

(4 weeks, 7 credits)

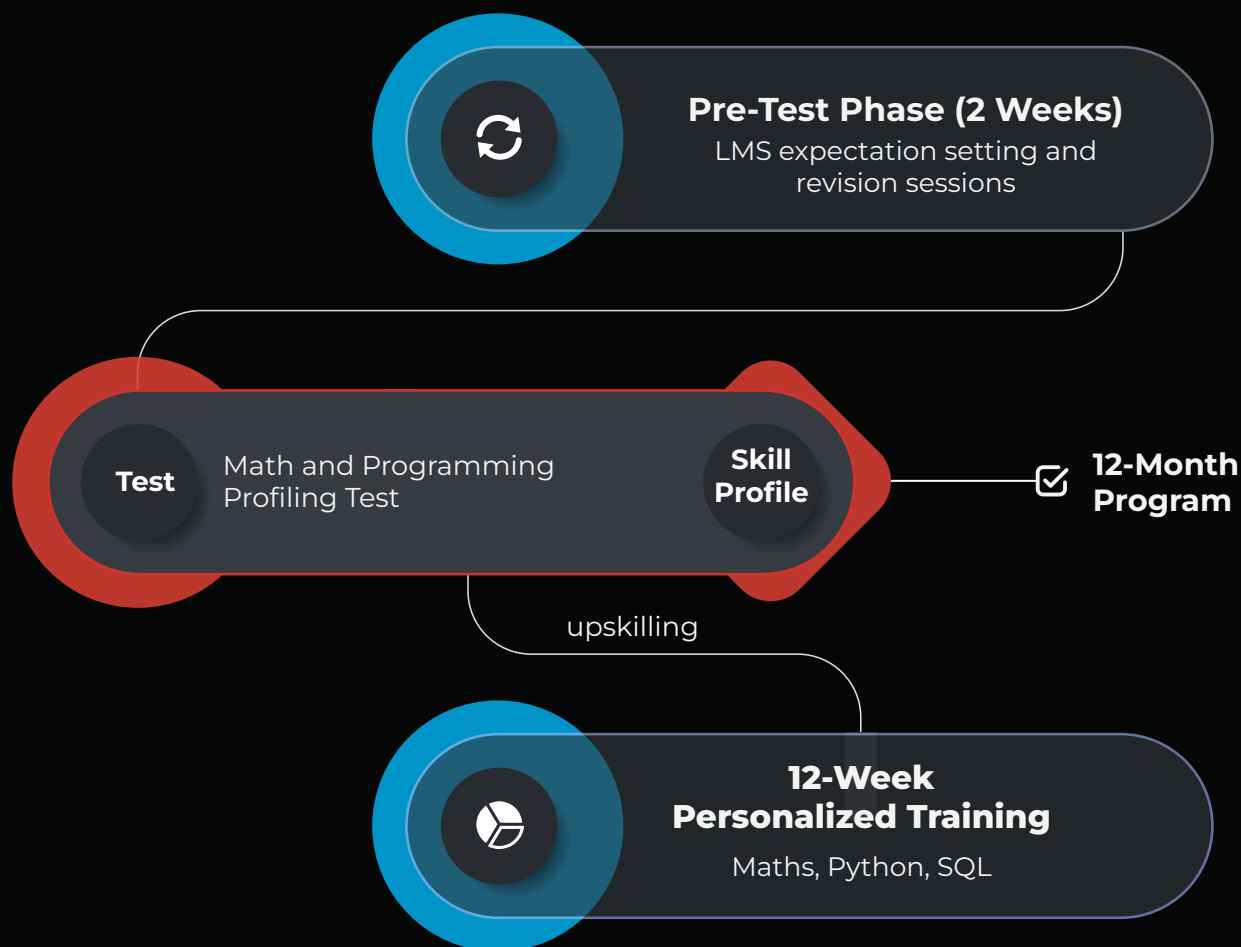
Phase II Specialisations MLOps/GenAI

(22 weeks, 14 credits)



Applied Math and Programming Bootcamp

Personalise the initial 3 months of the program to your profile



Topics: Sets, Combinatorics, Basics of Probability, Conditional Probability, Descriptive Statistics, Functions, Vector Algebra, Derivatives, Integrals, Coding Environments, Variables, Data Types, Syntax, Conditionals, Loops, Functions, Lists, Sets, Tuples, Dictionaries, Introduction to MySQL, Basic SQL Querying

Marks Structure: Total marks - 100

Section A - **40 marks** (basic mathematics)

Section B - **60 marks** (basic programming)

Passing marks - 25 marks in section A & 35 marks in section B

No added cost to be paid for the bootcamp

We make sure that you are well-equipped to draw the most benefit from the program!

Core Curriculum

The core phase of the curriculum will equip you with the most up-to-date and industry-relevant skills and technologies for data science and machine learning such as programming and mathematics, data analysis tools and techniques, cloud computing and big data analytics, and foundational topics in machine learning, deep learning, and natural language processing.

Topics

Advanced Mathematics for Data Science and Machine Learning

Master essential mathematical concepts to understand how to work with large amounts of data and train efficient machine learning models

- » Conditional Probability and Probability Distributions
- » Advanced Linear Algebra and Linear Transformations
- » Multivariate Calculus

Advanced Programming for Data Science and Machine Learning

Wrangle real-world data using universal programming languages such as Python and SQL, and use GenAI for generating and debugging code faster

- » GenAI for Coding and Problem-Solving
- » Object-Oriented Programming
- » Python Data Science Libraries
- » Database Design and SQL Querying with MySQL
- » Introduction to NoSQL Databases

Data Analysis and Exploration

Implement industry-standard statistical methods using tools such as Python, Tableau, and Power BI to analyse data and derive business insights

- » Data Analysis with Python
- » Exploratory Data Analysis
- » Inferential Statistics and Hypothesis Testing
- » Data Analysis and Visualisation with Power BI and Tableau

Cloud Computing and Big Data Fundamentals

Take your data processing and analysis workflows to the cloud and work with larger amounts of data to derive enterprise-scale business insights

- » Cloud Computing with AWS, GCP, Microsoft Azure
- » Big Data Analysis with PySpark

Foundations of Machine Learning

Train industry-standard machine learning models to automate insight generation and predict business metrics behaviour

- » Machine Learning Paradigms
- » Linear and Logistic Regression
- » K Nearest Neighbors
- » Regularisation and Hyperparameter Tuning
- » Decision Trees and Ensembles
- » Clustering Models

Deep Learning and Natural Language Processing

Build and train deep neural network models for different kinds of business data such as images and sequences

- » Artificial Neural Networks
- » Convolutional and Recurrent Neural Networks
- » Lexical, Syntactic, and Semantic Processing

Deployment Fundamentals

Share and deploy your insights and machine learning models so that other collaborators can work with your contributions

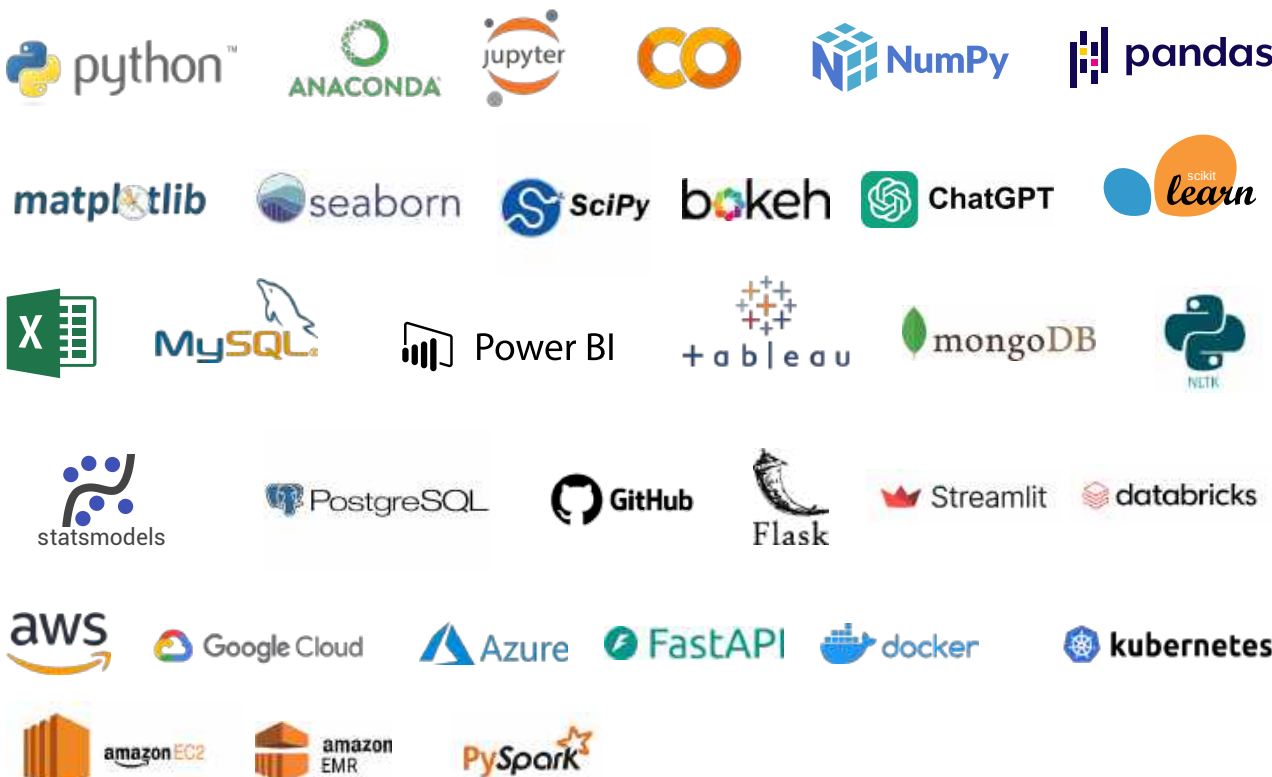
- » Containerisation and Deployment Tools
- » Version Control

Projects

- » **Querying with SQL**
Analyse Spotify music data for targeted recommendations or NDAP insurance data for risk assessment
- » **Exploratory Data Analysis**
Analyse NYC taxi operations for efficient taxi positioning or US beer production data for better brewery operation management
- » **Big Data Analysis**
Analyse Mercari products data for better targeted recommendations or customer interaction data to enhance customer engagement
- » **Linear Regression**
Predict household energy consumption using appliance energy readings data to increase power consumption efficiency or parcel delivery time for Porter using historical delivery data for better planning and management
- » **Deep Learning**
Predict stock prices of Microsoft, Amazon, Google, IBM, using their historical stock price variations or temperature/pressure readings in Morocco using historical weather data



Tools



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MLOps Specialisation

The machine learning operations (MLOps) specialisation of the curriculum will equip you with core in-demand and industry relevant skills and technologies essential for ML engineers such as advanced machine learning methods, modern deep learning architectures, real-time data processing and end-to-end data pipeline creation and monitoring, and model pipelining and monitoring at scale.

Topics

Advanced Machine Learning

Train advanced industry-oriented machine learning models for enhanced predictive power and stronger business insight generation

- Support Vector Machines and Naive Bayes
- Feature Engineering and Model Selection
- Dimensionality Reduction
- Time Series Analysis
- Association Rule Mining and Recommendation Systems
- Explainable AI

Advanced Deep Learning and Generative AI

Design and train advanced industry-standard deep learning architectures, and master core AI principles such as attention mechanisms, transformers, and prompt engineering

- Advanced CNN Architectures
- LSTMs and GRUs
- Transfer Learning Techniques
- Encoder-Decoder Architectures and Seq2Seq
- Machine Translation
- Attention Mechanisms and Transformers
- Fundamentals of Generative AI and Prompt Engineering
- Computer Vision, Variational Autoencoders, Generative Adversarial Networks
- Data and Model Security Principles

Large-Scale Data Pipelining

Build complete end-to-end data pipelines and automate them to generate both batch-wise and real-time business insights

- End-to-End Data Pipelining Fundamentals
- Pipeline Automation with AWS Lambda, GCP Functions, and Azure Automation
- Data Monitoring with Amazon CloudWatch, Google Cloud Monitoring, and Azure Monitor
- Feature Stores and Vector Databases
- Real-Time Analytics with Flink, Kafka, and Spark Streaming
- Real-Time Analytics with Amazon Kinesis, Google Cloud Pub/Sub and DataFlow, Azure Stream Analytics and Event Hubs
- Multicloud and Hybrid Cloud Operating Principles

Machine Learning Model Pipelining

Build end-to-end industry-ready ML model pipelines and design their functional behaviour such as training and inference

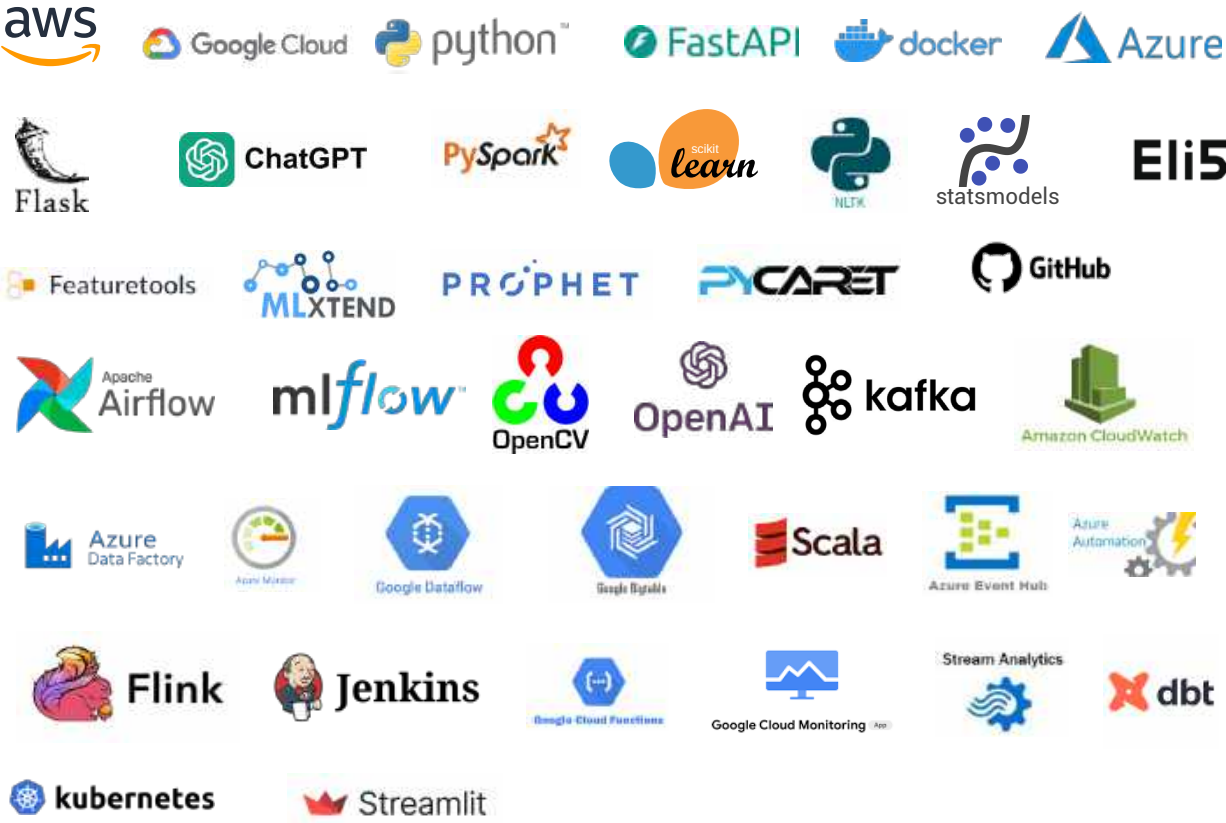
- Model Pipelining Principles
- Scheduling and Triggers
- Parallel Model Training and Real-Time Model Serving
- Data and Model Versioning
- Model Monitoring and System Design

Projects

- **Feature Engineering and Model Selection**
Predict fraudulent insurance claims using the Mendeley farmers insurance claims dataset or network intrusion events using historical network activity data
- **Semantic Classification**
Fake News Detection, Job Role Classification
- **Real-Time Data Analytics: Develop a**
real-time analytics pipeline for ecommerce data to enhance customer experience or a real-time patient health monitoring system for faster corrective actioning
- **Simulate and Retrigger Model Training Pipeline**



Tools



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Generative AI Specialisation

The generative artificial intelligence (GenAI) specialisation of the curriculum will equip you modern AI technologies and methods, particularly generative AI technologies, essential to data scientists and AI specialists, such as advanced machine learning methods, modern deep learning architectures, advanced prompt engineering and generative AI system design, information retrieval and retrieval-augmented generation, large language model (LLM) deployment, advanced computer vision and 3D vision, GenAI optimisations, and AI ethics.

Topics

Advanced Machine Learning

Train advanced industry-oriented machine learning models for enhanced predictive power and stronger business insight generation

- Support Vector Machines and Naive Bayes
- Feature Engineering and Model Selection
- Dimensionality Reduction
- Time Series Analysis
- Association Rule Mining and Recommendation Systems
- Explainable AI

Advanced Deep Learning for Generative AI

Design and train advanced industry-standard deep learning architectures, and master core AI principles such as attention mechanisms, transformers, and prompt engineering

- Advanced CNN Architectures
- LSTMs and GRUs
- Transfer Learning Techniques
- Encoder-Decoder Architectures and Seq2Seq
- Machine Translation
- Attention Mechanisms and Transformers
- Fundamentals of Generative AI and Prompt Engineering
- Computer Vision, Variational Autoencoders, Generative Adversarial Networks
- Data and Model Security Principles

GenAI System Design

Design and orchestrate generative AI systems to leverage the power of generative AI models and transform business operations

- Advanced Prompt Engineering and GenAI System Design
- Prompting Multimodal Models
- LLM Frameworks such as LangChain and LLaMa Index
- LLM Evaluation Methods
- Data Security and Governance
- AI Ethics

Advanced Generative AI

Develop AI-based cutting-edge industry-level systems for greater business efficiency such as retrieval-augmented generation (RAG) systems and multimodal GenAI model prompt engineering

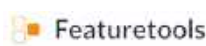
- Information Retrieval Principles
- Embeddings and Vector Databases
- RAG Architectures
- Agentic Systems and Multi-Agent Systems
- Advanced Multimodal GenAI Models
- LLM Deployment
- Advanced Computer Vision and 3D Vision
- GenAI Optimisations

Projects

- **Feature Engineering and Model Selection**
Predict fraudulent insurance claims using the Mendeley farmers insurance claims dataset or network intrusion events using historical network activity data
- **Semantic Classification**
Fake News Detection, Job Role Classification
- **GenAI System Design**
Analyse Amazon customer reviews to identify prevalent sentiments and themes to improve product offerings and enhance customer satisfaction or ChatGPT customer feedback to derive actionable insights for business improvement
- **RAG**
Develop an RAG system to transform Long Beach County Municipal meetings transcripts into actionable insights for better organisational communication and decision making or historical legal documentations to optimise legal workflows



Tools



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Capstone Projects

Capstone that Adapts to Your Preference

Infuse our Capstone with Your Data

Modify existing projects as per your industry data and problems

Bring Your Own Capstone

Work on a completely novel project of your choice and solve problems that excite you

Pre-Designed Industry Capstone

Choose one of our existing projects that cover in-demand trending industry domains

Bring Your Own Capstone

Design your own capstone project relevant to your domain and interest, and get feedback throughout your capstone stages



Identify a real-world problem relevant to your domain



Source datasets aligned with your business problem



Design and implement your solution



Document your efforts and present your findings



Continuous expert feedback at every step of capstone

Build A Strong Portfolio



- **Commits**
Demonstrate consistency, collaboration, and coding discipline
- **Code**
Showcase well-documented repositories
- **Projects**
Host end-to-end DS/ML/AI projects that highlight real-world problem-solving

- **Kernels**
Highlight data processing and EDA methodologies
- **Ranking**
Evaluate and reflect global standing among data science practitioners
- **Competitions**
Demonstrate problem-solving under tight constraints

- **Headline**
Concise summary of goals, competencies, and professional identity
- **Summary**
Engaging overview of learnin and career journey
- **Projects**
Showcase practical experience, outcomes, and skill application

GitHub helps with

- ✓ Validating coding skills
- ✓ Showing growth and consistency
- ✓ Being interview-ready for Tech roles

Kaggle helps with

- ✓ Building credibility in data science circles
- ✓ Applying learning to real datasets
- ✓ Speaking confidently in Tech interviews

LinkedIn helps with

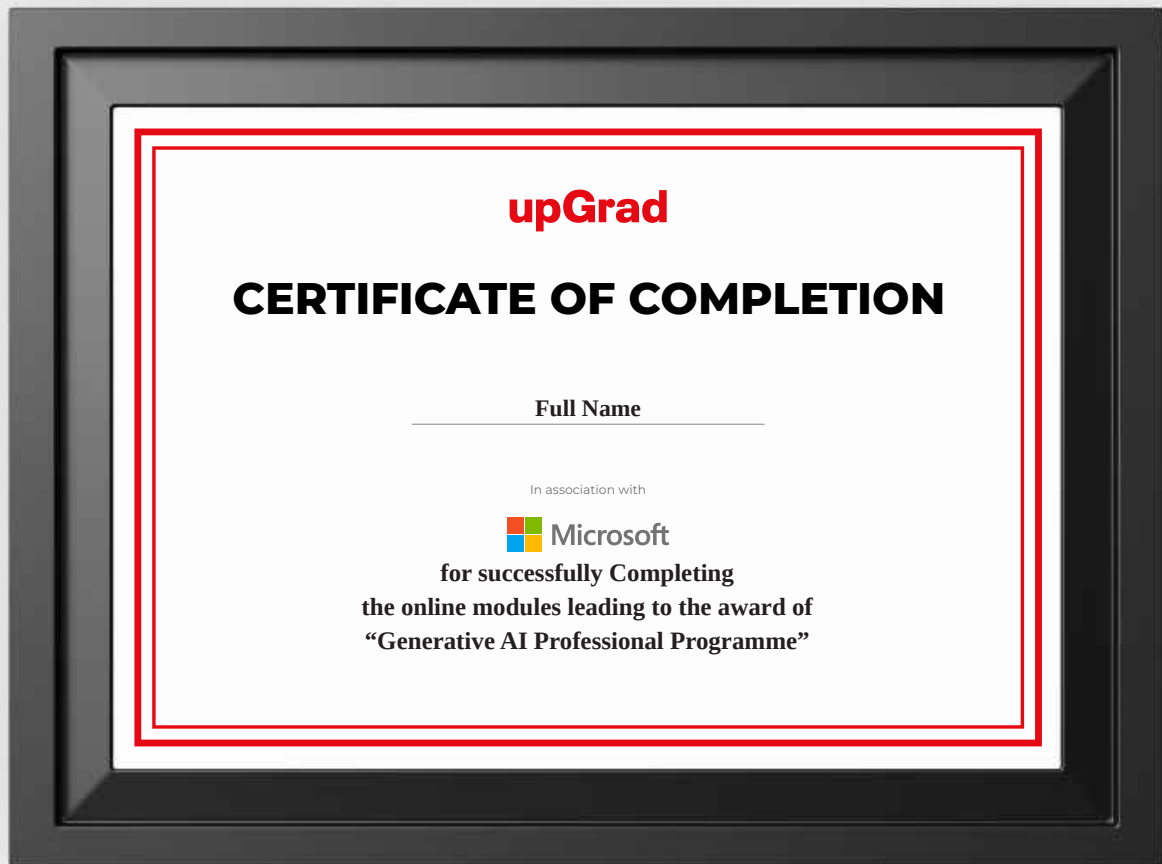
- ✓ Improving visibility with recruiters
- ✓ Positioning better for job openings
- ✓ Networking with peers and mentors in the field

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Microsoft-Certified Advantage –

This program doesn't just prepare you for the world of Data Science & AIML—it gives you the Microsoft edge.

Learners earn industry backed certification from upGrad in association with Microsoft by completing specially designed modules integrated into the program, boosting both credibility and career readiness.



Microsoft Learn content modules Certification Modules:

- Introduction to Generative AI Concepts
- Introduction to GitHub Copilot
- Design & Manage Analytics Solutions using Power BI
- Designing & Implementing a Data Science Solution on Azure

Rich and Dedicated **Live Support**

Industry Expert Sessions

Engage with industry practitioners as they help you master in-demand skills and concepts using a demonstrative hands-on approach



IIITB Faculty Sessions

Learn from some of the most accomplished academicians as they take your knowledge and understanding of data science to another level

Just-In-Time Interview Support

Participate in Technical and HR mock interviews designed to boost your confidence and prepare you to ace interviews.

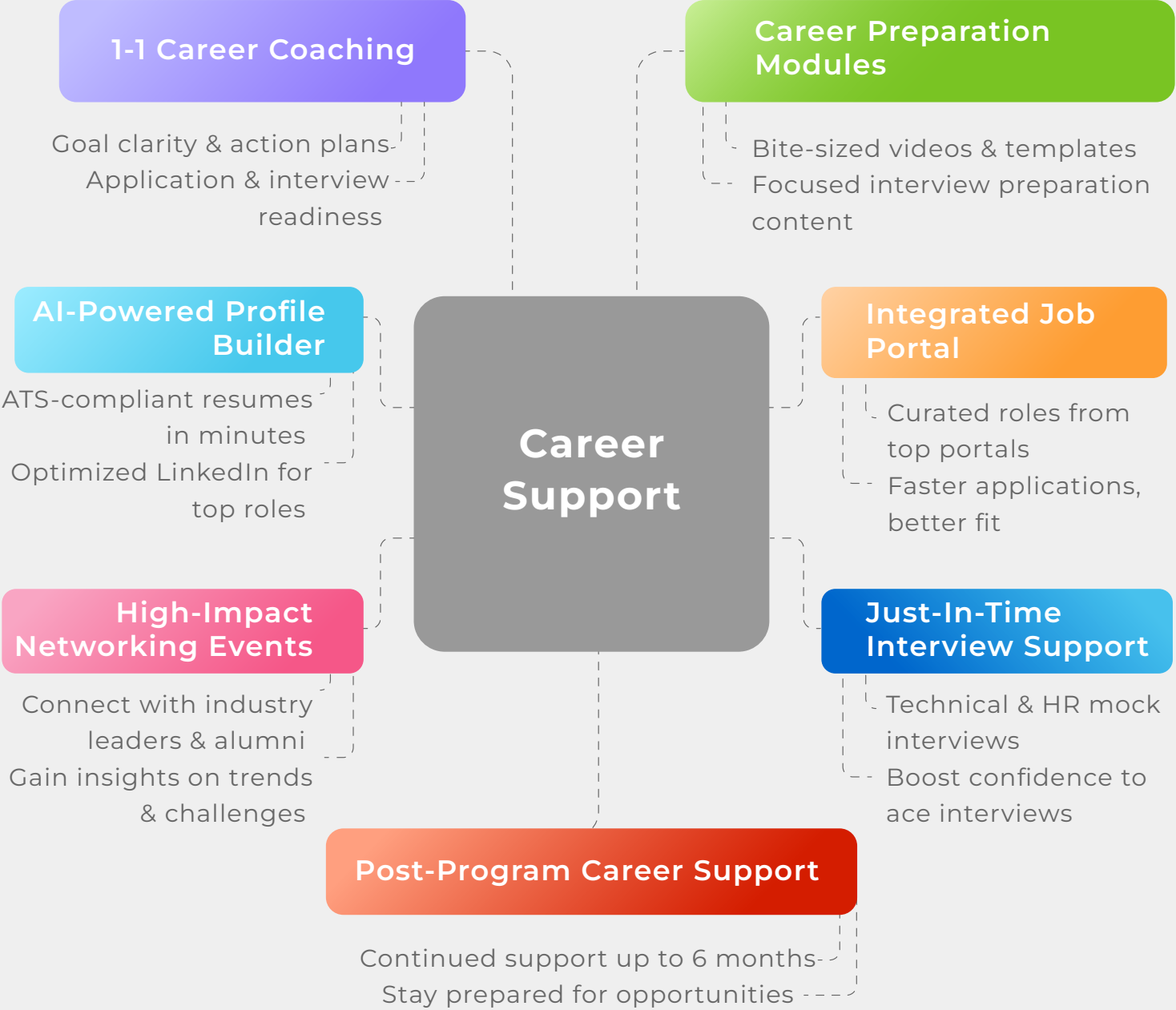
Career Coaching Sessions

Engage in Career Coaching Sessions via Career preparation modules, High-impact networking events and Just-in-time mock interviews

Daily doubt resolution sessions

Join doubt resolution session slots, that are available daily, and have an expert available to resolve your queries for a smooth learning journey

Effective and Complete Career Support



Student Support

Telegram channel for learner communications

Cohort Telegram channel for timely program updates and announcements.

Non-academic and non-technical query assistance

24 X 7 m-AI-ask bot support for non academic assistance.

Completion Support

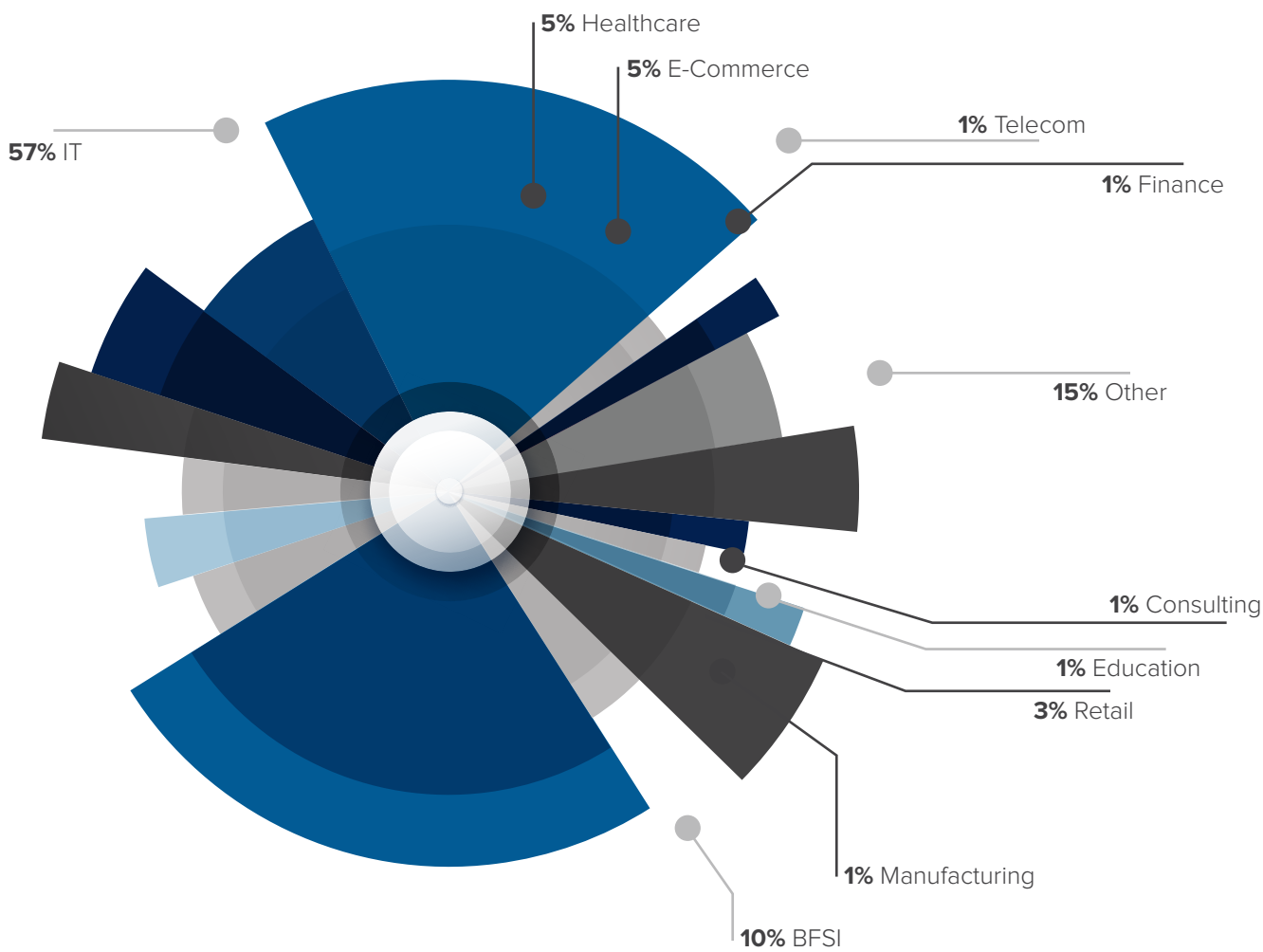
Personalised assistance for smooth program completion, managing backlogs, and cohort deferrals with free and paid waiver options.

Financial benefits

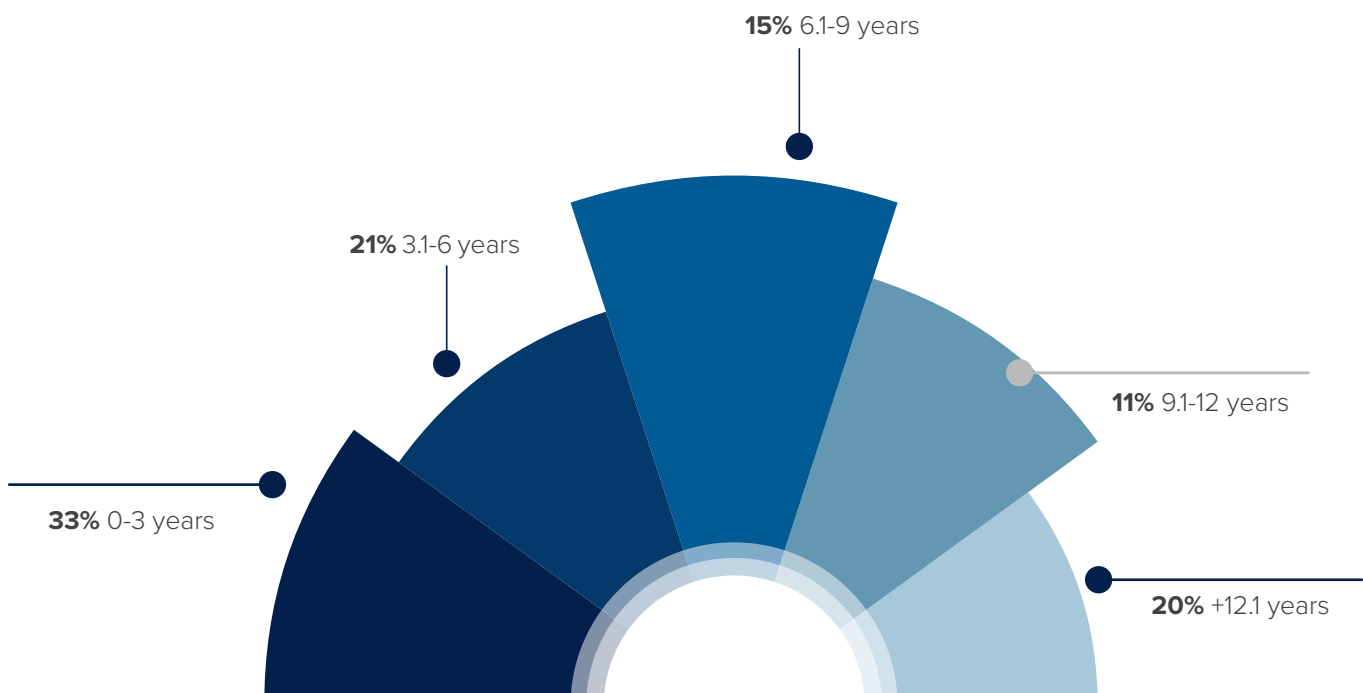
Access benefits like referrals and repeats by sharing details with your program co-ordinator or student support executive.

Meet the Class

Industries Our Students Come From



Work Experience





A single specialty is no longer enough.

For years, the T-shaped professional was considered the gold standard—someone with broad knowledge across multiple areas (the top of the “T”) and deep expertise in one core area (the vertical stem). This model worked well in stable environments where projects moved slowly, and roles were clearly divided

But today’s AI-driven, high-velocity world has exposed the limits of the T-shape. Knowledge silos, handoff delays, and rising complexity mean one deep skill is no longer enough

Today's employers need π -shaped professionals - someone with two deep, complementary specialisations while maintaining broad knowledge across domains, who can own entire project lifecycles without team dependencies, bridging the gap between development, operations, and analytics seamlessly

Introducing The π Professional Pack

Add an extra specialisation in just 3 months



STEP 1

Choose your primary foundation

Select your core specialisation:

Data Engineering, Data Analytics, Generative AI or MLOps

STEP 2

Add your strategic second specialty

In the latter half of your program, unlock a complete second track featuring:



Full curriculum coverage



Live interactive sessions with industry experts



Dedicated faculty support



Expert mentorship across both domains

Why is this a strategic career investment?



Expert in two complete specialisations



Industry-validated skill combinations based on hiring data



Hands-on experience in both domains



Portfolio projects demonstrating π -shaped expertise

Your Competitive Edge:

While others graduate with single specialisations, you emerge as a π -shaped leader ready to:



Own complete project lifecycles



Bridge team silos and communication gaps



Command premium positions in India's GCC & IT sectors



Adapt quickly as AI continues reshaping the industry

Popular specialisation combos

GenAI + MLOps

Why Gen AI + MLOps?

Enables end-to-end ownership of Gen AI models from fine-tuning to deployment, monitoring, and safety. This pairing creates a professional who can not only innovate but also reliably deliver and manage that innovation.

GenAI + Analytics

Why GenAI + Analytics?

Unlocks AI-driven business insights with a combination of natural language queries, auto-reporting, and data storytelling. This pairing is about creating smarter and more accessible business insights.

MLOps + Data Engineering

Why MLOps + Data Engineering?

This is the ultimate combination for building robust and scalable AI infrastructure. Enables you to build an automated, end-to-end machine learning factory that is efficient, reliable, and ready for production demands.

GenAI + Data Engineering

Why GenAI + Data Engineering?

Let's combine Gen AI innovation with solid data pipelines scalable AI products. This pairing ensures that innovative GenAI projects are built on a solid foundation, allowing them to be scalable and successful in the long run.

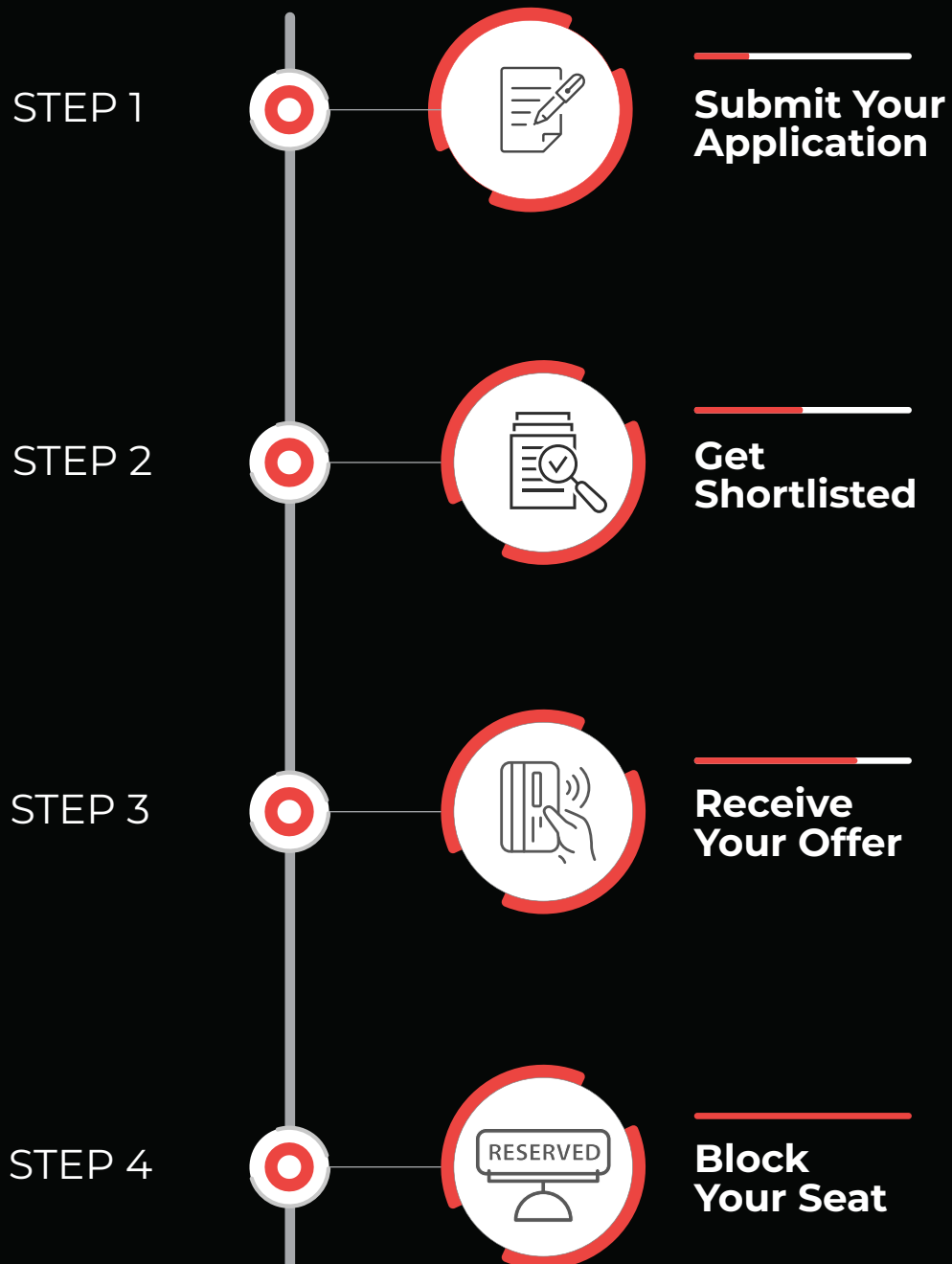


Option to articulate to a Master's degree
from Liverpool John Moores University after
successful completion of the program

LJMU



Enrol in 4 small steps,
Then take a giant leap.



Eligibility Criteria

Bachelor's or Master's Degree or its equivalent in any discipline with minimum 50% aggregate mark or equivalent CGPA.



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LET'S TALK



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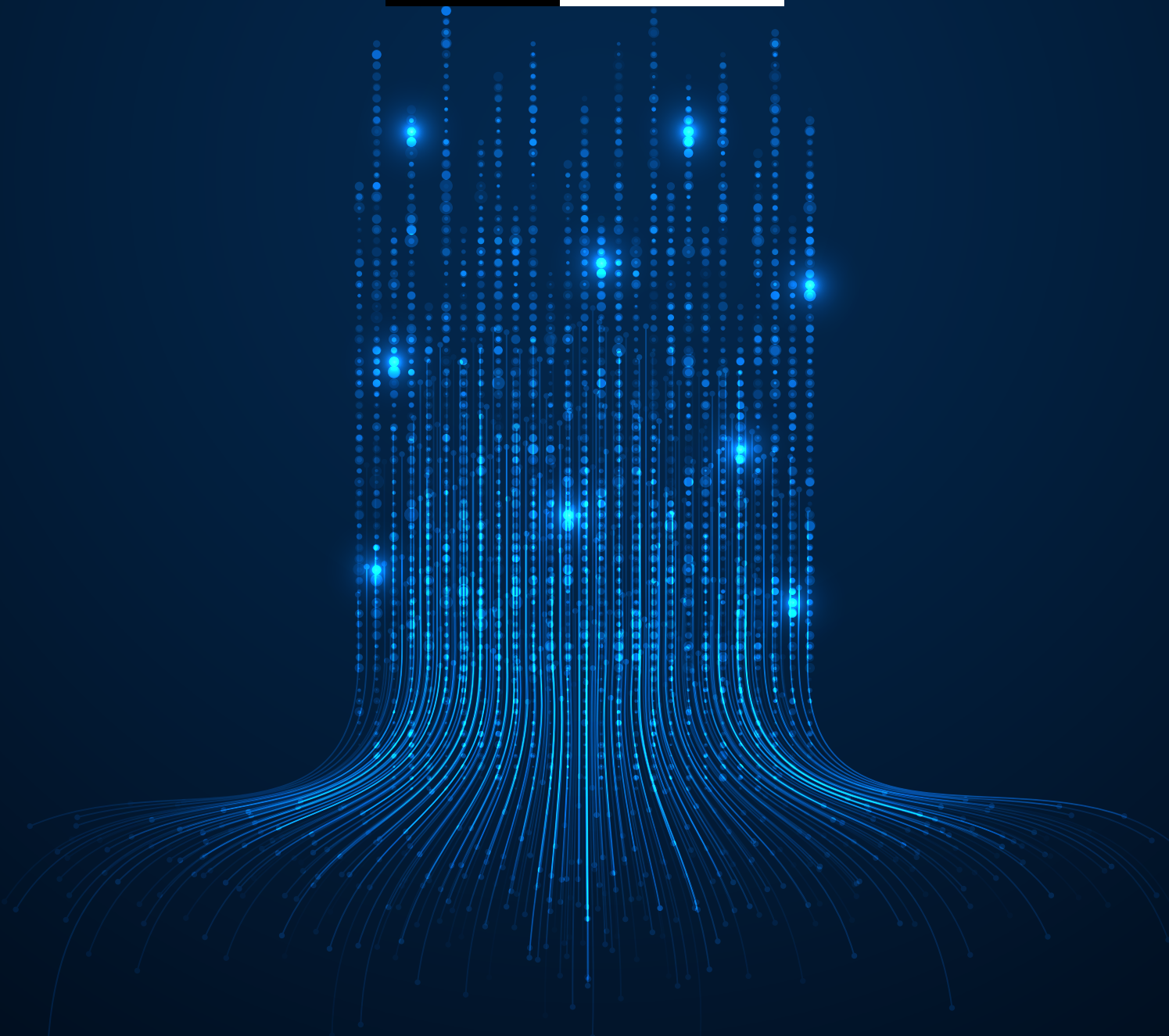
upGrad

Executive Post Graduate Certificate

Data Science & AI

Infused with GenAI

Duration 6-9 months



Largest DS & AI program in India

with 10+ Years of Legacy &
30,000+ Learners





About IIITB

The International Institute of Information Technology Bangalore (IIITB)

Established in 1998, IIITB is a premier institute known for its interdisciplinary approach, integrating technology with social sciences. Supported by the Government of Karnataka and the IT industry, IIITB fosters strong academic-industry partnerships and attracts top talent from across India and abroad through its merit-based selection process.

The institute has graduated over 3,500 students, many of whom work at leading IT companies globally. With a focus on research and development in fields like Artificial Intelligence (AI) and Machine Learning (ML), IIITB is recognised as a leader in AI education.

Ranked 74th in the Engineering category of the National Institutional Ranking Framework (NIRF) in 2022, IIITB continues to excel in education and research, making it a preferred destination for aspiring technologists and future leaders.



AICTE



UGC

ABOUT upGrad

“upGrad is a leading global learning and workforce development company. We’re on a single-minded mission of powering career success for every member of the global workforce as their trusted lifelong learning partner. Established in 2015, we have over 10 million learners who have upskilled in a range of online and offline programs from top universities in India and the world.”



Ronnie Screwvala

Co-founder & Executive Chairman

GSV EDTECH 150
Top 150 Most Transformative
Growth Companies

**BRANDON HALL
GOLD AWARDS**
15+ Across 2 Years

PROMISING BRAND 2022
The Economic Times

**EDUCATION COMPANY
OF THE YEAR**
VC Circle, 2022

Entrepreneur India
**EDUCATION INNOVATION
AWARDS 2020**
Best Communication &
Teaching Platform

Current Industry Trends

14 LPA

Average salary
for Data Scientists
in India (2025)

Source: Glassdoor

57.5%

CAGR of Indian
Data Science market
till 2028

Source: Milleniumpost

50%

Increase in search
volume for 'Data Science'
since 2020

Source: Exploding Topics

36%

Increase in growth of
Data Science jobs
by 2031

Source: Harvard Extension School

\$378.7B

Size of global
Data Science market
by 2030

Source: Milleniumpost

Program Highlights

Here are the
top reasons why
you should consider
this program



Future-Ready Curriculum

Master In-Demand and Trending Competencies



Personalised Learning Experience

Learning Experience Tailored to Your Needs



In-Demand Tools

30+ Industry Tools, Languages, Libraries



Outcome-Driven Learning Experience

Personalised Portfolio-Building Support and Career Preparation Sessions



Best-in-Industry Experts

Decorated IIITB Faculty and Top Industry Practitioners



Golden Learning Ratio

Perfect Blend of Mathematics, Technology, and Business Understanding



Hands-on Learning

Solve 15+ Domain-Focused Assignments and Case Studies

Career Transitions



Abhishek Singh
MIS Executive (F&B)

Senior Data Analyst, Myntra



Ayush Modi
Associate Consultant (IT)

Marketing Analyst,
Globalization Partners



Ashish Y
Asst. Manager
(Manufacturing/Production)

Business Analyst, Arvind



Jai Krishna
Fresher

Business Analyst, Quantzig



Shadab Hussain
Data Scientist (IT)

Analyst, TheMathCompany



Damodar Bandi
Global Supply Chain Analyst
(Software)

Sr. Business Analyst, Data
Semantics



Shubhadip B
Technical Project Manager
(Tech Solutions)

Sr. Technical Project Manager,
Aurionpro



Deepshikha
Senior Associate (Tech)

Data Analyst, Amazon



Rohit W
Quality Assurance Tester
(BFSI)

Python Automation Engineer,
Credit Suisse



Hariharan S
AVP (BFSI)

Project Manager, HSBC



Mohit Mamgain
Internship (IT)
Batch: March 2018

Data Analyst, Shine.com



Aakash Dusane
Software Engineer (Software)

Data Scientist, Quantzig

Program Completion Certificate



Earn valuable credentials with an Executive Post Graduate Certificate in Data Science and Artificial Intelligence accredited with NAAC A+ (2021), and join India's largest DS & AI alumni network of over 10,000 professionals

Faculty



Dr. Debabrata Das
Director of IIITB



He has received his PhD from IIT-KGP. His main areas of research are IoT and Wireless Access Network.



Prof. G. Srinivasaraghavan
Professor, IIITB



Prof. Srinivasaraghavan has a PhD in Computer Science from IIT-K and 18 years of experience with Infosys Technologies and several other companies.



Dr. Dinesh Babu Jayagopi
Professor, IIITB



Dr. Dinesh is currently an Associate Professor at IIIT-B where he heads the Multimodal Perception Lab. His research interests are in Audio-Visual Signal Processing, Machine Learning, and Social Computing. He obtained his doctorate from Ecole Polytechnic Federale Lausanne (EPFL), Switzerland.



Chandrashekar Ramanathan
Professor & Dean (Academics)



Prof. Chandrashekar is a faculty member at IIIT-B since 2007 serving as professor, researcher and administrator. He has been working in the field of Computing for over 25 years in various capacities across industry and academia.



Tricha Anjali
Ex-Associate Dean



Prof. Anjali has a PhD from Georgia Institute of Technology as well as an integrated MTech (EE) from IIT Bombay.

Industry Experts



Abhishek Vijayvargia
Senior Data Scientist



Having worked with Microsoft as a Senior Data Scientist, he is an alumnus of IIT Kharagpur with 10+ years of experience in a Data Science domain



Ex-Senior Data Scientist



Anand
CEO



CEO, Gramener A gold medallist from IIM Bangalore, an alumnus of IIT Madras and London Business School, Anand is among the top 10 data scientists in India with 20 years of experience.



Faculty



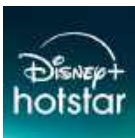
Principal



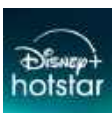
Ex-Consultant



Sajjan Kedia
Sr. Engineering manager



Senior Engineering Manager, Hotstar Sajjan has extensive experience in the field of ML, Big Data, Data Science, and AI.



Sr. Engineering manager



Machine Learning Engineer



Machine Learning Research Engineer



Mirza Rahim Baig
Startup Mentor



Analytics Lead, Zalando Mirza is a veteran professional with 10+ years of experience in application of data science, machine learning in e-commerce and healthcare.



Team Lead - Product



Marketing Analytics



Ex-Analytics Lead

More than 30+ In-Demand Industry Tools and Technologies

Python for Data Science



Data Analytics and Business Intelligence



Cloud Computing, Big Data, Software Fundamentals



Foundations of Machine Learning



Deep Learning and Natural Language Processing



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Assignments and Case Studies from 6+ In-Demand Business Domains



InfoSec

Feature Engineering
using Sklearn



Media & Entertainment

Data Analysis using SQL



Transportation

EDA
using Python



Manufacturing

Regularisation using
Sklearn



Civil Engineering

Classification using
CNNs



BFSI

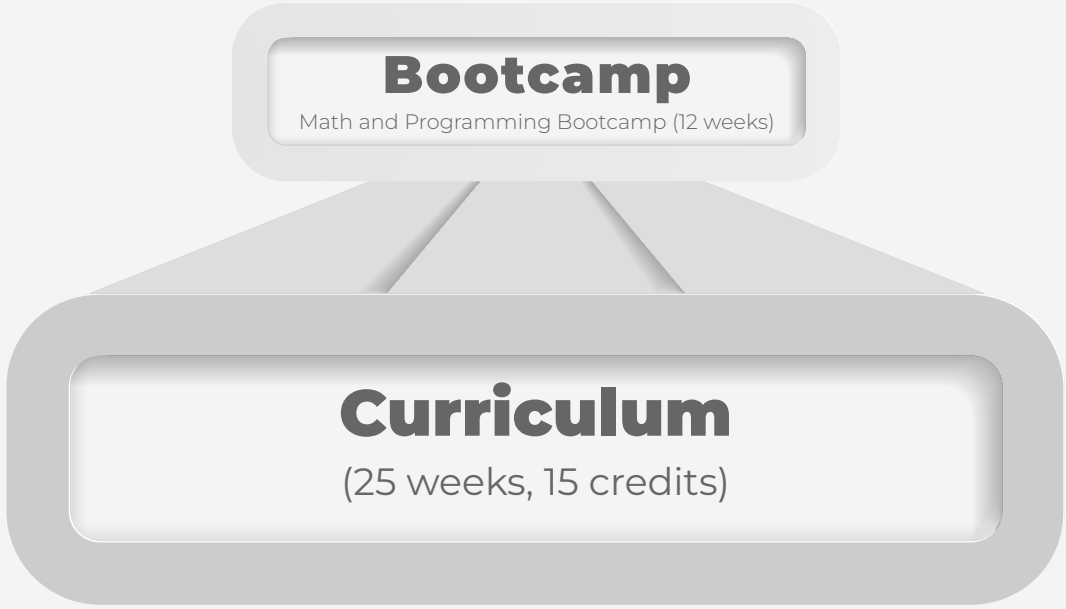
Sequence Data Prediction
using RNN



FMCG

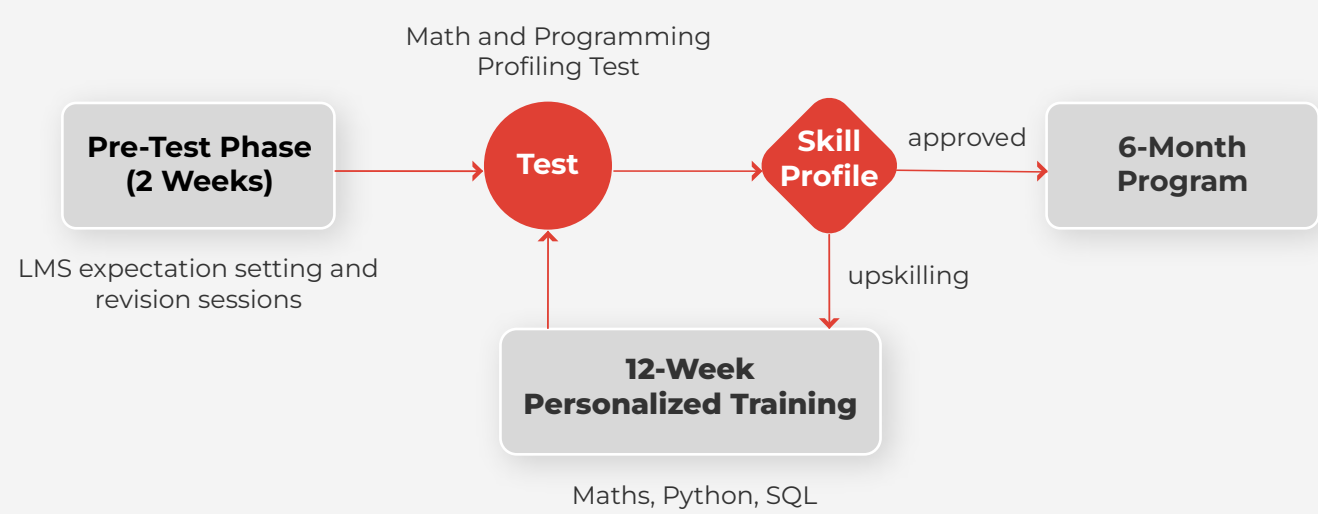
Big Data Analysis
using Spark

Your Program Journey



Applied Math and Programming Bootcamp

Personalise the initial 3 months of the program to your profile



Topics: Sets, Combinatorics, Basics of Probability, Conditional Probability, Descriptive Statistics, Functions, Vector Algebra, Derivatives, Integrals, Coding Environments, Variables, Data Types, Syntax, Conditionals, Loops, Functions, Lists, Sets, Tuples, Dictionaries, Introduction to MySQL, Basic SQL Querying

Marks Structure: Total marks - 100

- **Section A - 40 marks**, basic mathematics **Section B - 60 marks**, basic programming
- Passing marks - 25 marks in section A & 35 marks in section B

*Unlimited attempts at the test and the bootcamp at no added cost!
We make sure that you are well-equipped to draw the most benefit from the program!*

Curriculum

The curriculum will equip you with the most up-to-date and industry-relevant skills and technologies for data science and machine learning such as programming and mathematics, data analysis tools and techniques, cloud computing and big data analytics, and foundational topics in machine learning, deep learning, and natural language processing.

Topics

Advanced Mathematics for Data Science and Machine Learning

Master essential mathematical concepts to understand how to work with large amounts of data and train efficient machine learning models

- Conditional Probability and Probability Distributions
- Advanced Linear Algebra and Linear Transformations
- Multivariate Calculus

Advanced Programming for Data Science and Machine Learning

Wrangle real-world data using universal programming languages such as Python and SQL, and use GenAI for generating and debugging code faster

- GenAI for Coding and Problem-Solving
- Object-Oriented Programming
- Python Data Science Libraries
- Database Design and SQL Querying with MySQL
- Introduction to NoSQL Databases

Data Analysis and Exploration

Implement industry-standard statistical methods using tools such as Python, Tableau, and Power BI to analyse data and derive business insights

- Data Analysis with Python
- Exploratory Data Analysis
- Inferential Statistics and Hypothesis Testing
- Data Analysis and Visualisation with Power BI and Tableau

Cloud Computing and Big Data Fundamentals

Take your data processing and analysis workflows to the cloud and work with larger amounts of data to derive enterprise-scale business insights

- Cloud Computing with AWS, GCP, Microsoft Azure
- Big Data Analysis with PySpark

Foundations of Machine Learning

Train industry-standard machine learning models to automate insight generation and predict business metrics behaviour

- Machine Learning Paradigms
- Linear and Logistic Regression
- K Nearest Neighbors
- Regularisation and Hyperparameter Tuning
- Decision Trees and Ensembles
- Clustering Models

Deep Learning and Natural Language Processing

Build and train deep neural network models for different kinds of business data such as images and sequences

- Artificial Neural Networks
- Convolutional and Recurrent Neural Networks
- Lexical, Syntactic, and Semantic Processing

Deployment Fundamentals

Share and deploy your insights and machine learning models so that other collaborators can work with your contributions

- Containerisation and Deployment Tools
- Version Control

Projects

- **Querying with SQL**
Analyse Spotify music data for targeted recommendations or NDAP insurance data for risk assessment
- **Exploratory Data Analysis**
Analyse NYC taxi operations for efficient taxi positioning or US beer production data for better brewery operation management
- **Big Data Analysis**
Analyse Mercari products data for better targeted recommendations or customer interaction data to enhance customer engagement
- **Linear Regression**
Predict household energy consumption using appliance energy readings data to increase power consumption efficiency or parcel delivery time for Porter using historical delivery data for better planning and management
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Predict stock prices of Microsoft, Amazon, Google, IBM, using their historical stock price variations or temperature/pressure readings in Morocco using historical weather data



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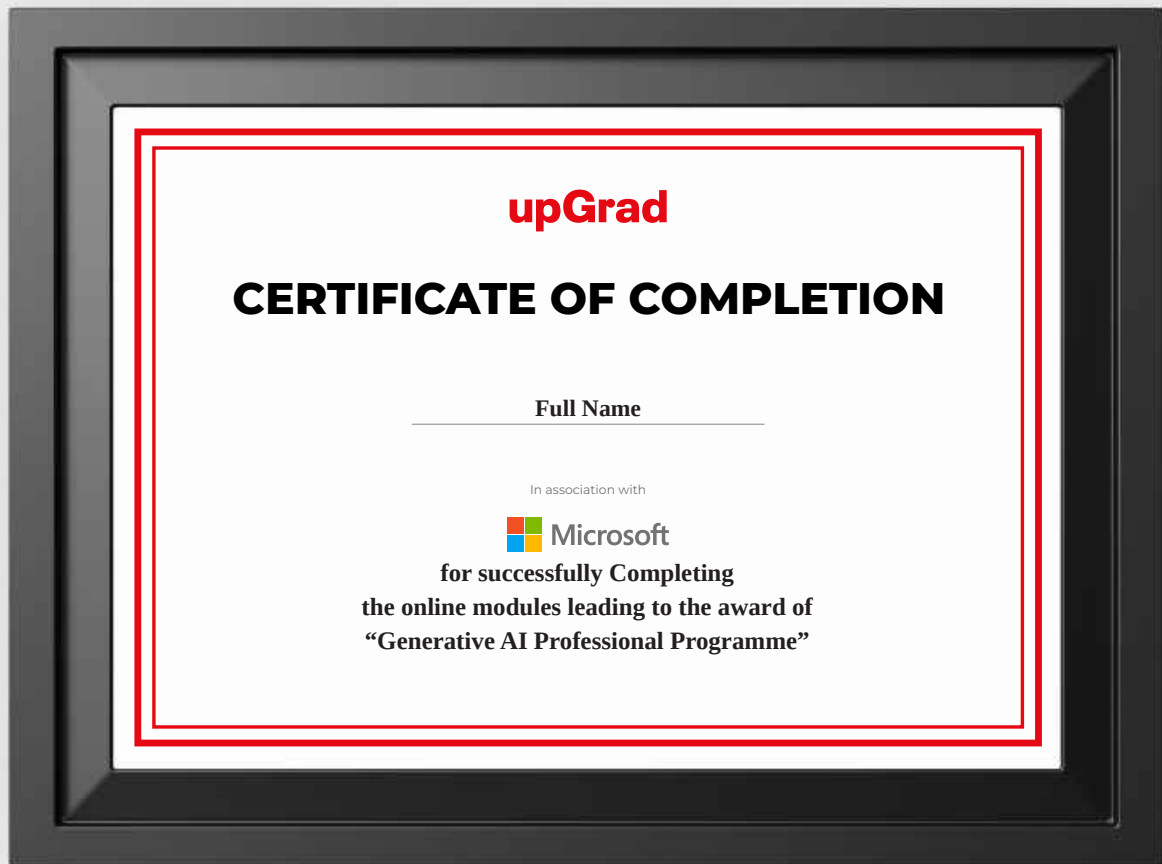
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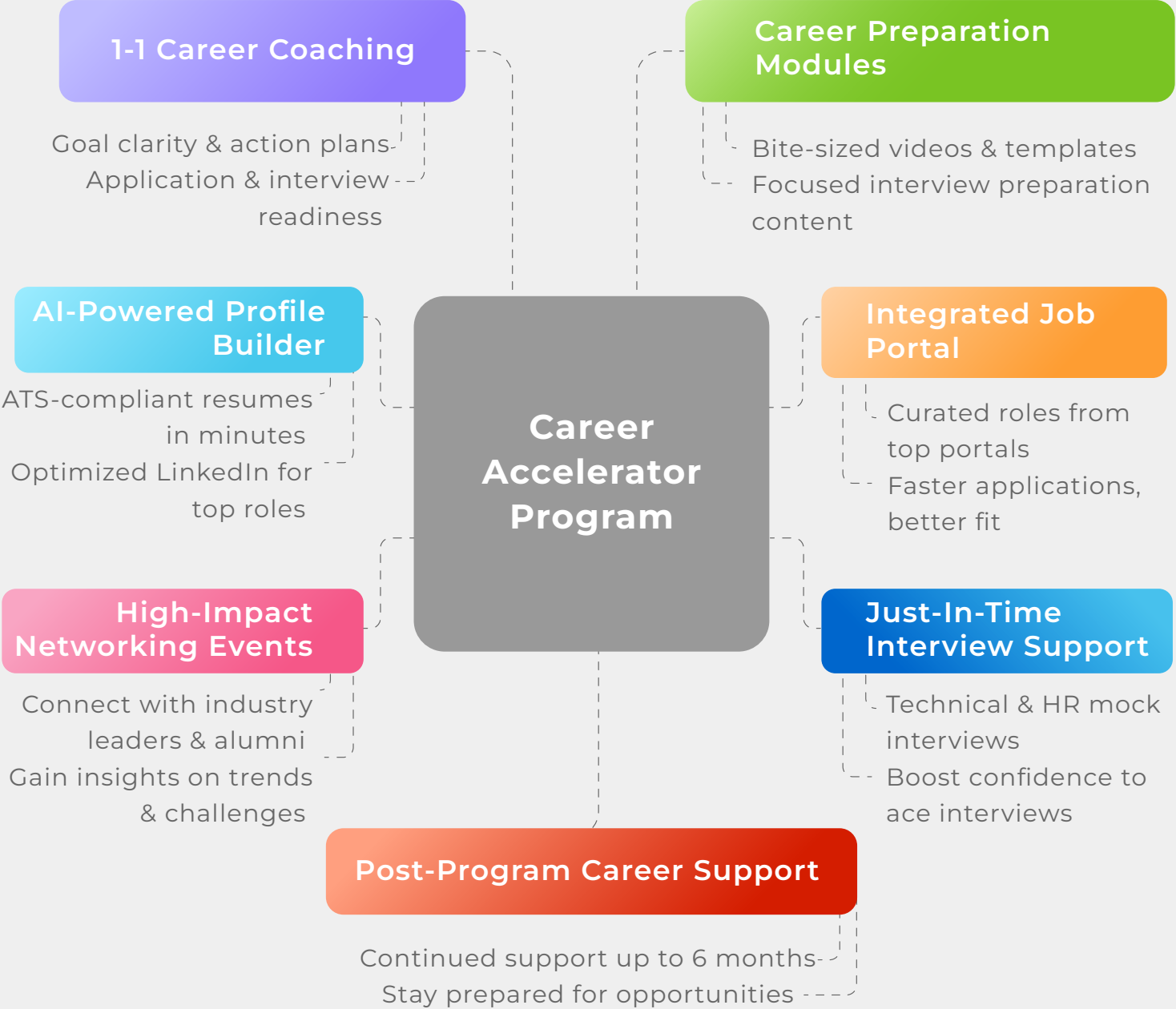
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24 X 7 m-AI-ask bot support for non academic assistance.

Completion Support

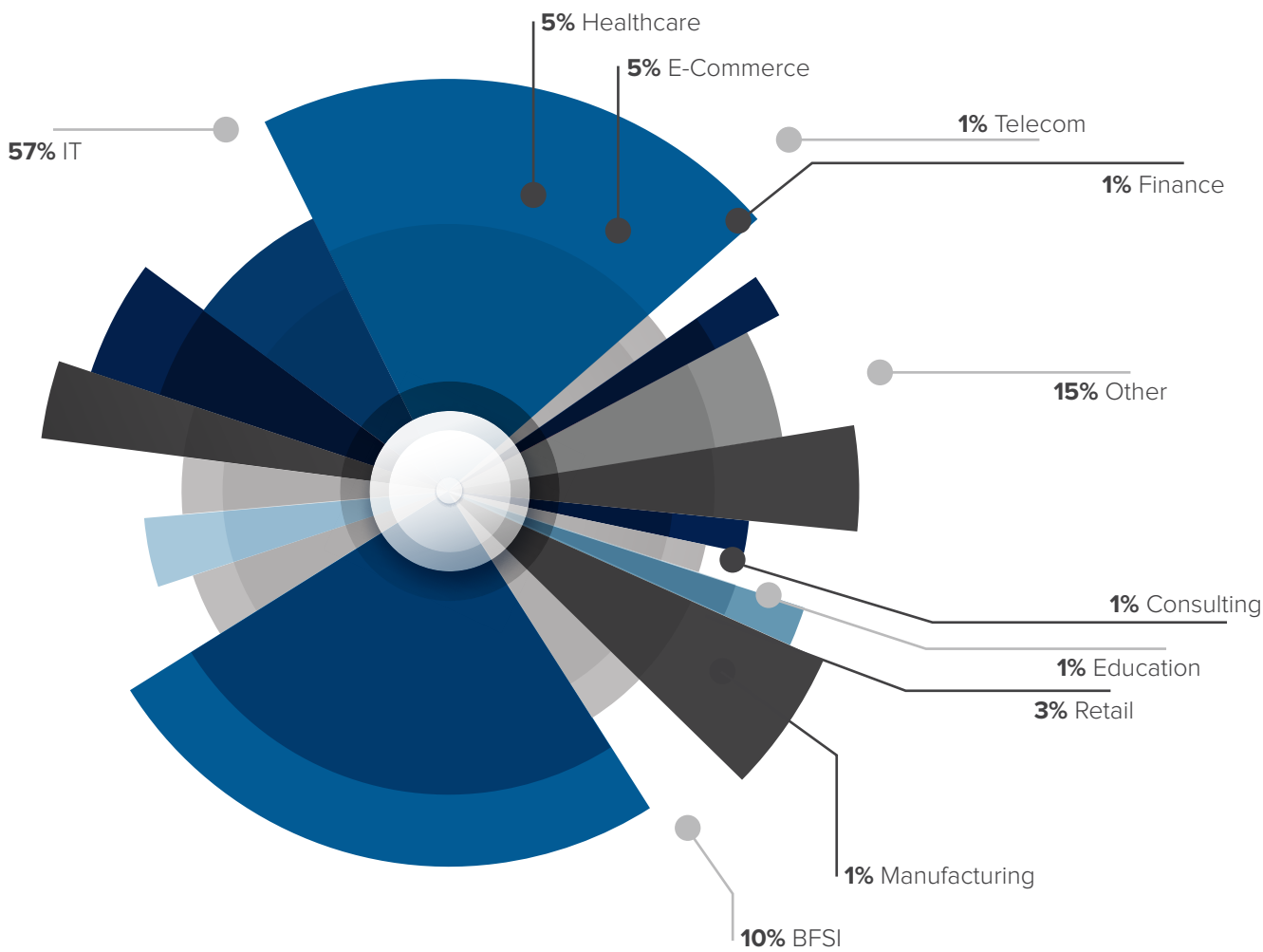
Personalised assistance for smooth program completion, managing backlogs, and cohort deferrals with free and paid waiver options.

Financial benefits

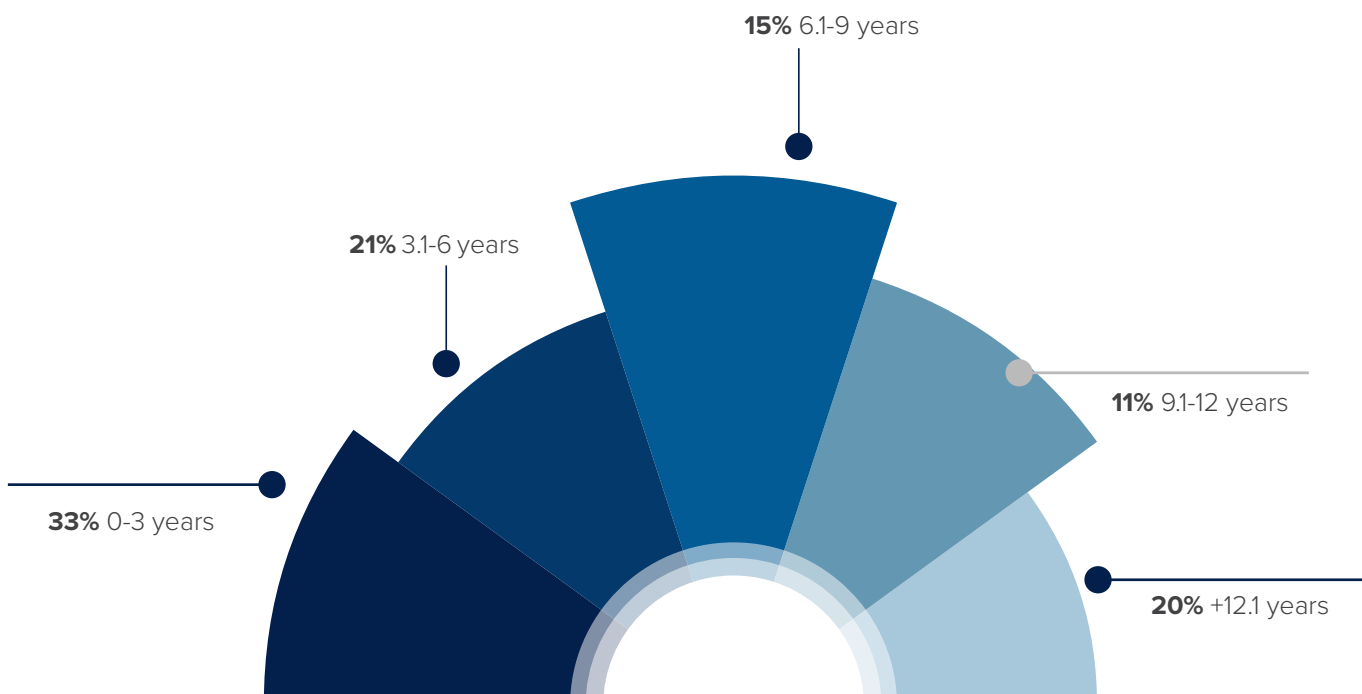
Access benefits like referrals and repeats by sharing details with your program co-ordinator or student support executive.

Meet the Class

Industries Our Students Come From



Work Experience





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from Liverpool John Moores University

LJMU



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Generative AI

Duration 18-21 months



The Era Of Generative AI



Usually, this first page is reserved for "About upGrad". But the world is at the cusp of Generative AI rapidly changing the world as we know it. At upGrad, we've always believed in imparting learners the skills necessary to thrive in the fast-evolving world of technology. We are hence quite thrilled to pioneer Generative AI as an elective in the Master of Science in Data Science.

With this key inclusion of Generative AI, learners will delve deeper into the fascinating realm of using Data Science to build practical applications like conversational AI chat bots, Image creators, and content recommenders amongst others, to solve real-world challenges. So dive into this brave new world of Generative AI and Large Language Models with us, and watch yourself transform into a 10x Data Scientist.

"IIT Bangalore prides itself in constantly updating cutting-edge topics to its curriculum. Our faculty has shaped this exciting Generative AI elective along with upGrad's industry experts, thus ensuring both academic rigour as well as incorporating the latest advancements in tech"

Dr. V. Sridhar,
Head-Faculty, IITB

"As an organisation that asks professionals to stay updated with the latest skills, we had to be one of the first to teach Generative AI. With this move, we are excited to witness the impact that Generative AI will have on the future, as well as the value our learners will bring to the field with this essential skill."

Mayank Kumar, Co-founder & MD
upGrad

About upGrad, IIITB, and LJMU, UK

upGrad has delivered over 20 million hours of learning, delivering programs by collaborating with universities across the world including Duke CE, IIIT Bangalore and Golden Gate University, among others

Online education is a fundamental disruption that will have a far-reaching impact. **upGrad** was founded taking this into consideration. upGrad is an online education platform to help individuals develop their professional potential in the most engaging learning environment. Since inception, upGrad is focussed on helping working professionals in their bid to learn, grow and move up in their career through a wide-range of programs designed to improve their expertise.

IIITB is a renowned university offering programs specialising in data science, machine learning and artificial intelligence. IIITB's faculty bring with them an average of 15 years of experience. The faculty covers the conceptual depths of topics such as Data Science, Machine Learning and AI, and Big Data Analytics. These will be complemented by industry relevant case studies from major industry verticals by industry leaders with 10+ yrs of experience from upGrad's industry network. The Executive Diplomas in DS and ML has been developed with the experienced faculty of IIITB in collaboration with Industry experts & upGrad to bring you cutting edge-curriculum with industry relevance. The strong placement network, industry mentorship and the credibility of this Executive Diploma from IIITB will provide you with just the right push to accelerate your career in Machine Learning and AI!

With a heritage that stretches back to **Liverpool John Moores University** UK is now one of the largest and well-established universities in the UK. It has been ranked in the Top 100 World Young Universities & Top 50 in the UK by Student Satisfaction. There are two faculties within the University - Faculty of Society and Culture and the Faculty of Health, Innovation, Science and Technology. The university is well regarded for its esteemed faculty & teaching as well as research & also for student satisfaction. With an M.Sc. from this university, you are sure to be able to access global job opportunities.

Program Highlights

Here are the
top reasons why
you should consider
this program



Future-Ready Curriculum

Master In-Demand and Trending Competencies



Personalised Learning Experience

Learning Experience Tailored to Your Needs



Specialisations

Specialise in Two In-Demand Data Science Specialisations



In-Demand Tools

100+ Industry Tools, Languages, Libraries



Outcome-Driven Learning Experience

Personalised Portfolio-Building Support and Career Preparation Sessions



Best-in-Industry Experts

Decorated Faculty and Top Industry Practitioners



Golden Learning Ratio

Perfect Blend of Mathematics, Technology, and Business Understanding



Hands-on Learning

Solve 30+ Domain-Focused Assignments and Case Studies

Offline Graduation Function

On-Campus IIITB Graduation Ceremony

Double Accreditation and Alumni Status

Get Executive Diploma from IIITB and MSc Data Science from LJMU and gain double alumni status

Live LJMU Classroom Hour

Live sessions with LJMU staff for dissertation related queries

LJMU Immersion

Option to Participate in On-Campus Immersion at LJMU, UK

Program Impact Success Stories

Comprises success stories from the MSc Data Science and MSc Machine Learning & Artificial Intelligence Programs

Before upGrad

Tejasvy Gunturu

accenture

Application Development Associate



After upGrad

accenture

Application Development Analyst

Before upGrad

Vijeet Ved

tcs TATA CONSULTANCY SERVICES

System Engineer



After upGrad

tcs TATA CONSULTANCY SERVICES

IT Analyst

Before upGrad

Krish

MARUTI SUZUKI

Deputy Manager



After upGrad

SAMSUNG

Lead Engineer

Before upGrad

Vikas Kumar Gupta

TATA POWER

Data Scientist



After upGrad

TIBIL SOLUTIONS

Senior Data Scientist

Before upGrad

Sunil Acharya

intel

Software Engineer



After upGrad

intel

AI Engineer

Our Transition Statistics

INR 1.23 Cr Highest Salary

433% Highest Hike

50% Average Hike

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Faculty in the LJMU Dissertation Phase



Prof. Dhiya Al-Jumeily

Program Leader, Professor - School of Computer Science and Mathematics, LJMU



A senior member of the IEEE, a chartered IT professional, and a member of the UK Higher Education Academy

Faculty in the IIIT Bangalore Executive Diploma Phase



Prof. Debabrata Das

Director, IIIT Bangalore



He has received his PhD from IIT-KGP. His main areas of research are IoT and Wireless Access Network.

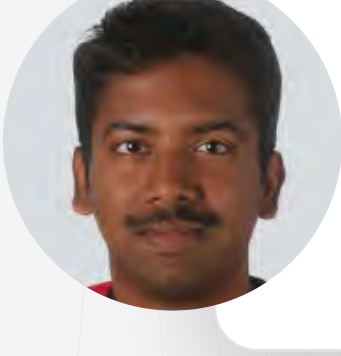


Prof. G. Srinivasaraghavan

IIIT Bangalore



Prof. Srinivasaraghavan has a PhD in Computer Science from IIT-K and 18 years of experience with Infosys Technologies and several other companies.



Prof. Dinesh Babu Jayagopi

IIIT Bangalore



Dr. Dinesh is currently an Associate Professor at IIIT-B where he heads the Multimodal Perception Lab. His research interests are in Audio-Visual Signal Processing, Machine Learning, and Social Computing. He obtained his doctorate from Ecole Polytechnic Federale Lausanne (EPFL), Switzerland.



Prof. Chandrashekar Ramanathan

Dean (Academics), IIIT Bangalore



Prof. Chandrashekar is a faculty member at IIIT-B since 2007 serving as professor, researcher and administrator. He has been working in the field of Computing for over 25 years in various capacities across industry and academia.



Prof. Tricha Anjali

Ex-Associate Dean, IIIT Bangalore



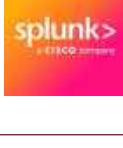
Prof. Anjali has a PhD from Georgia Institute of Technology as well as an integrated MTech (EE) from IIT Bombay.

Industry Experts



Abhishek Vijayvargia

Senior Data Scientist



Having worked with Microsoft as a Senior Data Scientist, he is an alumnus of IIT Kharagpur with 10+ years of experience in a Data Science domain



Ex-Senior Data Scientist



Anand

CEO



A Straive Company

CEO, Gramener A gold medallist from IIM Bangalore, an alumnus of IIT Madras and London Business School, Anand is among the top 10 data scientists in India with 20 years of experience.



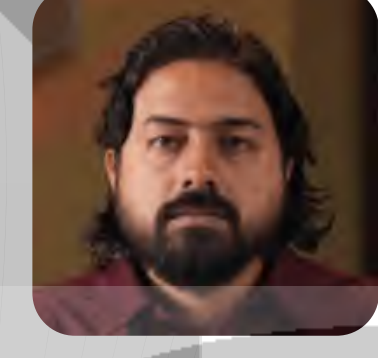
Faculty



Principal



Ex-Consultant



Manish Shukla

Head of Generative AI



Leading cutting-edge GenAI platform development at NatWest Group. Expertise in OpenAI products and MLOps for optimisation of operational efficiency and seamless project delivery with high user satisfaction.



Release Manager



Release Manager



Certified Scrum Master



Deependra Singh

VP & Head of Data Science

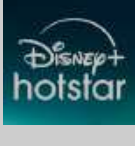


Over 15 years of experience in leading analytics practices, data science, deep learning, and AI product development. Successfully led teams at Jungle Games, American Express Digital Business, and National Insurance Company, pioneering key projects like the analytics engine for the GOI PMJAY policy. Respected speaker at top educational institutes like IMT Hyderabad, BIT Mesra, and NMIMS.

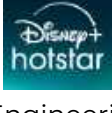


Sajan Kedia

Sr. Engineering manager



Senior Engineering Manager, Hotstar Sajan has extensive experience in the field of ML, Big Data, Data Science, and AI.



Sr. Engineering manager



Machine Learning Engineer



Machine Learning Research Engineer



Mirza Rahim Baig

Startup Mentor



Analytics Lead, Zalando Mirza is a veteran professional with 10+ years of experience in application of data science, machine learning in e-commerce and healthcare.



Team Lead - Product



Marketing Analytics



Ex-Analytics Lead

More than 100+ In-Demand Industry Tools and Technologies

To be covered in the Executive Diploma phase from IIIT Bangalore

Python for Data Science



Data Analytics and Business Intelligence



Cloud Computing, Big Data, Software Fundamentals



Foundations of Machine Learning



Deep Learning and Natural Language Processing



Generative AI



Data Engineering & Big Data Analytics



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Assignments and Case Studies from 12+ In-Demand Business Domains

To be covered in the Executive Diploma phase from IIIT Bangalore



Retail & Ecommerce

ETL Pipelining with Spark



Media & Entertainment

Data Analysis using SQL



Transportation

EDA
using Python



Education

Model Selection
using Sklearn



Civil Engineering

Classification using
CNNs



HR

Semantic Classification
using Word2Vec



Manufacturing

Regularisation using
Sklearn



Healthcare

Classification using
Sklearn



Law

RAG using
LangChain



InfoSec

Feature Engineering
using Sklearn



FMCG

Big Data Analysis
using Spark



BFSI

Sequence Data Prediction
using RNN

Your Program Journey

Phase 0

Math and Programming Bootcamp (12 weeks)

Phase I

Core Curriculum

(28 weeks, 16 IIITB credits)



Phase II

Specialisation Tracks

(22 weeks, 20 IIITB credits)



Phase III

Capstone

(4 weeks, 4 IIITB credits)



Phase IV

Thesis Dissertation

(6 months, 70 LJMU credits)

Note: 40 IIITB credits are considered equivalent to 110 LJMU credits.

Applied Math and Programming Bootcamp

Bootcamp delivered by upGrad

Personalise the initial 3 months of the program to your profile



Topics: Sets, Combinatorics, Basics of Probability, Conditional Probability, Descriptive Statistics, Functions, Vector Algebra, Derivatives, Integrals, Coding Environments, Variables, Data Types, Syntax, Conditionals, Loops, Functions, Lists, Sets, Tuples, Dictionaries, Introduction to MySQL, Basic SQL Querying

Marks Structure: Total marks - 100

Section A - 40 marks (basic mathematics)

Section B - 60 marks (basic programming)

Passing marks - 25 marks in section A & 35 marks in section B

No added cost to be paid for the bootcamp

We make sure that you are well-equipped to draw the most benefit from the program!

Core Curriculum

This is a part of Executive Diploma in DS & AI delivered by IIIT Bangalore

The core phase of the curriculum will equip you with the most up-to-date and industry-relevant skills and technologies for data science and machine learning such as programming and mathematics, data analysis tools and techniques, cloud computing and big data analytics, and foundational topics in machine learning, deep learning, and natural language processing.

Topics

Advanced Mathematics for Data Science and Machine Learning

Master essential mathematical concepts to understand how to work with large amounts of data and train efficient machine learning models

- Conditional Probability and Probability Distributions
- Advanced Linear Algebra and Linear Transformations
- Multivariate Calculus

Advanced Programming for Data Science and Machine Learning

Wrangle real-world data using universal programming languages such as Python and SQL, and use GenAI for generating and debugging code faster

- GenAI for Coding and Problem-Solving
- Object-Oriented Programming
- Python Data Science Libraries
- Database Design and SQL Querying with MySQL
- Introduction to NoSQL Databases

Data Analysis and Exploration

Implement industry-standard statistical methods using tools such as Python, Tableau, and Power BI to analyse data and derive business insights

- Data Analysis with Python
- Exploratory Data Analysis
- Inferential Statistics and Hypothesis Testing
- Data Analysis and Visualisation with Power BI and Tableau

Cloud Computing and Big Data Fundamentals

Take your data processing and analysis workflows to the cloud and work with larger amounts of data to derive enterprise-scale business insights

- Cloud Computing with AWS, GCP, Microsoft Azure
- Big Data Analysis with PySpark

Foundations of Machine Learning

Train industry-standard machine learning models to automate insight generation and predict business metrics behaviour

- Machine Learning Paradigms
- Linear and Logistic Regression
- K Nearest Neighbors
- Regularisation and Hyperparameter Tuning
- Decision Trees and Ensembles
- Clustering Models

Deep Learning and Natural Language Processing

Build and train deep neural network models for different kinds of business data such as images and sequences

- Artificial Neural Networks
- Convolutional and Recurrent Neural Networks
- Lexical, Syntactic, and Semantic Processing

Deployment Fundamentals

Share and deploy your insights and machine learning models so that other collaborators can work with your contributions

- Containerisation and Deployment Tools
- Version Control

Projects

➤ Querying with SQL

Analyse Spotify music data for targeted recommendations or NDAP insurance data for risk assessment

➤ Exploratory Data Analysis

Analyse NYC taxi operations for efficient taxi positioning or US beer production data for better brewery operation management

➤ Big Data Analysis

Analyse Mercari products data for better targeted recommendations or customer interaction data to enhance customer engagement

➤ Linear Regression

Predict household energy consumption using appliance energy readings data to increase power consumption efficiency or parcel delivery time for Porter using historical delivery data for better planning and management

➤ Deep Learning

Predict stock prices of Microsoft, Amazon, Google, IBM, using their historical stock price variations or temperature/pressure readings in Morocco using historical weather data



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Data Analysis Specialisation

DA

This is a part of Executive Diploma in DS & AI delivered by IIIT Bangalore

The data analysis (DA) specialisation of the curriculum will focus on essential modern skills for data and business analysts such as advanced machine learning techniques, advanced analytics and dashboarding technologies, AI integrations in analytics tools, generative AI for data analysis, and core business analysis and project management principles.

Topics

Advanced Machine Learning

Train advanced industry-oriented machine learning models for enhanced predictive power and stronger business insight generation

- » Support Vector Machines and Naive Bayes
- » Feature Engineering and Model Selection
- » Dimensionality Reduction
- » Time Series Analysis
- » Association Rule Mining and Recommendation Systems
- » Explainable AI

Advanced Analytics

Wrangle with enterprise-level data using advanced analytics tools such as Tableau and Power BI, and use GenAI integrations to automate analytics and storytelling workflows

- » Advanced Excel and Power BI with Copilot
- » Advanced Tableau
- » Data Storytelling Principles
- » Machine Learning with PySpark

GenAI for Data Analysis

Prompt large language models (LLMs) for simplifying and automating analytics tasks and understand the advantages and disadvantages of GenAI-based methods

- » Fundamentals of GenAI and Prompt Engineering
- » Advanced Prompt Engineering
- » Large Language Model (LLM) Frameworks such as LangChain
- » GenAI ChatBot System Design and Development
- » Data Security and Governance
- » AI Ethics and LLM Security

Business Analytics Essentials

Transform your analytics insights into actual actionable business statements by translating mathematical language into realistic business metrics

- » Requirements Gathering and Guesstimates
- » Business Problem Solving and Project Management
- » Data Science Applications in Finance and Ecommerce

Projects

- » **Feature Engineering and Model Selection**
Predict fraudulent insurance claims using the Mendeley farmers insurance claims dataset or network intrusion events using historical network activity data
- » **Advanced Data Analytics**
Analyse Namma Yatri travels data to understand customer behaviour and route utilisation for optimisation or Blinkit customer transactions and purchases data to improve product recommendations and enhance shopping experience
- » **Advanced GenAI for Analytics**
Analyse Amazon customer reviews to identify prevalent sentiments and themes to improve product offerings and enhance customer satisfaction or ChatGPT customer feedback to derive actionable insights for business improvement
- » **Essentials of Business Analytics**
Analyse HDFC Bank's annual reports and create a BCG matrix to provide strategic business recommendations or Snapdeal app feedback data to create a business requirement document for improving app functionality

amazon

ChatGPT

namma yatri

snapdeal

blinkit

HDFC BANK

Data Engineering Specialisation

DE

The data engineering (DE) specialisation of the curriculum will focus on essential modern skills for data engineers around the world such as distributed data processing frameworks, cloud native big data processing frameworks and technologies, large-scale data warehousing principles and technologies, real-time data processing, end-to-end data pipeline creation and monitoring, and data architecting principles.

Topics

Large-Scale Distributed Data Processing

Design robust distributed frameworks, both server-based and cloud-based, for big data processing to handle vast amounts of enterprise data

- » Distributed Data Processing with Hadoop Framework
- » Data Ingestion with Sqoop/Flume and HBase Data Querying with Hive
- » Cloud Native SQL Databases such as Amazon Aurora, Google Spanner, and Azure SQL
- » Cloud Native NoSQL Databases such as Amazon DynamoDB, Google BigTable, and Azure Cosmos DB
- » Linux and Java Programming

Data Warehousing Principles and Methodologies

Understand industry-standard data warehousing and ETL/ELT pipelining principles

- » Datawarehousing and Cloud Data Warehousing
- » Cloud Data Warehouses such as Amazon Redshift, Google BigQuery, and Azure Synapse Analytics
- » Understanding ETL and ELT Pipelines
- » Advanced Data Modeling Concepts and Techniques

Large-Scale Data Pipelining

Build complete end-to-end data pipelines and automate them to generate both batch-wise and real-time business insights

- » End-to-End Data Pipelining Fundamentals
- » Pipeline Automation with AWS Lambda, GCP Functions, and Azure Automation
- » Data Monitoring with Amazon CloudWatch, Google Cloud Monitoring, and Azure Monitor
- » Feature Stores and Vector Databases
- » Real-Time Analytics with Flink, Kafka, and Spark Streaming
- » Real-Time Analytics with Amazon Kinesis, Google Cloud Pub/Sub and DataFlow, Azure Stream Analytics and Event Hubs
- » Multicloud and Hybrid Cloud Operating Principles

Modern Data Engineering Technologies

Work with some of the most in-demand advanced data engineering technologies such as modern databases and designing data infrastructures using code

- » Modern NoSQL Databases
- » Infrastructure as Code (IaC) with Terraform
- » Data Architecting Principles
- » Data Security and Governance
- » Decentralized Governance and Data Mesh

Projects

- » **Hadoop Processing**
Analyse Iowa liquor sales data to identify purchasing trends and customer preferences or Los Angeles crime data to identify crime hotspots and trends
- » **Hadoop Processing**
Analyse Iowa liquor sales data to identify purchasing trends and customer preferences or Los Angeles crime data to identify crime hotspots and trends
- » **Data Pipelining and Warehousing**
Develop an ETL pipeline to aggregate and standardise banking data to enhance financial decision making using Wikipedia and currency exchange rates data or aggregate and analyse California traffic collision data to drive road safety improvement measures
- » **Real-Time Data Analytics**
Develop a real-time analytics pipeline for ecommerce data to enhance customer experience or a real-time patient health monitoring system for faster corrective actioning
- » **Data Architecting**
Develop a multi-cloud system using Terraform and cloud services of your choice to provide a layer of redundancy while working with critical healthcare data or website monitoring data

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LOS ANGELES POLICE DEPARTMENT

OTS TRAFFIC SAFETY

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Capstone that Adapts to Your Preference

Infuse our Capstone with Your Data

Modify existing projects as per your industry data and problems

Bring Your Own Capstone

Work on a completely novel project of your choice and solve problems that excite you

Pre-Designed Industry Capstone

Choose one of our existing projects that cover in-demand trending industry domains

Bring Your Own Capstone

Design your own capstone project relevant to your domain and interest, and get feedback throughout your capstone stages



Identify a real-world problem relevant to your domain



Source datasets aligned with your business problem



Design and implement your solution



Document your efforts and present your findings



Continuous expert feedback at every step of capstone

Thesis Dissertation

This is part of LJMU phase and delivered by LJMU

The thesis dissertation phase of your program experience focuses on working on your master's thesis in the domain of data science, machine learning, and artificial intelligence, so that you can truly master this high-impact domain.

Topics

Introduction to Research

Understand different types of research, formulate your research question, and learn to study and cite research papers

- Aspects of Research and Formulating a Research Question
- Understanding Various Research Designs
- Reading and Citing Research Papers
- Research Project Management
- Report Writing and Presentation Design
- Scientific Ethics in Research

Sample Thesis Topics

Study sample thesis topics to get a concrete understanding of what a research thesis entails

- Investigate dietary patterns and metabolite fingerprints of takeaway (fast) food consumers using PCA and clustering methods
- Investigate a diagnosis of eye diseases using imaging ophthalmic data
- Structure medical images with information geometry
- Using social media feed to place tweets regarding natural disasters on a map
- Preventing credit card fraud through pattern recognition
- Developing a recommender system for a media giant
- Risk modelling for financial activities and investment banking

Final Thesis Report

Submit your in-depth research work in a final thesis report and present your findings

Research of Our Learners

A Glimpse

1

Thesis Topic

Build a prediction model to accurately detect

Abstract

Background

Damage to peripheral nerves causes Peripheral neuropathy (PN). Patients complain of pain, numbness and loss of balance. If not identified early and treated adequately, PN could progress rapidly and lead to fatal complications. Defining factors to classify PN accurately has remained challenging. This research proposes a model to detect and classify PN into axonal, demyelinating, mixed and normal types from clinical and nerve conduction study (NCS) data using the Random Forest algorithm.

Data and methods

Clinical and NCS data of 304 Indian patients, 229 affected by PN and 75 normal was collected with ethical approval from Kauvery hospital, Chennai. Exploratory data analysis and the Random Forest Algorithm was used to build a model.

Results

Random Forest model was able to predict and classify PN with an accuracy of 96%. In axonal cases, sensory and motor nerves showed a drop in amplitudes of greater than 40% compared to normal patients. Reduced amplitude (>40%) in motor nerves of lower limbs and missing values (>90%) in sensory nerves of lower limbs identified axonal PN. Delayed onset latency (>40%) in motor nerves of upper limbs, decreased conduction velocity (>60%) in sensory nerves of upper limbs and increased onset latency (>40%) in F-waves of upper limbs delineated the demyelinating type. Median ages of patients were mixed (65), demyelinating (51) and axonal (61). Axonal (18.75%) was significant in diabetic patients and demyelinating (14.8%) in non-diabetic patients. Both axonal and mixed (16.78%) types were greater in hypertensive patients, and demyelinating (17.11%) type was higher in patients without hypertension. Reflex was depressed more in mixed (17.49%) than axonal (15.51%) and demyelinating (11.89%). Mixed (37.06%) type showed more insensitivity to pin-prick than axonal (29.37%) and demyelinating (24.48%) types. Mixed (45%) patients tested positive for Romberg's test more than axonal (31%) and demyelinating (21%). Mixed (34.65%) patients complained of numbness more than axonal (23.62%) and demyelinating (26.77%) types.

Conclusion

Random forest algorithm identified and classified PN well using clinical and NCS features. Clinical features (age, diabetes, hypertension, reflex, Romberg's test, numbness and perception to pin-prick) were useful in detecting PN. Nerve conduction study features (amplitude, onset latency, conduction velocity, F-wave response and missing sensory values) were instrumental in classifying PN. Reduced amplitudes of sensory and motor nerves identified the axonal condition. Delayed onset latency and low conduction velocities along with missing and delayed F-wave responses identified the demyelinating type.

2

Thesis Topic

Automatic network coding of traffic junctions using

Abstract

Before any traffic simulation can be performed, the network of roads and junctions is modeled. Assigning attributes to the roadway network, such as the road length and width, the junction type, number of arms, and lanes, is a crucial task while building the network. This research is an attempt to develop an efficient traffic junction classifier using machine learning and deep learning algorithms on satellite images. Three junction categories, Priority, Roundabout, and Signal, are considered for analysis. As this is a novel research idea, the required image dataset of junctions is created using the Google Maps API. By using robotic process automation, the downloading of the images is automated. Two approaches are taken to build the classifiers: a machine-learning approach and a deep-learning approach. The machine learning approach is split into two phases: the feature extraction phase and the classification phase. In the feature extraction phase, a Histogram of Oriented Gradients (HOG) descriptors is used to extract features from the images. Furthermore, in the classification phase, several classification algorithms are applied to the HOG features to build classifiers. In the deep-learning approach, taking advantage of powerful pre-trained models and transfer learning, a Convolutional Neural Network (CNN) is developed for classifying the junctions. The models are evaluated, and in the end, a comparison between the various classification models is performed. The results showed that the CNN classifier modeled had the best accuracy and AUC compared to the other models with scores of 0.81 and 0.94 respectively. Among the machine learning models that were trained on the HOG features, the Extreme Gradient Boosting model has the best accuracy of 0.62. The ultimate aim of this work is to use this junction-classifier model on real projects to aid the process of finding the type of junctions and reduce the effort and time required to model the roadway networks.

Build A Strong Portfolio



Commits

Demonstrate consistency, collaboration, and coding discipline

Code

Showcase well-documented repositories

Projects

Host end-to-end DS/ML/AI projects that highlight real-world problem-solving

Kernels

Highlight data processing and EDA methodologies

Ranking

Evaluate and reflect global standing among data science practitioners

Competitions

Demonstrate problem-solving under tight constraints

Headline

Concise summary of goals, competencies, and professional identity

Summary

Engaging overview of learnin and career journey

Projects

Showcase practical experience, outcomes, and skill application

GitHub helps with

- ✓ Validating coding skills
- ✓ Showing growth and consistency
- ✓ Being interview-ready for Tech roles

Kaggle helps with

- ✓ Building credibility in data science circles
- ✓ Applying learning to real datasets
- ✓ Speaking confidently in Tech interviews

LinkedIn helps with

- ✓ Improving visibility with recruiters
- ✓ Positioning better for job openings
- ✓ Networking with peers and mentors in the field

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Microsoft-Certified Advantage –

This program doesn't just prepare you for the world of Data Science & AIML—it gives you the Microsoft edge.

Learners earn industry backed certification from upGrad in association with Microsoft by completing specially designed modules integrated into the program, boosting both credibility and career readiness.



Microsoft Learn content modules covered

- Introduction to Generative AI Concepts
- Introduction to GitHub Copilot
- Design & Manage Analytics Solutions using Power BI
- Designing & Implementing a Data Science Solution on Azure

Rich and Dedicated **Live Support**

Industry Expert Sessions

Engage with industry practitioners as they help you master in-demand skills and concepts using a demonstrative hands-on approach



IIITB Faculty Sessions

Learn from some of the most accomplished academicians as they take your knowledge and understanding of data science to another level



Just-In-Time Interview Support

Participate in Technical and HR mock interviews designed to boost your confidence and prepare you to ace interviews.



Career Coaching Sessions

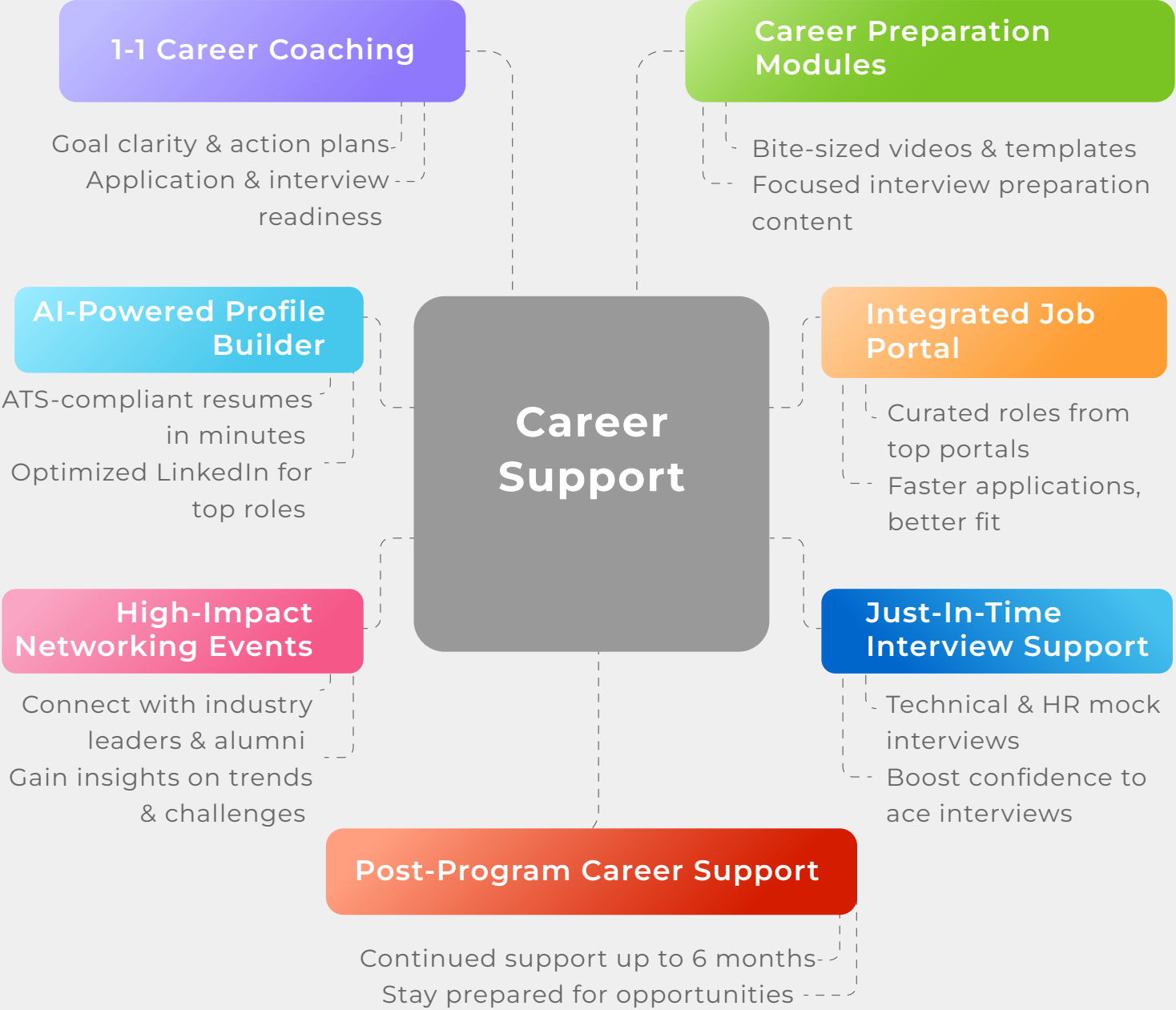
Engage in Career Coaching Sessions via Career preparation modules, High-impact networking events and Just-in-time mock interviews



Daily doubt resolution sessions

Join doubt resolution session slots, that are available daily, and have an expert available to resolve your queries for a smooth learning journey

Effective and Complete Career Support

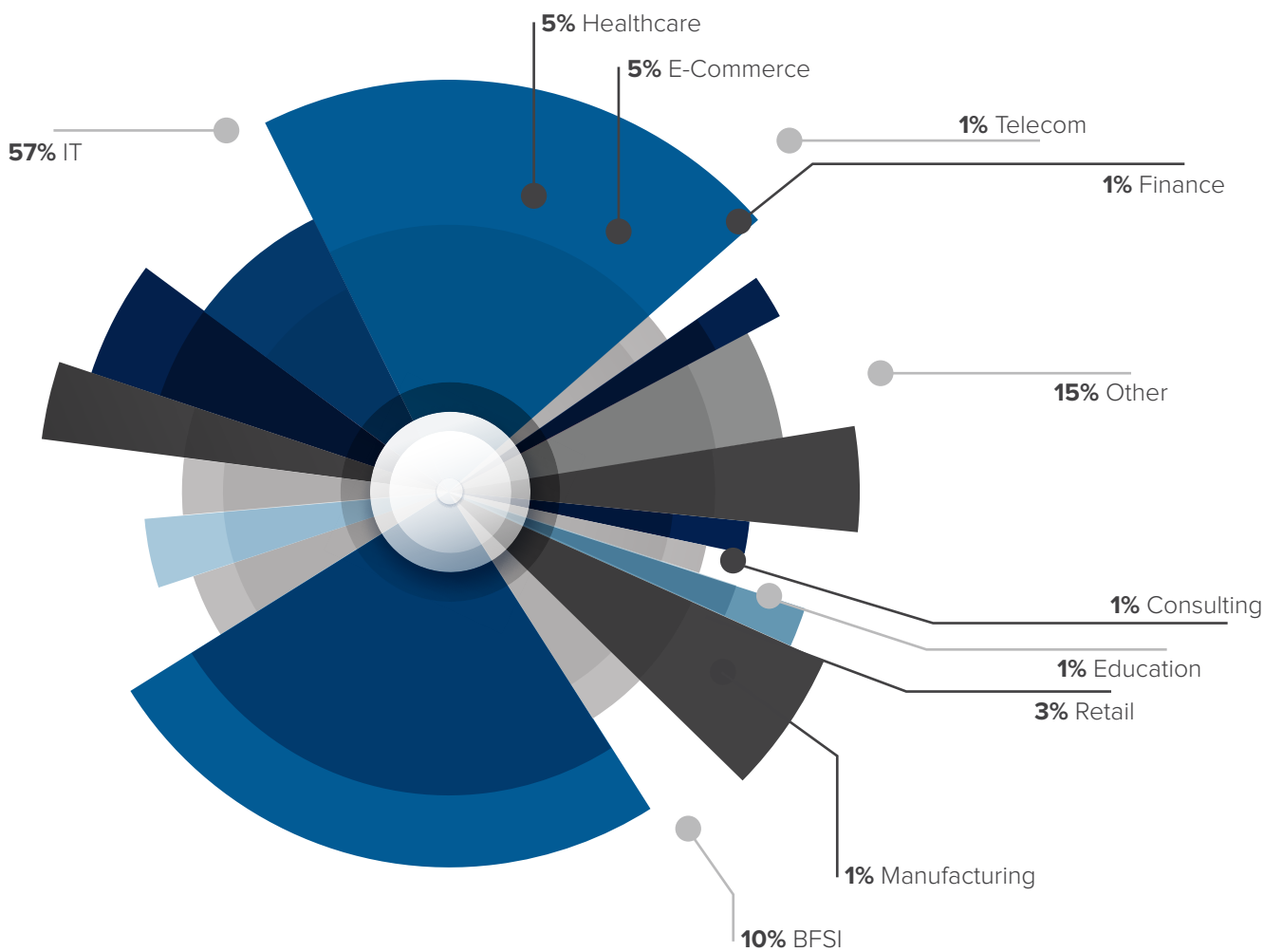


Student Support

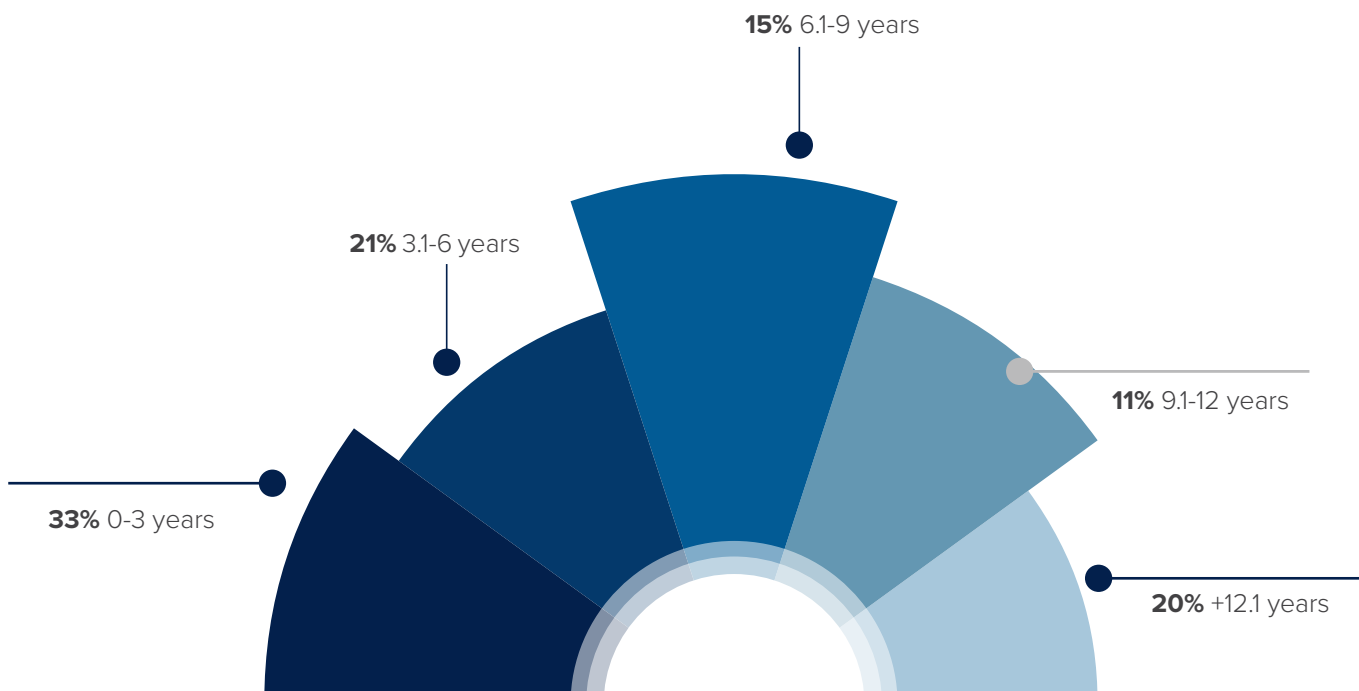
Telegram channel for learner communications Cohort Telegram channel for timely program updates and announcements.	Non-academic and non-technical query assistance 24 X 7 m-AI-ask bot support for non academic assistance.
Completion Support Personalised assistance for smooth program completion, managing backlogs, and cohort deferrals with free and paid waiver options.	Financial benefits Access benefits like referrals and repeats by sharing details with your program co-ordinator or student support executive.

Meet the Class

Industries Our Students Come From



Work Experience





A single specialty is no longer enough.

For years, the T-shaped professional was considered the gold standard—someone with broad knowledge across multiple areas (the top of the “T”) and deep expertise in one core area (the vertical stem). This model worked well in stable environments where projects moved slowly, and roles were clearly divided

But today’s AI-driven, high-velocity world has exposed the limits of the T-shape. Knowledge silos, handoff delays, and rising complexity mean one deep skill is no longer enough

Today's employers need π -shaped professionals - someone with two deep, complementary specialisations while maintaining broad knowledge across domains, who can own entire project lifecycles without team dependencies, bridging the gap between development, operations, and analytics seamlessly

Introducing The π Professional Pack

Add an extra specialisation in just 3 months

STEP 1

Choose your primary foundation

Select your core specialisation:

Data Engineering, Data Analytics, Generative AI or MLOps

STEP 2

Add your strategic second specialty

In the latter half of your program, unlock a complete second track featuring:



Full curriculum coverage



Live interactive sessions with industry experts



Dedicated faculty support



Expert mentorship across both domains

Why is this a strategic career investment?



Expert in two complete specialisations



Industry-validated skill combinations based on hiring data



Hands-on experience in both domains



Portfolio projects demonstrating π -shaped expertise

Your Competitive Edge:

While others graduate with single specialisations, you emerge as a π -shaped leader ready to:



Own complete project lifecycles



Bridge team silos and communication gaps



Command premium positions in India's GCC & IT sectors



Adapt quickly as AI continues reshaping the industry

Popular specialisation combos

GenAI + MLOps

Why Gen AI + MLOps?

Enables end-to-end ownership of Gen AI models from fine-tuning to deployment, monitoring, and safety. This pairing creates a professional who can not only innovate but also reliably deliver and manage that innovation.

GenAI + Analytics

Why GenAI + Analytics?

Unlocks AI-driven business insights with a combination of natural language queries, auto-reporting, and data storytelling. This pairing is about creating smarter and more accessible business insights.

MLOps + Data Engineering

Why MLOps + Data Engineering?

This is the ultimate combination for building robust and scalable AI infrastructure. Enables you to build an automated, end-to-end machine learning factory that is efficient, reliable, and ready for production demands.

GenAI + Data Engineering

Why GenAI + Data Engineering?

Let's combine Gen AI innovation with solid data pipelines scalable AI products. This pairing ensures that innovative GenAI projects are built on a solid foundation, allowing them to be scalable and successful in the long run.

Earn Second Certificate

π Professional pack for dual specialisations

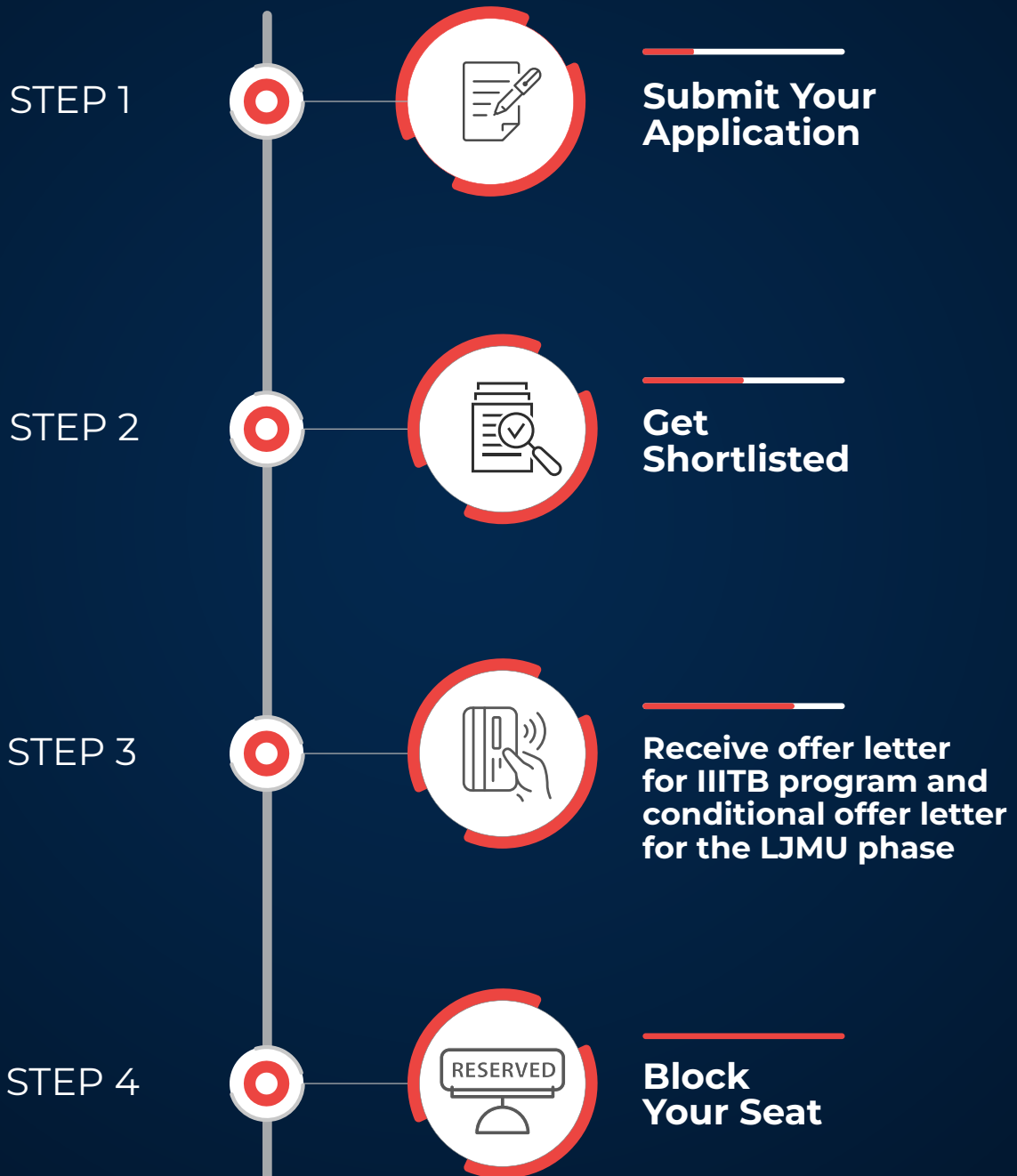


In just 3 more months learners can earn 1 more specialization across both Data Science & Machine learning specializations

Enrol in 4 small steps, **Then take a giant leap.**

Eligibility Criteria

Bachelor's or Master's Degree or its equivalent in any discipline with minimum 50% aggregate mark or equivalent CGPA.



Please note that your admission to the LJMU's MSc in Data Science Program is strictly provisional and conditional upon your successful completion of the IIITB Executive Diploma in Data Science & AI



Write to us:

admissions@upgrad.com



Visit us at:

www.upgrad.com



Call: 1800-210-2020

upGrad

LET'S TALK

Disclaimer: This program is not a dual, joint, or twinning degree. upGrad is not a college or university and does not grant academic credit. Credit recognition or transfer is at the sole discretion of the receiving institution. The Universities offering the program are accredited in their home countries but are not recognised by the UGC/AIU in India. Learners must independently verify the program's suitability for further studies, public-sector roles, immigration requirements, or any employment needing recognised credits. upGrad and the Universities do not make any representation regarding the standing, recognition, or acceptance of the program for such purposes. Tuition fees, eligibility criteria, and academic requirements are determined solely by the University and may change; learners are responsible for tracking modifications and all University communications. upGrad and the Universities will not be responsible for any consequences arising from the learner's failure to ensure the program meets their academic, professional, or regulatory needs. Please reach out to us if you have any queries. Terms & Conditions apply.

upGrad



Master of Science in

Machine Learning & Artificial Intelligence



Now integrated with
Generative AI

Duration **18-21 months**



The Era Of Generative AI



The world is at the cusp of Generative AI rapidly changing the world as we know it. At upGrad, we've always believed in imparting learners the skills necessary to thrive in the fast-evolving world of technology. We are hence quite thrilled to pioneer Generative AI as an elective in the Master of Science in Machine Learning & Artificial Intelligence.

With this key inclusion of Generative AI, learners will delve deeper into the fascinating realm of using Data Science to build practical applications like conversational AI chat bots, Image creators, and content recommenders amongst others, to solve real-world challenges. So dive into this brave new world of Generative AI and Large Language Models with us, and watch yourself transform into a 10x Data Scientist.

"IIT Bangalore prides itself in constantly updating cutting-edge topics to its curriculum. Our faculty has shaped this exciting Generative AI elective along with upGrad's industry experts, thus ensuring both academic rigour as well as incorporating the latest advancements in tech"

Dr. V. Sridhar,
Head-Faculty, IITB

"As an organisation that asks professionals to stay updated with the latest skills, we had to be one of the first to teach Generative AI. With this move, we are excited to witness the impact that Generative AI will have on the future, as well as the value our learners will bring to the field with this essential skill."

Mayank Kumar, Co-founder & MD
upGrad

About upGrad, IIITB, and LJMU, UK

upGrad has delivered over 20 million hours of learning, delivering programs by collaborating with universities across the world including Duke CE, IIIT Bangalore and Golden Gate University, among others

Online education is a fundamental disruption that will have a far-reaching impact. **upGrad** was founded taking this into consideration. upGrad is an online education platform to help individuals develop their professional potential in the most engaging learning environment. Since inception, upGrad is focussed on helping working professionals in their bid to learn, grow and move up in their career through a wide-range of programs designed to improve their expertise.

IIITB is a renowned university offering programs specialising in data science, machine learning and artificial intelligence. IIITB's faculty bring with them an average of 15 years of experience. The faculty covers the conceptual depths of topics such as Data Science, Machine Learning and AI, and Big Data Analytics. These will be complemented by industry relevant case studies from major industry verticals by industry leaders with 10+ yrs of experience from upGrad's industry network. The Executive Diplomas in DS and ML has been developed with the experienced faculty of IIITB in collaboration with Industry experts & upGrad to bring you cutting edge-curriculum with industry relevance. The strong placement network, industry mentorship and the credibility of this Executive Diploma from IIITB will provide you with just the right push to accelerate your career in Machine Learning and AI!

With a heritage that stretches back to **Liverpool John Moores University** UK is now one of the largest and well-established universities in the UK. It has been ranked in the Top 100 World Young Universities & Top 50 in the UK by Student Satisfaction. There are two faculties within the University - Faculty of Society and Culture and the Faculty of Health, Innovation, Science and Technology. The university is well regarded for its esteemed faculty & teaching as well as research & also for student satisfaction. With an M.Sc. from this university, you are sure to be able to access global job opportunities.

Program Highlights

Here are the
top reasons why
you should consider
this program



Future-Ready Curriculum

Master In-Demand and Trending Competencies



Personalised Learning Experience

Learning Experience Tailored to Your Needs



Specialisations

Specialise in Two In-Demand Machine Learning Specialisations



In-Demand Tools

80+ Industry Tools, Languages, Libraries



Outcome-Driven Learning Experience

Personalised Portfolio-Building Support and Career Preparation Sessions



Best-in-Industry Experts

Decorated Faculty and Top Industry Practitioners



Golden Learning Ratio

Perfect Blend of Mathematics, Technology, and Business Understanding



Hands-on Learning

Solve 30+ Domain-Focused Assignments and Case Studies

Offline Graduation Function

On-Campus IIITB Graduation Ceremony

Double Accreditation and Alumni Status

Get Executive Diploma from IIITB and MSc Data Science from LJMU and gain double alumni status

Live LJMU Classroom Hour

Live sessions with LJMU staff for dissertation related queries

LJMU Immersion

Option to Participate in On-Campus Immersion at LJMU, UK

Program Impact Success Stories

Comprises success stories from the MSc Data Science and MSc Machine Learning & Artificial Intelligence Programs

Before upGrad

Tejasvy Gunturu

accenture

Application Development Associate



After upGrad

accenture

Application Development Analyst

Before upGrad

Vijeet Ved

tcsTATA CONSULTANCY SERVICES

System Engineer



After upGrad

tcsTATA CONSULTANCY SERVICES

IT Analyst

Before upGrad

Krish

MARUTI SUZUKI

Deputy Manager



After upGrad

SAMSUNG

Lead Engineer

Before upGrad

Vikas Kumar Gupta

TATA POWER

Data Scientist



After upGrad

TIBIL SOLUTIONS

Senior Data Scientist

Before upGrad

Sunil Acharya

intel

Software Engineer



After upGrad

intel

AI Engineer

Our Transition Statistics

INR 1.23 Cr Highest Salary

433% Highest Hike

50% Average Hike

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Faculty in the IIIT Bangalore Executive Diploma Phase



Prof. Debabrata Das
Director, IIIT Bangalore



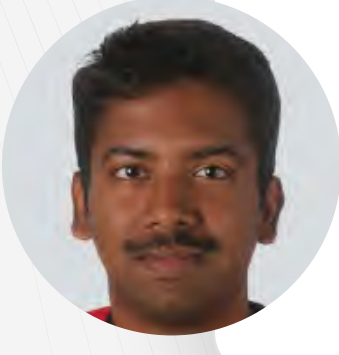
He has received his PhD from IIT-KGP. His main areas of research are IoT and Wireless Access Network.



Prof. G. Srinivasaraghavan
IIIT Bangalore



Prof. Srinivasaraghavan has a PhD in Computer Science from IIT-K and 18 years of experience with Infosys Technologies and several other companies.



Prof. Dinesh Babu Jayagopi
IIIT Bangalore



Dr. Dinesh is currently an Associate Professor at IIIT-B where he heads the Multimodal Perception Lab. His research interests are in Audio-Visual Signal Processing, Machine Learning, and Social Computing. He obtained his doctorate from Ecole Polytechnic Federale Lausanne (EPFL), Switzerland.



Prof. Chandrashekar Ramanathan
Dean (Academics), IIIT Bangalore



Prof. Chandrashekar is a faculty member at IIIT-B since 2007 serving as professor, researcher and administrator. He has been working in the field of Computing for over 25 years in various capacities across industry and academia.



Prof. Tricha Anjali
Ex-Associate Dean, IIIT Bangalore



Prof. Anjali has a PhD from Georgia Institute of Technology as well as an integrated MTech (EE) from IIT Bombay.

Industry Experts



Abhishek Vijayvargia
Senior Data Scientist



Having worked with Microsoft as a Senior Data Scientist, he is an alumnus of IIT Kharagpur with 10+ years of experience in a Data Science domain



Ex-Senior Data Scientist



Anand
CEO



CEO, Gramener A gold medallist from IIM Bangalore, an alumnus of IIT Madras and London Business School, Anand is among the top 10 data scientists in India with 20 years of experience.



Faculty



Principal



Ex-Consultant



Manish Shukla
Head of Generative AI



Leading cutting-edge GenAI platform development at NatWest Group. Expertise in OpenAI products and MLOps for optimisation of operational efficiency and seamless project delivery with high user satisfaction.



Release Manager



Release Manager



Certified Scrum Master



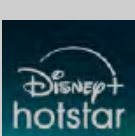
Deependra Singh
VP & Head of Data Science



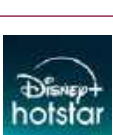
Over 15 years of experience in leading analytics practices, data science, deep learning, and AI product development. Successfully led teams at Jungle Games, American Express Digital Business, and National Insurance Company, pioneering key projects like the analytics engine for the GOI PMJAY policy. Respected speaker at top educational institutes like IMT Hyderabad, BIT Mesra, and NMIMS.



Sajjan Kedia
Sr. Engineering manager



Senior Engineering Manager, Hotstar Sajjan has extensive experience in the field of ML, Big Data, Data Science, and AI.



Sr. Engineering manager



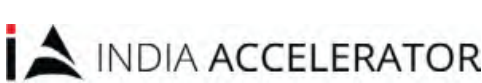
Machine Learning Engineer



Machine Learning Research Engineer



Mirza Rahim Baig
Startup Mentor



Analytics Lead, Zalando Mirza is a veteran professional with 10+ years of experience in application of data science, machine learning in e-commerce and healthcare.



Team Lead - Product



Marketing Analytics



Ex-Analytics Lead

Assignments and Case Studies from 12+ In-Demand Business Domains

To be covered in the Executive Diploma phase from IIIT Bangalore



Retail & Ecommerce

ETL Pipelining with Spark



Media & Entertainment

Data Analysis using SQL



Transportation

EDA
using Python



Education

Model Selection
using Sklearn



Civil Engineering

Classification using
CNNs



HR

Semantic Classification
using Word2Vec



Manufacturing

Regularisation using
Sklearn



Healthcare

Classification using
Sklearn



Law

RAG using
LangChain



InfoSec

Feature Engineering
using Sklearn



FMCG

Big Data Analysis
using Spark



BFSI

Sequence Data Prediction
using RNN

Your Program Journey

Phase 0

Math and Programming Bootcamp (12 weeks)

Phase I

Core Curriculum

(28 weeks, 16 IIITB credits)



Phase II

Specialisation Tracks

(22 weeks, 20 IIITB credits)



Phase III

Capstone

(4 weeks, 4 IIITB credits)



Phase IV

Thesis Dissertation

(6 months, 70 LJMU credits)

Note: 40 IIITB credits are considered equivalent to 110 LJMU credits.

Applied Math and Programming Bootcamp

Bootcamp delivered by upGrad

Personalise the initial 3 months of the program to your profile



Topics: Sets, Combinatorics, Basics of Probability, Conditional Probability, Descriptive Statistics, Functions, Vector Algebra, Derivatives, Integrals, Coding Environments, Variables, Data Types, Syntax, Conditionals, Loops, Functions, Lists, Sets, Tuples, Dictionaries, Introduction to MySQL, Basic SQL Querying

Marks Structure: Total marks - 100

Section A - 40 marks (basic mathematics)

Section B - 60 marks (basic programming)

Passing marks - 25 marks in section A & 35 marks in section B

No added cost to be paid for the bootcamp

We make sure that you are well-equipped to draw the most benefit from the program!

Core Curriculum

This is a part of Executive Diploma in DS & AI delivered by IIIT Bangalore

The core phase of the curriculum will equip you with the most up-to-date and industry-relevant skills and technologies for data science and machine learning such as programming and mathematics, data analysis tools and techniques, cloud computing and big data analytics, and foundational topics in machine learning, deep learning, and natural language processing.

Topics

Advanced Mathematics for Data Science and Machine Learning

Master essential mathematical concepts to understand how to work with large amounts of data and train efficient machine learning models

- Conditional Probability and Probability Distributions
- Advanced Linear Algebra and Linear Transformations
- Multivariate Calculus

Advanced Programming for Data Science and Machine Learning

Wrangle real-world data using universal programming languages such as Python and SQL, and use GenAI for generating and debugging code faster

- GenAI for Coding and Problem-Solving
- Object-Oriented Programming
- Python Data Science Libraries
- Database Design and SQL Querying with MySQL
- Introduction to NoSQL Databases

Data Analysis and Exploration

Implement industry-standard statistical methods using tools such as Python, Tableau, and Power BI to analyse data and derive business insights

- Data Analysis with Python
- Exploratory Data Analysis
- Inferential Statistics and Hypothesis Testing
- Data Analysis and Visualisation with Power BI and Tableau

Cloud Computing and Big Data Fundamentals

Take your data processing and analysis workflows to the cloud and work with larger amounts of data to derive enterprise-scale business insights

- Cloud Computing with AWS, GCP, Microsoft Azure
- Big Data Analysis with PySpark

Foundations of Machine Learning

Train industry-standard machine learning models to automate insight generation and predict business metrics behaviour

- Machine Learning Paradigms
- Linear and Logistic Regression
- K Nearest Neighbors
- Regularisation and Hyperparameter Tuning
- Decision Trees and Ensembles
- Clustering Models

Deep Learning and Natural Language Processing

Build and train deep neural network models for different kinds of business data such as images and sequences

- Artificial Neural Networks
- Convolutional and Recurrent Neural Networks
- Lexical, Syntactic, and Semantic Processing

Deployment Fundamentals

Share and deploy your insights and machine learning models so that other collaborators can work with your contributions

- Containerisation and Deployment Tools
- Version Control

Projects

- **Querying with SQL**
Analyse Spotify music data for targeted recommendations or NDAP insurance data for risk assessment
- **Exploratory Data Analysis**
Analyse NYC taxi operations for efficient taxi positioning or US beer production data for better brewery operation management
- **Big Data Analysis**
Analyse Mercari products data for better targeted recommendations or customer interaction data to enhance customer engagement
- **Linear Regression**
Predict household energy consumption using appliance energy readings data to increase power consumption efficiency or parcel delivery time for Porter using historical delivery data for better planning and management
- **Deep Learning**
Predict stock prices of Microsoft, Amazon, Google, IBM, using their historical stock price variations or temperature/pressure readings in Morocco using historical weather data

 Microsoft

 Google

 amazon

 IBM

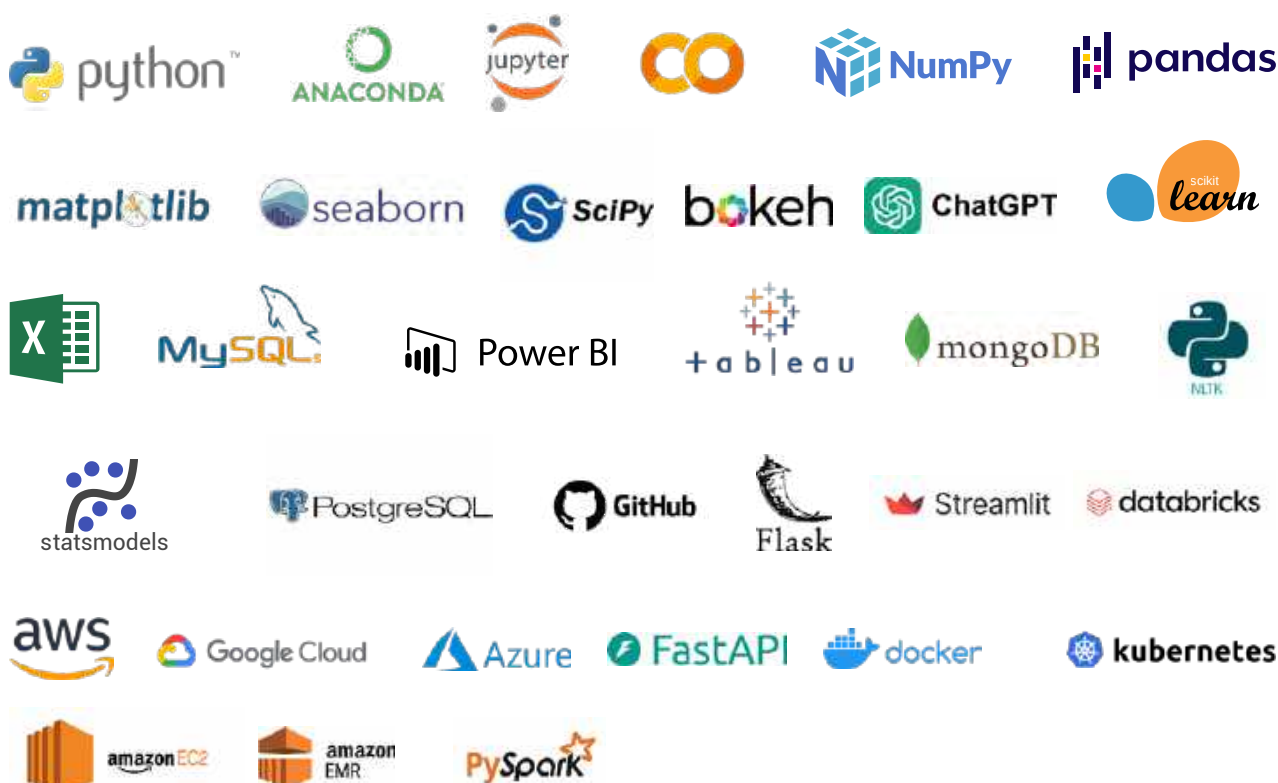
 mercari

 NYC TAXI

 ndap

 Spotify

Tools



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MLOps Specialisation

This is a part of Executive Diploma in DS & AI delivered by IIIT Bangalore

The machine learning operations (MLOps) specialisation of the curriculum will equip you with core in-demand and industry relevant skills and technologies essential for ML engineers such as advanced machine learning methods, modern deep learning architectures, real-time data processing and end-to-end data pipeline creation and monitoring, and model pipelining and monitoring at scale.

Topics

Advanced Machine Learning

Train advanced industry-oriented machine learning models for enhanced predictive power and stronger business insight generation

- Support Vector Machines and Naive Bayes
- Feature Engineering and Model Selection
- Dimensionality Reduction
- Time Series Analysis
- Association Rule Mining and Recommendation Systems
- Explainable AI

Advanced Deep Learning and Generative AI

Design and train advanced industry-standard deep learning architectures, and master core AI principles such as attention mechanisms, transformers, and prompt engineering

- Advanced CNN Architectures
- LSTMs and GRUs
- Transfer Learning Techniques
- Encoder-Decoder Architectures and Seq2Seq
- Machine Translation
- Attention Mechanisms and Transformers
- Fundamentals of Generative AI and Prompt Engineering
- Computer Vision, Variational Autoencoders, Generative Adversarial Networks
- Data and Model Security Principles

Large-Scale Data Pipelining

Build complete end-to-end data pipelines and automate them to generate both batch-wise and real-time business insights

- End-to-End Data Pipelining Fundamentals
- Pipeline Automation with AWS Lambda, GCP Functions, and Azure Automation
- Data Monitoring with Amazon CloudWatch, Google Cloud Monitoring, and Azure Monitor
- Feature Stores and Vector Databases
- Real-Time Analytics with Flink, Kafka, and Spark Streaming
- Real-Time Analytics with Amazon Kinesis, Google Cloud Pub/Sub and DataFlow, Azure Stream Analytics and Event Hubs
- Multicloud and Hybrid Cloud Operating Principles

Machine Learning Model Pipelining

Build end-to-end industry-ready ML model pipelines and design their functional behaviour such as training and inference

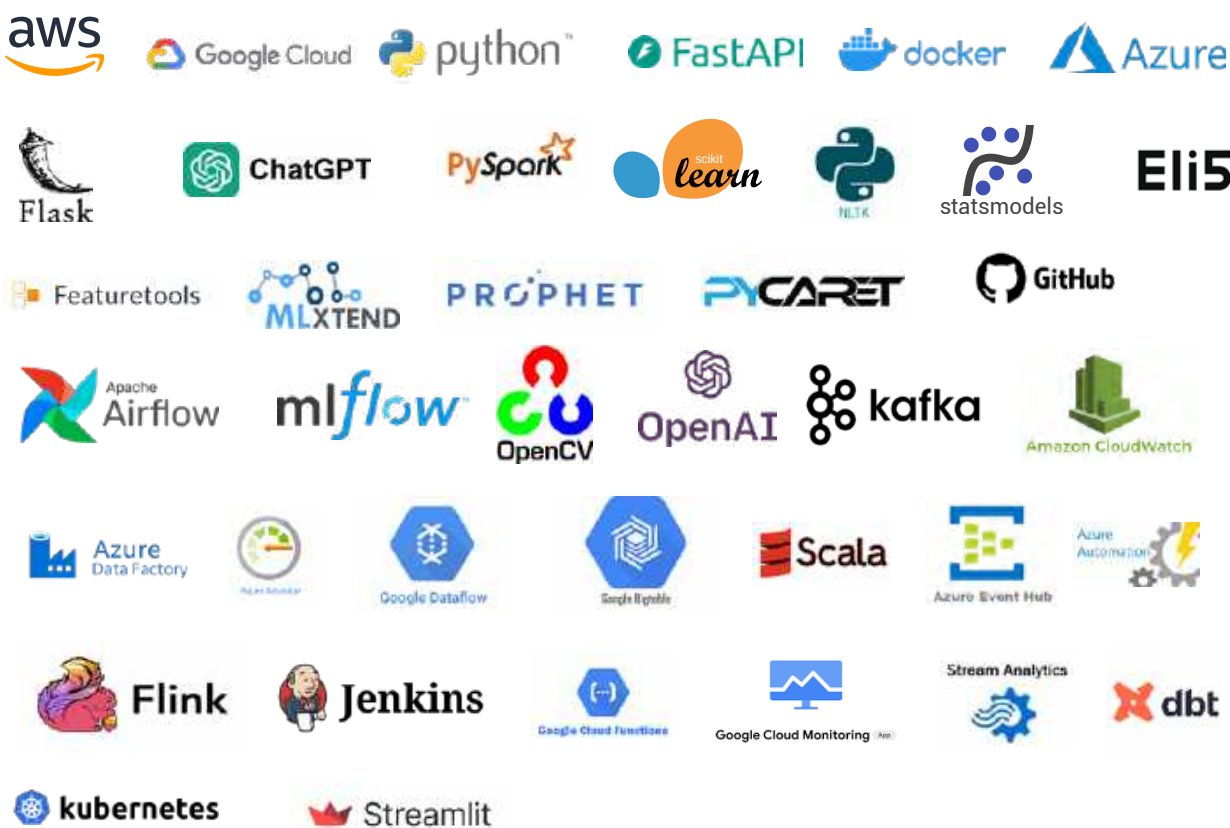
- Model Pipelining Principles
- Scheduling and Triggers
- Parallel Model Training and Real-Time Model Serving
- Data and Model Versioning
- Model Monitoring and System Design

Projects

- **Feature Engineering and Model Selection**
Predict fraudulent insurance claims using the Mendeley farmers insurance claims dataset or network intrusion events using historical network activity data
- **Semantic Classification**
Fake News Detection, Job Role Classification
- **Real-Time Data Analytics: Develop a**
real-time analytics pipeline for ecommerce data to enhance customer experience or a real-time patient health monitoring system for faster corrective actioning
- **Simulate and Retrigger Model Training Pipeline**



Tools



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Generative AI Specialisation

The generative artificial intelligence (GenAI) specialisation of the curriculum will equip you modern AI technologies and methods, particularly generative AI technologies, essential to data scientists and AI specialists, such as advanced machine learning methods, modern deep learning architectures, advanced prompt engineering and generative AI system design, information retrieval and retrieval-augmented generation, large language model (LLM) deployment, advanced computer vision and 3D vision, GenAI optimisations, and AI ethics.

Topics

Advanced Machine Learning

Train advanced industry-oriented machine learning models for enhanced predictive power and stronger business insight generation

- Support Vector Machines and Naive Bayes
- Feature Engineering and Model Selection
- Dimensionality Reduction
- Time Series Analysis
- Association Rule Mining and Recommendation Systems
- Explainable AI

Advanced Deep Learning for Generative AI

Design and train advanced industry-standard deep learning architectures, and master core AI principles such as attention mechanisms, transformers, and prompt engineering

- Advanced CNN Architectures
- LSTMs and GRUs
- Transfer Learning Techniques
- Encoder-Decoder Architectures and Seq2Seq
- Machine Translation
- Attention Mechanisms and Transformers
- Fundamentals of Generative AI and Prompt Engineering
- Computer Vision, Variational Autoencoders, Generative Adversarial Networks
- Data and Model Security Principles

GenAI System Design

Design and orchestrate generative AI systems to leverage the power of generative AI models and transform business operations

- Advanced Prompt Engineering and GenAI System Design
- Prompting Multimodal Models
- LLM Frameworks such as LangChain and LLaMa Index
- LLM Evaluation Methods
- Data Security and Governance
- AI Ethics

Advanced Generative AI

Develop AI-based cutting-edge industry-level systems for greater business efficiency such as retrieval-augmented generation (RAG) systems and multimodal GenAI model prompt engineering

- Information Retrieval Principles
- Embeddings and Vector Databases
- RAG Architectures
- Agentic Systems and Multi-Agent Systems
- Advanced Multimodal GenAI Models
- LLM Deployment
- Advanced Computer Vision and 3D Vision
- GenAI Optimisations

Projects

- **Feature Engineering and Model Selection**
Predict fraudulent insurance claims using the Mendeley farmers insurance claims dataset or network intrusion events using historical network activity data
- **Semantic Classification**
Fake News Detection, Job Role Classification
- **GenAI System Design**
Analyse Amazon customer reviews to identify prevalent sentiments and themes to improve product offerings and enhance customer satisfaction or ChatGPT customer feedback to derive actionable insights for business improvement
- **RAG**
Develop an RAG system to transform Long Beach County Municipal meetings transcripts into actionable insights for better organisational communication and decision making or historical legal documentations to optimise legal workflows



Tools



Hugging Face



Chroma



PINECONE



Milvus



deepseek



Gemini



LlamaIndex



LangChain



Stable Diffusion



Claude



pgvector



OpenCV



Eli5



IMAGEN



Featuretools



MLXTEND



PROPHET



crewai



Ollama



LangGraph



CLIP



OpenAI CLIP



Autogen



DALL-E

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Capstone Projects

This is a part of Executive Diploma in DS & AI delivered by IIIT Bangalore

CAPSTONE PROJECTS

Capstone that Adapts to Your Preference

Infuse our Capstone with Your Data

Modify existing projects as per your industry data and problems

Bring Your Own Capstone

Work on a completely novel project of your choice and solve problems that excite you

Pre-Designed Industry Capstone

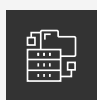
Choose one of our existing projects that cover in-demand trending industry domains

Bring Your Own Capstone

Design your own capstone project relevant to your domain and interest, and get feedback throughout your capstone stages



Identify a real-world problem relevant to your domain



Source datasets aligned with your business problem



Design and implement your solution



Document your efforts and present your findings



Continuous expert feedback at every step of capstone

Thesis Dissertation

This is part of LJMU phase and delivered by LJMU

The thesis dissertation phase of your program experience focuses on working on your master's thesis in the domain of data science, machine learning, and artificial intelligence, so that you can truly master this high-impact domain.

Topics

Introduction to Research

Understand different types of research, formulate your research question, and learn to study and cite research papers

- Aspects of Research and Formulating a Research Question
- Understanding Various Research Designs
- Reading and Citing Research Papers
- Research Project Management
- Report Writing and Presentation Design
- Scientific Ethics in Research

Sample Thesis Topics

Study sample thesis topics to get a concrete understanding of what a research thesis entails

- Investigate dietary patterns and metabolite fingerprints of takeaway (fast) food consumers using PCA and clustering methods
- Investigate a diagnosis of eye diseases using imaging ophthalmic data
- Structure medical images with information geometry
- Using social media feed to place tweets regarding natural disasters on a map
- Preventing credit card fraud through pattern recognition
- Developing a recommender system for a media giant
- Risk modelling for financial activities and investment banking
- Using social media feed to place tweets regarding natural disasters on a map

Final Thesis Report

Submit your in-depth research work in a final thesis report and present your findings

Research of our learners

A Glimpse

1

Thesis Topic

Build a prediction model to accurately detect

Abstract

Background

Damage to peripheral nerves causes Peripheral neuropathy (PN). Patients complain of pain, numbness and loss of balance. If not identified early and treated adequately, PN could progress rapidly and lead to fatal complications. Defining factors to classify PN accurately has remained challenging. This research proposes a model to detect and classify PN into axonal, demyelinating, mixed and normal types from clinical and nerve conduction study (NCS) data using the Random Forest algorithm.

Data and methods:

Clinical and NCS data of 304 Indian patients, 229 affected by PN and 75 normal was collected with ethical approval from Kauvery hospital, Chennai. Exploratory data analysis and the Random Forest Algorithm was used to build a model.

Results:

Random Forest model was able to predict and classify PN with an accuracy of 96%. In axonal cases, sensory and motor nerves showed a drop in amplitudes of greater than 40% compared to normal patients. Reduced amplitude (>40%) in motor nerves of lower limbs and missing values (>90%) in sensory nerves of lower limbs identified axonal PN. Delayed onset latency (>40%) in motor nerves of upper limbs, decreased conduction velocity (>60%) in sensory nerves of upper limbs and increased onset latency (>40%) in F-waves of upper limbs delineated the demyelinating type. Median ages of patients were mixed (65), demyelinating (51) and axonal (61). Axonal (18.75%) was significant in diabetic patients and demyelinating (14.8%) in non-diabetic patients. Both axonal and mixed (16.78%) types were greater in hypertensive patients, and demyelinating (17.11%) type was higher in patients without hypertension. Reflex was depressed more in mixed (17.49%) than axonal (15.51%) and demyelinating (11.89%). Mixed (37.06%) type showed more insensitivity to pin-prick than axonal (29.37%) and demyelinating (24.48%) types. Mixed (45%) patients tested positive for Romberg's test more than axonal (31%) and demyelinating (21%). Mixed (34.65%) patients complained of numbness more than axonal (23.62%) and demyelinating (26.77%) types.

Conclusion:

Random forest algorithm identified and classified PN well using clinical and NCS features. Clinical features (age, diabetes, hypertension, reflex, Romberg's test, numbness and perception to pin-prick) were useful in detecting PN. Nerve conduction study features (amplitude, onset latency, conduction velocity, F-wave response and missing sensory values) were instrumental in classifying PN. Reduced amplitudes of sensory and motor nerves identified the axonal condition. Delayed onset latency and low conduction velocities along with missing and delayed F-wave responses identified the demyelinating type.

2

Thesis Topic

Automatic network coding of traffic junctions using

Abstract

Before any traffic simulation can be performed, the network of roads and junctions is modeled. Assigning attributes to the roadway network, such as the road length and width, the junction type, number of arms, and lanes, is a crucial task while building the network. This research is an attempt to develop an efficient traffic junction classifier using machine learning and deep learning algorithms on satellite images. Three junction categories, Priority, Roundabout, and Signal, are considered for analysis. As this is a novel research idea, the required image dataset of junctions is created using the Google Maps API. By using robotic process automation, the downloading of the images is automated. Two approaches are taken to build the classifiers: a machine-learning approach and a deep-learning approach. The machine learning approach is split into two phases: the feature extraction phase and the classification phase. In the feature extraction phase, a Histogram of Oriented Gradients (HOG) descriptors is used to extract features from the images. Furthermore, in the classification phase, several classification algorithms are applied to the HOG features to build classifiers. In the deep-learning approach, taking advantage of powerful pre-trained models and transfer learning, a Convolutional Neural Network (CNN) is developed for classifying the junctions. The models are evaluated, and in the end, a comparison between the various classification models is performed. The results showed that the CNN classifier modeled had the best accuracy and AUC compared to the other models with scores of 0.81 and 0.94 respectively. Among the machine learning models that were trained on the HOG features, the Extreme Gradient Boosting model has the best accuracy of 0.62. The ultimate aim of this work is to use this junction-classifier model on real projects to aid the process of finding the type of junctions and reduce the effort and time required to model the roadway networks.

Build A Strong Portfolio



Commits

Demonstrate consistency, collaboration, and coding discipline

Code

Showcase well-documented repositories

Projects

Host end-to-end DS/ML/AI projects that highlight real-world problem-solving

Kernels

Highlight data processing and EDA methodologies

Ranking

Evaluate and reflect global standing among data science practitioners

Competitions

Demonstrate problem-solving under tight constraints

Headline

Concise summary of goals, competencies, and professional identity

Summary

Engaging overview of learnin and career journey

Projects

Showcase practical experience, outcomes, and skill application

GitHub helps with

- ✓ Validating coding skills
- ✓ Showing growth and consistency
- ✓ Being interview-ready for Tech roles

Kaggle helps with

- ✓ Building credibility in data science circles
- ✓ Applying learning to real datasets
- ✓ Speaking confidently in Tech interviews

LinkedIn helps with

- ✓ Improving visibility with recruiters
- ✓ Positioning better for job openings
- ✓ Networking with peers and mentors in the field

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Microsoft-Certified Advantage –

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Microsoft Learn content modules covered

- Introduction to Generative AI Concepts
- Introduction to GitHub Copilot
- Design & Manage Analytics Solutions using Power BI
- Designing & Implementing a Data Science Solution on Azure

Microsoft AI Immersion | Bangalore

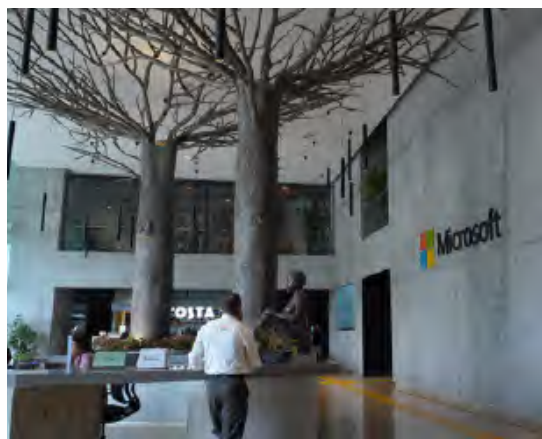
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- Master advanced AI-powered collaboration & automation tools
- Build muscle memory with real-world Microsoft applications
- Gain insider strategies directly from the creators of the technology

Why It Matters:

-  Global Recognition – Microsoft badges showcase your skills worldwide
-  Career Differentiator – Employers value certified Microsoft professionals
-  Practical Edge – Learn tools you'll use daily in analytics, AI, and cloud

With industry backed certification from upGrad in association with Microsoft certification + IITB's academic rigour, you graduate with **dual credibility**: academic excellence and industry validation.



Rich and Dedicated **Live Support**

Industry Expert Sessions

Engage with industry practitioners as they help you master in-demand skills and concepts using a demonstrative hands-on approach



IIITB Faculty Sessions

Learn from some of the most accomplished academicians as they take your knowledge and understanding of data science to another level



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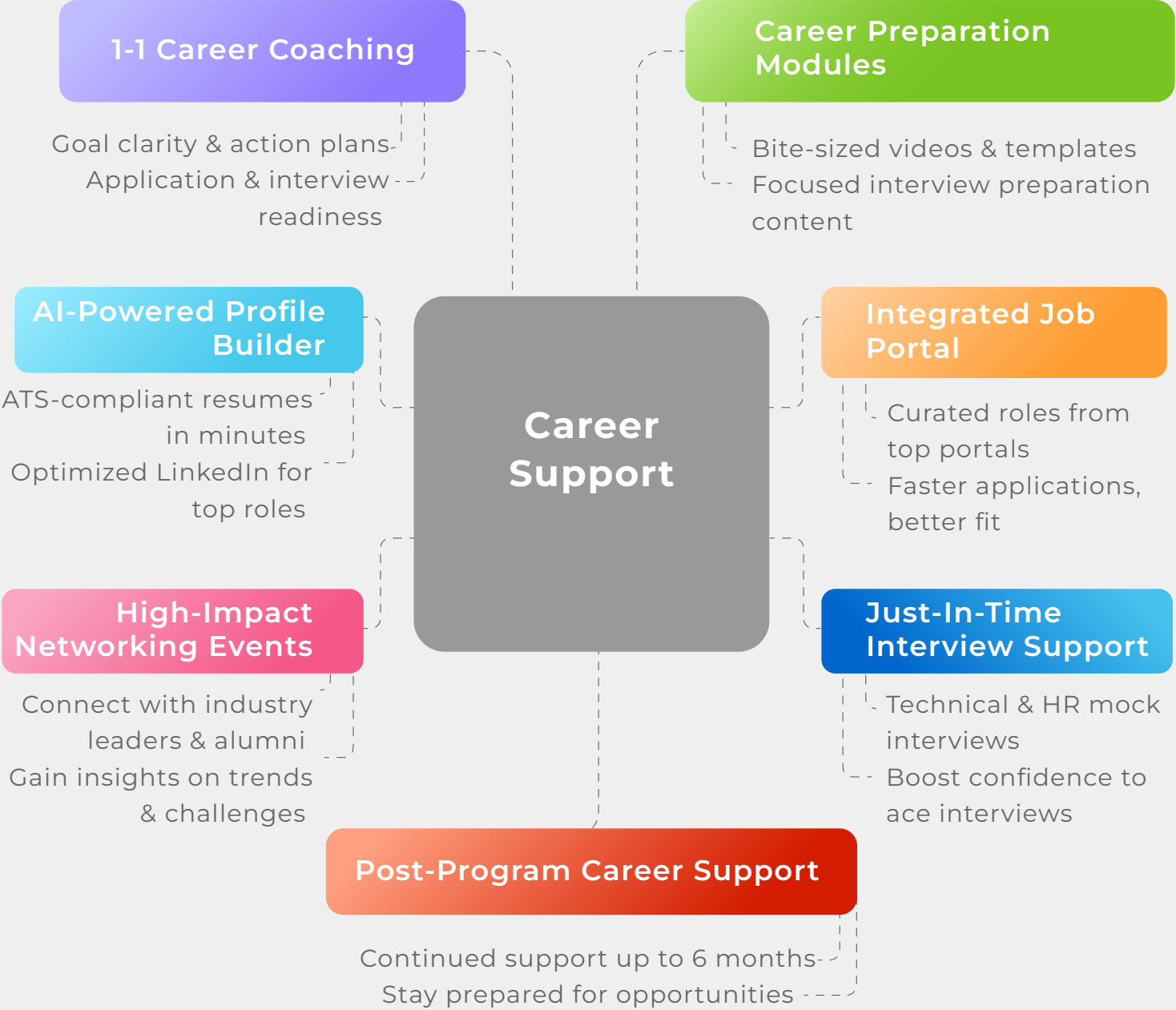
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Join doubt resolution session slots, that are available daily, and have an expert available to resolve your queries for a smooth learning journey

Effective and Complete Career Support

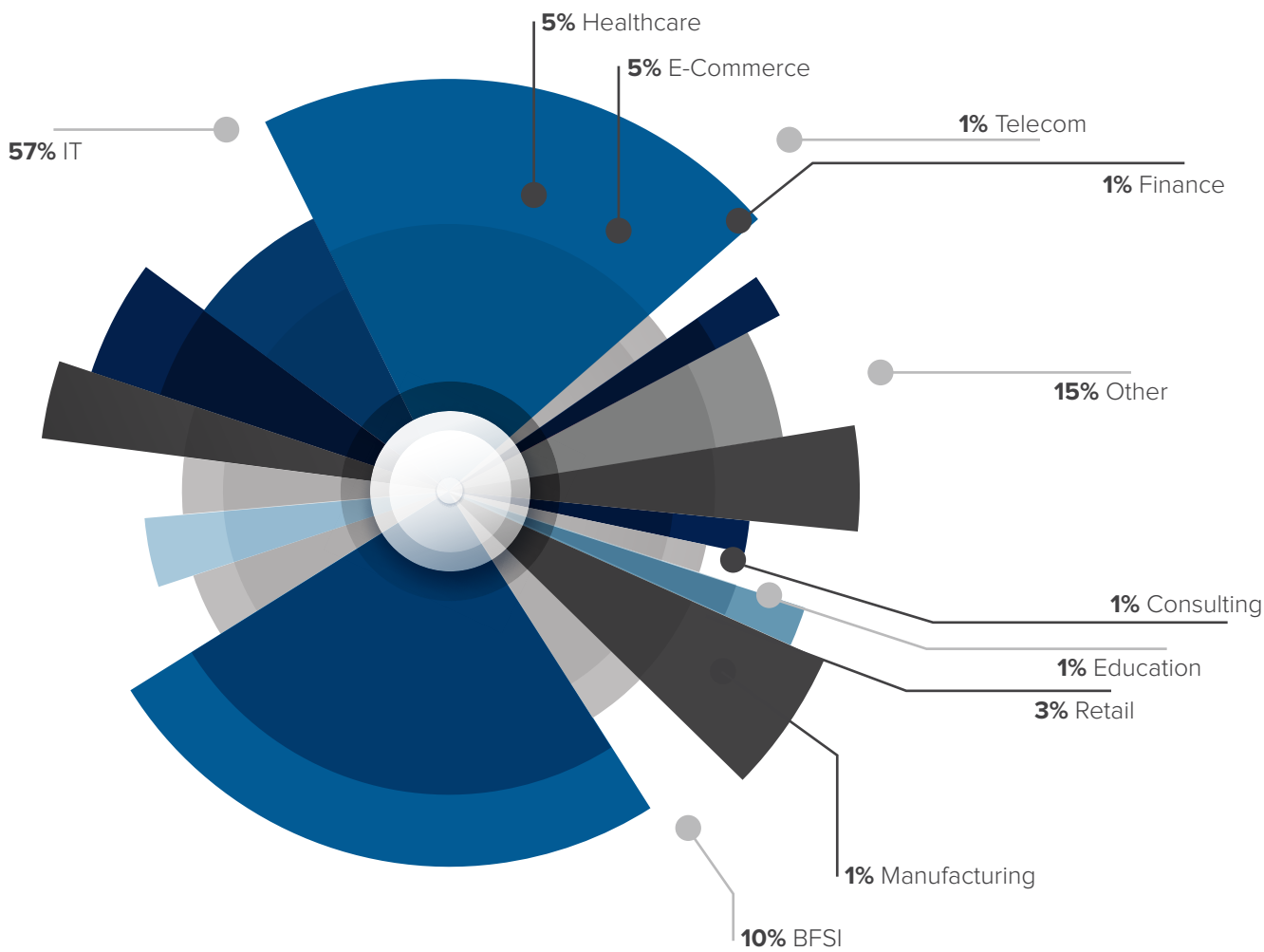


Student Support

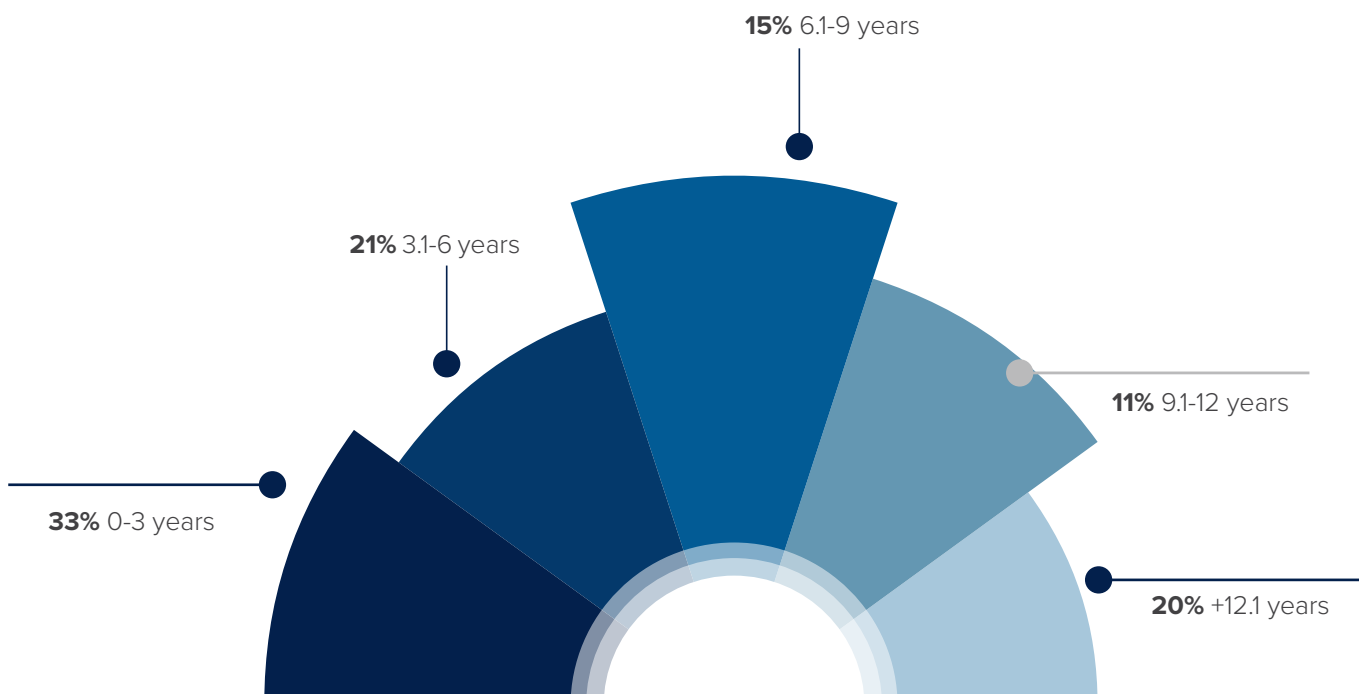
Telegram channel for learner communications Cohort Telegram channel for timely program updates and announcements.	Non-academic and non-technical query assistance 24 X 7 m-AI-ask bot support for non academic assistance.
Completion Support Personalised assistance for smooth program completion, managing backlogs, and cohort deferrals with free and paid waiver options.	Financial benefits Access benefits like referrals and repeats by sharing details with your program co-ordinator or student support executive.

Meet the Class

Industries Our Students Come From



Work Experience





A single specialty is no longer enough.

For years, the T-shaped professional was considered the gold standard—someone with broad knowledge across multiple areas (the top of the “T”) and deep expertise in one core area (the vertical stem). This model worked well in stable environments where projects moved slowly, and roles were clearly divided

But today’s AI-driven, high-velocity world has exposed the limits of the T-shape. Knowledge silos, handoff delays, and rising complexity mean one deep skill is no longer enough

Today's employers need π -shaped professionals - someone with two deep, complementary specialisations while maintaining broad knowledge across domains, who can own entire project lifecycles without team dependencies, bridging the gap between development, operations, and analytics seamlessly

Introducing The π Professional Pack

Add an extra specialisation in just 3 months

STEP 1

Choose your primary foundation

Select your core specialisation:

Data Engineering, Data Analytics, Generative AI or MLOps

STEP 2

Add your strategic second specialty

In the latter half of your program, unlock a complete second track featuring:

-  **Full curriculum coverage**
-  **Live interactive sessions with industry experts**
-  **Dedicated faculty support**
-  **Expert mentorship across both domains**

Why is this a strategic career investment?

-  **Expert in two complete specialisations**
-  **Industry-validated skill combinations based on hiring data**
-  **Hands-on experience in both domains**
-  **Portfolio projects demonstrating π -shaped expertise**

Your Competitive Edge:

While others graduate with single specialisations, you emerge as a π -shaped leader ready to:

-  **Own complete project lifecycles**
-  **Bridge team silos and communication gaps**
-  **Command premium positions in India's GCC & IT sectors**
-  **Adapt quickly as AI continues reshaping the industry**

Popular specialisation combos

GenAI + MLOps

Why Gen AI + MLOps?

Enables end-to-end ownership of Gen AI models from fine-tuning to deployment, monitoring, and safety. This pairing creates a professional who can not only innovate but also reliably deliver and manage that innovation.

GenAI + Analytics

Why GenAI + Analytics?

Unlocks AI-driven business insights with a combination of natural language queries, auto-reporting, and data storytelling. This pairing is about creating smarter and more accessible business insights.

MLOps + Data Engineering

Why MLOps + Data Engineering?

This is the ultimate combination for building robust and scalable AI infrastructure. Enables you to build an automated, end-to-end machine learning factory that is efficient, reliable, and ready for production demands.

GenAI + Data Engineering

Why GenAI + Data Engineering?

Let's combine Gen AI innovation with solid data pipelines scalable AI products. This pairing ensures that innovative GenAI projects are built on a solid foundation, allowing them to be scalable and successful in the long run.

Earn Second Certificate

π Professional pack for dual specialisations

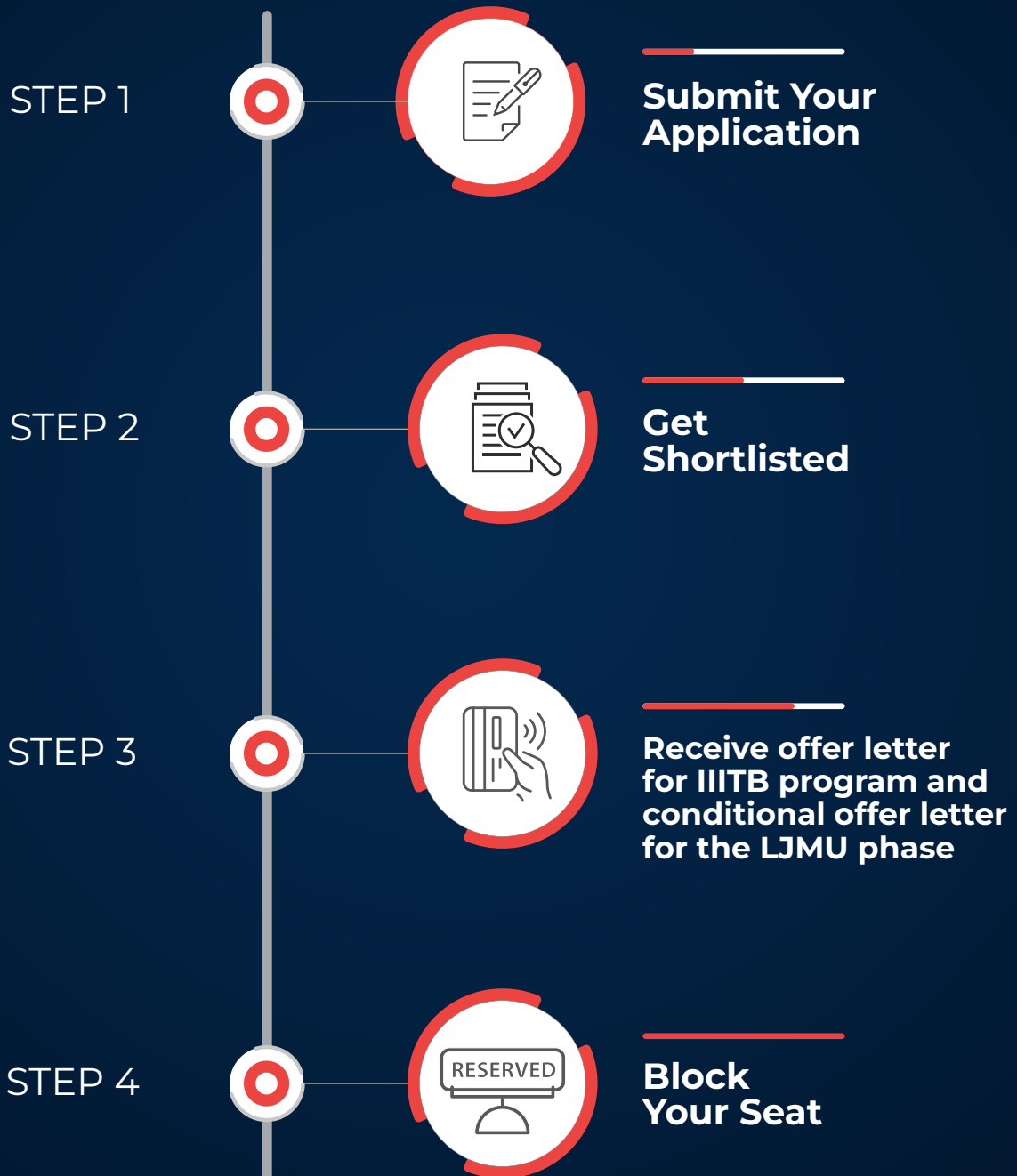


In just 3 more months learners can earn 1more specialization across both Data Science & Machine learning specializations

Enrol in 4 small steps, **Then take a giant leap.**

Eligibility Criteria

Bachelor's or Master's Degree or its equivalent in any discipline with minimum 50% aggregate mark or equivalent CGPA.



Please note that your admission to the LJMU's MSc in Data Science Program is strictly provisional and conditional upon your successful completion of the IIITB Executive Diploma in Data Science & AI



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LET'S TALK

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Professional Certificate Programme in

Data Science With Generative AI

Become an AI-augmented data science professional in 24 weeks. Build dashboards leaders use, models that move revenue, and AI workflows that make you 10× faster.



24 Weeks

8–10 hrs/week | Live + Recorded + Workshops | Capstone project

Get Certified by IIIT Bangalore | 10+ Years Legacy Data Science and AI Programs

The AI-Augmented Data Analytics Revolution

Today, every business decision is a data decision. Most teams still struggle to bridge traditional BI with modern AI. This programme closes that gap.



Traditional Analyst

Excel reports
Days to insight
Reactive →



Modern Data Scientist

Dashboards
Days to insight
Reactive →



AI-Augmented Data Scientist

Predictive + GenAI Near
real-time Prescriptive &
explainable →

Your 24-Week Transformation

Weeks 1–8 Foundation

Python, SQL,
Statistics

Weeks 9–16 Application

Dashboards,
Models, ETL

Weeks 17–24 Innovation

GenAI, RAG,
Automation

Who This Programme Is For

- ▶ Early-career professionals moving into data science/analytics
- ▶ Business professionals looking to add technical depth
- ▶ Analysts preparing for the AI shift
- ▶ Graduates building industry-ready portfolios

The convergence of BI and AI is no longer future—it's now.



Programme Structure & Learning Path



Learn → Apply → Build → Deploy
The IIITB approach

Program at a Glance



Duration
24 weeks



Effort
8–10 hrs/week



Format

Live + recorded + labs



Capstone

End-to-end real world project



Masterclass

10+ Live deep dives



Workshops

10+ Hands on workshops

Learning Methodology

10+ Masterclasses

10+ Workshops

30+ Industry cases

Daily support sessions

Peer community

1-to-1 capstone mentorship

Technology Stack

Python | SQL | Power BI/Tableau
scikit-learn | MLflow (basics)
Daily support sessions
OpenAI/Claude | LangChain
Vector DBs
Cloud (AWS/Azure)

Tools may change; equivalents used as needed. Certificate programme, not a degree. Career services are not a placement guarantee.

What You'll Build & Where It Leads

Your 24-Week Portfolio

Technical Assets	Business Outputs
✓ 6 Projects	Executive Deck
✓ 15+ Notebooks	ROI Framework
✓ GitHub Repos	A/B Test Reports
✓ Live Dashboards	Model Cards
✓ Kaggle Kernels	GenAI Blueprint

Industry Applications

BFSI	E-Commerce	Healthcare	Tech
Risk Analytics	Recommenders & Personal	Patient Analytics	Product Analytics



Your 6-Module Journey To Mastery

Module 1 Foundation (Weeks 1–5)

- Python for data
- SQL for analysis
- KPI datasets



Module 2 Statistics (Weeks 6–9)

- A/B tests that drive decisions
- Cohorts & funnels



Module 3 BI & Visual Storytelling (Weeks 6–9)

- Dashboards execs actually use
- Narratives & RLS



Module 4 Machine Learning (Weeks 14–17)

- Models that move metrics
- ROI & monitoring

Module 5 GenAI (Weeks 18–20)

- AI-assisted SQL & analysis
- RAG over knowledge



Module 6 CAPSTONE (Weeks 21–24)

- Your signature project end-to-end
- Repo + dashboards

How You'll Learn — And Get Support

TOUCHPOINTS

- 1-2 in-depth 4-hour Bootcamps Every Month (Apply, Q&A, code-along)
- 120+ hours of live session, 2 hours/week of recorded content
- Projects and Assessments for Practice

DAILY HELP & COMMUNITY

- Discussion forum / chat with <24h weekday response target
- m-AI-Ask AI Chatbot Support

MENTORING & CHECKPOINTS

- Module-end grading and feedback (M1→M5)
- Capstone group mentoring (W21–W24): plan → build → test → defend

LEARNING PLATFORM & TOOLS

- LMS for videos, quizzes, submissions
- GitHub for repos & reviews • BI sandbox (Power BI/Tableau)
- Jupyter/Colab for notebooks • Email/slack alerts for deadlines

PROGRESS & INTERVENTIONS

- Dashboards for grades & attendance
- “Nudge” emails if you fall behind

Detailed Curriculum

Module 1: Foundations of Applied Data Science

(Weeks 1–5)

Build your data science tool kit: Python, SQL, business thinking

Outcome

Join, clean, and prepare messy business data. Ship a reproducible pipeline and a SQL backend that powers a dashboard.

Topics

- Pandas loading/cleaning
- Merges
- KPI aggregations
- Reusable
- Pipelines
- SQL joins/CTEs/windows
- Query optimisation

Sample projects

- Weekly sales & margin pipeline
- Cohort table (retention)
- SQL backend for revenue dashboard

Tools



Module 2: Statistical Intelligence & Business Analytics

(Weeks 6–9)

Turn analysis into confident business decisions

Outcome

Choose the right metrics and tests, run clean analyses, and present evidence-backed recommendations leaders can act on.

Topics

- KPI science (trees, North Star)
- Cohorts & funnels
- RFM
- Market
- Basket
- Time-series decomposition
- Attribution
- Hypothesis testing & power
- CIs
- Bayesian thinking
- Pitfalls & insight documentation

Sample projects

- A/B test plan & results
- Cohort retention study
- Executive insight report

Tools



Module 3: Data visualisation & Storytelling

(Weeks 10–13)

Create executive-ready dashboards and stories

Outcome

Build fast, reliable dashboards and present the “so what” with clear narratives that influence decisions.

Topics

- Visual theory (chart choice, colour/accessibility, Gestalt, mobile)
- BI modelling
- Calculated fields
- Parameters/filters
- Drill-through & RLS
- Performance optimisation
- Python viz & automation (Plotly, scheduled email)
- Style guide
- Pyramid Principle

Sample projects

- C-suite revenue dashboard
- Operations KPI board
- Automated
- Weekly distribution

Tools



Module 4: Machine Learning

(Weeks 14–17)

Predictive analytics that drive action

Outcome

Train and explain models for churn, demand, pricing, or risk; deploy and monitor them with business-first metrics.

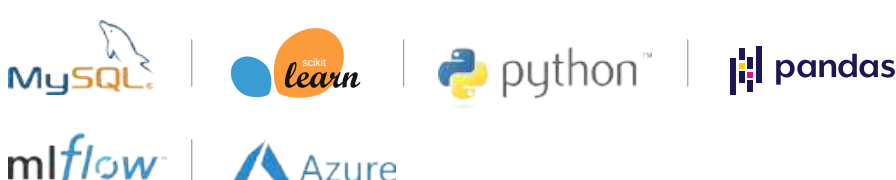
Topics

- When ML vs rules
- Feature engineering
- Supervised/unsupervised
- Business evaluation (cost of error, ROI)
- Model cards
- AutoML (PyCaret/Vertex/Azure)
- Deployment endpoints
- MLflow tracking
- Monitoring
- Dashboards & A/B of models

Sample projects

- Churn prediction & playbook
- Demand forecast with CI
- Price elasticity & optimisation

Tools



Module 5: Generative AI

(Weeks 18–20)

GenAI as your analytics multiplier

Outcome

Use AI to draft SQL/code, speed up EDA, explain anomalies, and power safe Q&A over business knowledge with guardrails.

Topics

- SQL/code generation
- Automated EDA
- Anomaly explanations
- NL Q&A for dashboards
- RAG over company knowledge
- Workflow automation (orchestration)
- Governance: validation, bias, privacy,
- Cost control

Sample projects

- AI-augmented analysis pipeline
- RAG Q&A over metrics
- Time-reduction evidence note

Tools



Capstone AI driven business solution

(Weeks 21–24)

Prove your mastery with a real business solution

Outcome

Deliver an end-to-end business solution (repo + dashboards + model card + exec deck) ready for portfolio and review.

Phases

- W21 charter/data
- W22 analysis & model
- W23 BI build + GenAI + user test
- W24 exec presentation, adoption & monitoring plan

Track options

- AI-augmented analysis pipeline
- RAG Q&A over metrics
- Time-reduction evidence note

Tools



Faculty



Chandrashekar Ramanathan
Professor & Dean (Academics)



Prof. Chandrashekar is a faculty member at IIT-B since 2007 serving as professor, researcher and administrator. He has been working in the field of Computing for over 25 years in various capacities across industry and academia.



Dr. V. Sridhar
Professor



Dr. V. Sridhar, a professor at IIT-Bangalore's CITAPP, specializes in telecommunications technology, management, and regulation. He has co-authored two books, published numerous articles in leading journals, and written about 300 articles on Telecom and IT in major business newspapers and magazines.

Industry Experts



Deependra Singh
VP & Head of Data Science



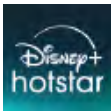
Over 15 years of experience in leading analytics practices, data science, deep learning, and AI product development. Successfully led teams at Jungle Games, American Express Digital Business, and National Insurance Company, pioneering key projects like the analytics engine for the GOI PMJAY policy. Respected speaker at top educational institutes like IMT Hyderabad, BIT Mesra, and NMIMS.



Sajan Kedia
Sr. Engineering manager



Senior Engineering Manager, Hotstar Sajan has extensive experience in the field of ML, Big Data, Data Science, and AI.



Sr. Engineering manager



Machine Learning Engineer



Machine Learning Research Engineer



Mirza Rahim Baig
Startup Mentor



Analytics Lead, Zalando Mirza is a veteran professional with 10+ years of experience in application of data science, machine learning in e-commerce and healthcare.



Team Lead - Product



Marketing Analytics



Ex-Analytics Lead

Program Completion Certificate



Earn valuable credentials with a certificate and accredited with NAAC A+ (2021).
Join India's largest DS & AI alumni network of over 10,000 professionals.

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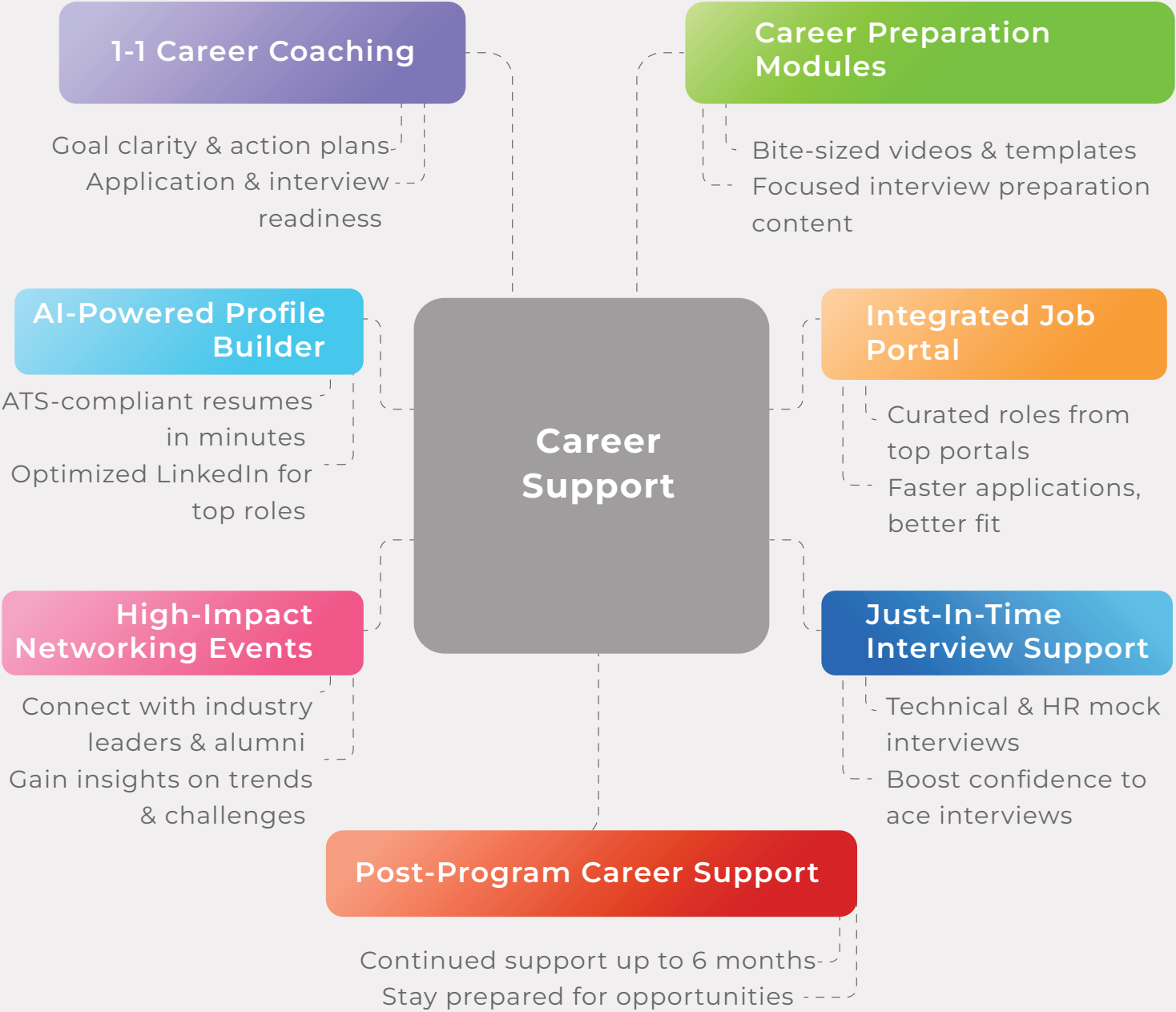
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About IIITB

The International Institute of Information Technology Bangalore (IIITB)

Established in 1998, IIITB is a premier institute known for its interdisciplinary approach, integrating technology with social sciences. Supported by the Government of Karnataka and the IT industry, IIITB fosters strong academic-industry partnerships and attracts top talent from across India and abroad through its merit-based selection process.

The institute has graduated over 3,500 students, many of whom work at leading IT companies globally. With a focus on research and development in fields like Artificial Intelligence (AI) and Machine Learning (ML), IIITB is recognised as a leader in AI education.

Ranked 69th in the Engineering category of the National Institutional Ranking Framework (NIRF) in 2025, IIITB continues to excel in education and research, making it a preferred destination for aspiring technologists and future leaders.



AICTE



UGC

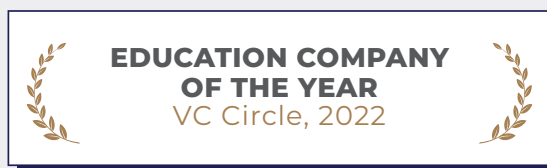
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Ronnie Screwvala

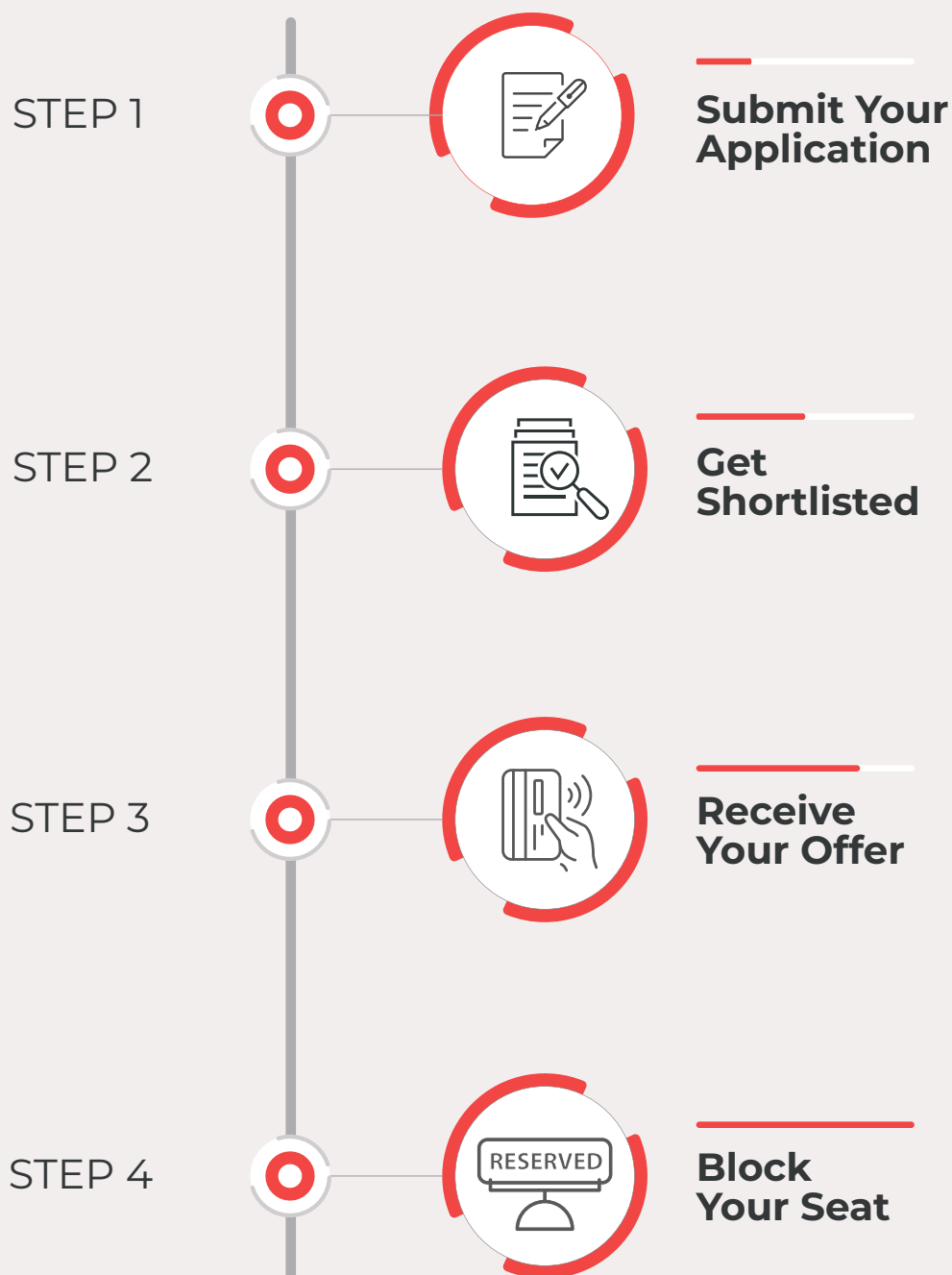
Co-founder & Executive Chairman



Admissions & How To Apply

Eligibility Criteria

- Graduates & Undergraduates (preferably in the final year)





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admissions@upgrad.com



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Call: 1800-210-2020

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LET'S TALK

The Agentic AI Revolution is Here

2023: ChatGPT changes the conversation
2024: Companies race to adopt AI
2025: Winners deploy autonomous agent systems



The question isn't whether AI will transform your industry. It's whether you'll build the systems, or be replaced by them.

Executive Post Graduate Programme in **Applied AI & Agentic AI**

30 Weeks | LIVE + Labs

From ML Foundations to Autonomous Agent Systems



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Ronnie Screwvala

Co-founder & Executive Chairman



GSV EDTECH 150
Top 150 Most Transformative
Growth Companies



**BRANDON HALL
GOLD AWARDS**
15+ awards across 2 years



PROMISING BRAND 2022
The Economic Times



**EDUCATION COMPANY
OF THE YEAR**
VC Circle, 2022



Entrepreneur India
**EDUCATION INNOVATION
AWARDS 2020**
Best Communication &
Teaching Platform



The Skills Gap

(And Your Opportunity)

Companies Need

- ✓ Deploy LLMs in production
- ✓ Build autonomous agents
- ✓ Scale AI economically
- ✓ RAG with enterprise data
- ✓ Measure ROI, control costs

Most Engineers Have

- ✗ Theoretical ML knowledge
- ✗ Isolated deep learning
- ✗ Notebooks that never deploy
- ✗ No production experience
- ✗ Academic projects only

Market **Reality** (2023–2025)

88%

of AI projects fail to reach production

IDC Enterprise Survey

Agentic AI roles command

+30-40%

Salary premiums

Typical time to build deployment skills

18-24

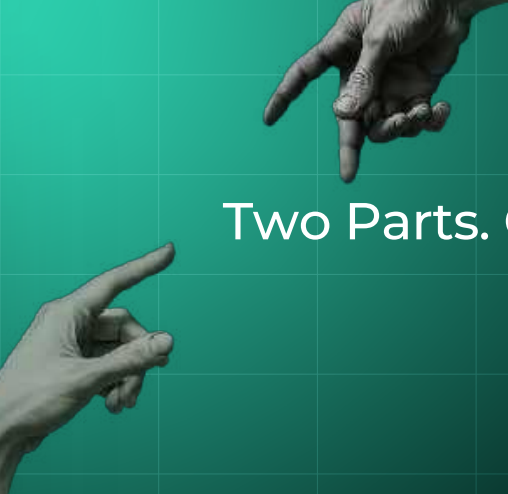
Months

The Programme Advantage



30 weeks to production

Live systems with monitoring, ROI docs, and live URLs



Two Parts. One Transformation

Part A: Applied AI

Master Production ML

Module 1 (W1-4)
Foundations & Data
Python • SQL • Features

Module 2 (W5-9)
ML & Deep Learning
Models • CNNs • Transformers

Module 3 (W10-14)
MLOps & Production
AWS • Docker • Monitor

Capstone A (W15)
End-to-End ML System
Deploy + Monitor + ROI

- **Part A Outcome:**
ML API in production with monitoring & cost controls

Part B: Agentic AI

Build Autonomous Systems

Module 4 (W16-18)
LLM Foundations & RAG
APIs • Embeddings • Vectors

Module 5 (W19-23)
Agent Frameworks & orchestration • LangChain • LlamaIndex • LangGraph

Module 6 (W24-28)
Fine-tuning & Scale LoRA • Bedrock • Security

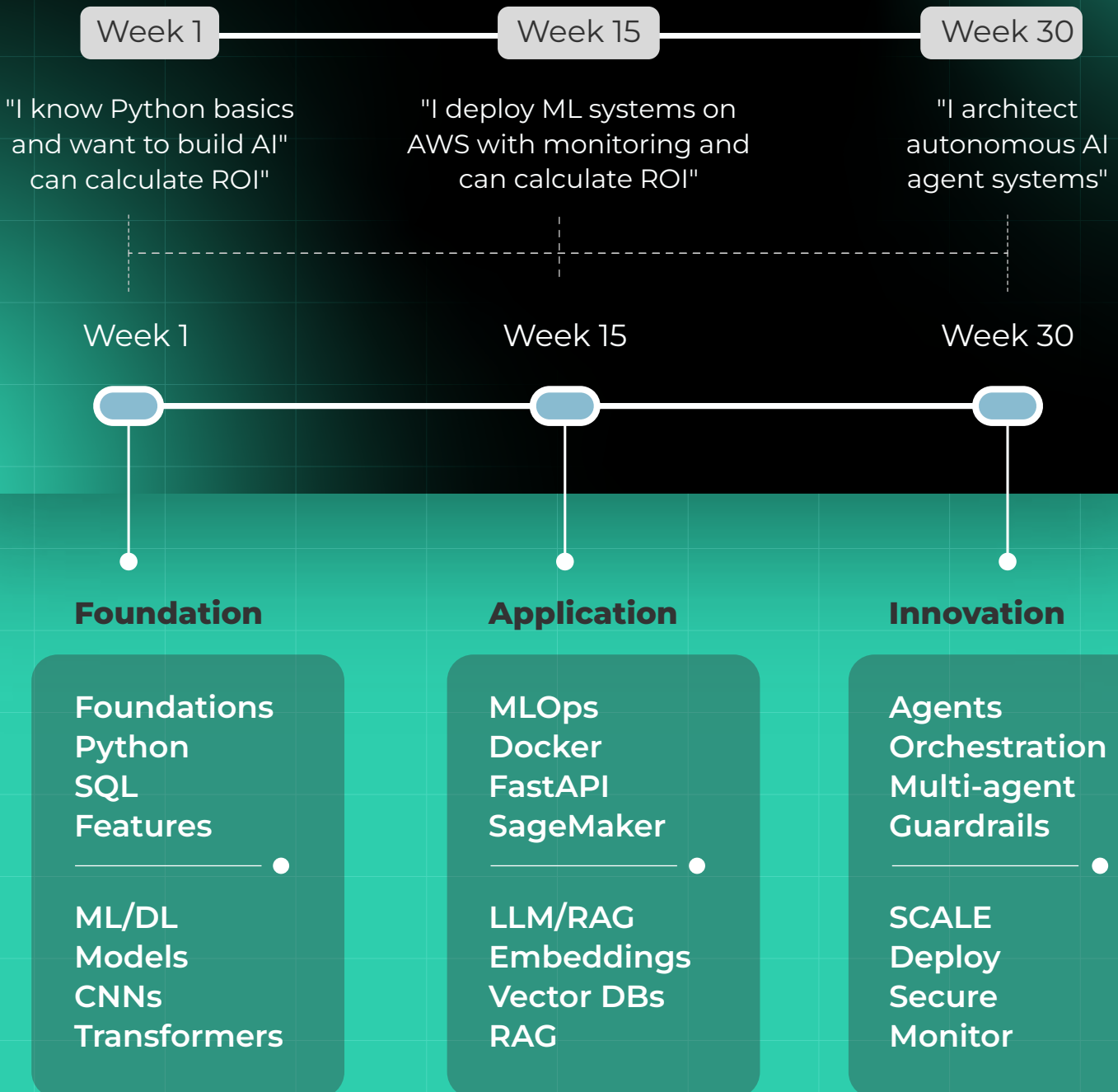
Capstone B (W29-30)
Production Agent Platform
Multi-Agent + RAG + Deploy

- **Part B Outcome:**
Multi-agent RAG platform with security & ROI report

Tools evolve

we teach orchestration patterns first (planning, memory, tool-use, safety), with hands-on in current leaders like LangChain, LlamaIndex, LangGraph.

Your 30-Week Transformation



Why This Programme Vs Everything Else

Typical Programmes

- Theory-heavy ML
- 2020-era techniques
- Academic toy projects
- One capstone (maybe)
- Cloud mentioned, rarely used
- "Here's scikit-learn"
- LLMs as theory
- Cost ignored
- Generic certificate

This Programme

- Applied + Agentic in one
- 2024–25 frontier: agents, RAG
- Production systems, live APIs
- Two capstones, two portfolios
- AWS hands-on every week
- "Here's deployment + monitoring"
- Orchestration in prod (patterns)
- Token economics + ROI built-in
- Deployed systems you can demo

The Three Competitive Advantages

Dual Architecture

- Applied AI (W1–15) + Agentic AI (W16–30) in one programme
- Build churn models AND autonomous agent crews
- Prove versatility hiring managers want

Production-First

Every module ends with deployment
W15: ML API on AWS with monitoring
W30: Multi-agent platform in production
Portfolio = Live URLs, not notebooks

Agentic AI Depth

15 weeks on autonomous agents (most programmes: 0–2 weeks)
LangChain · LangGraph · AutoGen · CrewAI
Production patterns for 2025 job market (30–40% premium roles)

What You'll Ship

8+

Projects

Pipeline » Agents
Every Week Ships

2

Capstones

A: ML System (W15)
B: Agent Platform (W30)

90+

Live Hours

Lectures + Labs +
Clinics + Support

Portfolio Assets



Github Repos

Clean, Tested Code | CI/Cd Pipelines | Docs | Code Reviews



Deployed Sys

ML Apis On Aws | Fes; Endpoints | Agent Platforms | Monito



Business Docs

Roi Calculations | Model Cards | Architecture | Exec Deck

Capstone

A

● (Week 15)

End-To-End ML System

Track

Predictive Maintenance OR Customer Intelligence OR Project form your domain

Deliver

Repo - Aws Deploy - Monitoring - Roi Brief - P,E,G,Tacg Deck

Capstone

B

● (Week 29-30)

PRODUCTION AGENTIC AI

Track

Enterprise Assistant OR Autonomous Research OR Ops Crews OR
Develop a custom AI agent for your field

Deliver

Architecture | Repo | Bedrock deploy | Monitoring | Cost report
Security review | ROI | Defence presentation

Detailed Curriculum

Module 1: Foundations & Data Engineering

(Weeks 1–4)

Build production-ready data pipelines and SQL backends

Key Topics

Python OOP, Pandas advanced, SQL joins/CTEs/windows, Feature engineering, Scaling/encoding, Pipeline automation, EDA methodology, Matplotlib/Seaborn, Git/GitHub, Business communication, Stakeholder presentations

Sample projects

Weekly sales & margin pipeline | Cohort retention table | Feature engineering competition | SQL backend for revenue dashboard

Tools



Module 2: Machine Learning & Deep Learning

(Weeks 5-9)

Train, interpret, and deploy ML/DL models for business impact

Key Topics

Linear/logistic regression, Trees/boosting (XGBoost/LightGBM), Model eval, SHAP/LIME interpretability, SVM/ensembles, Clustering/PCA/UMAP, Neural nets from scratch, Backpropagation, CNNs/transfer learning, RNNs/LSTMs, Transformers/attention, BERT/GPT, Time series (ARIMA/Prophet)

Sample projects

Churn model with SHAP report | Image classifier via transfer learning | NLP sentiment with transformers | Time series forecasting | Customer segmentation with clustering

Tools



Module 3: Mlops & Production Systems

(Weeks 10-14)

Deploy ML with monitoring, cost control, and governance

Key Topics

AWS (S3/IAM/VPC/Glue/Athena/Redshift), SageMaker training/HPO, MLflow tracking, Docker containerization, FastAPI serving, A/B testing, Model drift detection, Monitoring dashboards, Fairness/bias, Governance/audit, ROI calculation, Stakeholder communication

Sample projects

Cloud data pipeline | Distributed training | Dockerized API | MLOps pipeline with monitoring | Executive ROI presentation

Tools



Capstone A End-To-End ML System

(Weeks 15)

Prove your mastery with a real business solution

Tracks:

- Predictive Maintenance
- Customer Intelligence

Deliverables

GitHub repo | AWS deployment | Monitoring dashboard | Defense deck | ROI document | Security & cost checklists (must pass)

Module 4: LLM Foundations & Rag Systems

(Weeks 16-18)

Build production RAG with governance and cost optimization

Key Topics

LLM architecture/scaling laws, OpenAI/Anthropic/Google APIs, Token economics/cost optimization, Prompt engineering (few-shot/ReAct), Embeddings/similarity, Vector DBs (Pinecone/Weaviate/Chroma), Hybrid search, Chunking strategies, RAG patterns, Query rewriting, Multi-hop reasoning, Citation/attribution, Retrieval governance, RAG evaluation

Sample projects

Prompt engineering challenge | RAG prototype with metrics | Production RAG API with citations and governance

Tools



Module 5: Agent Frameworks & Orchestration

(Weeks 15-23)

Design multi-agent systems with planning and supervision

Key Topics

Agent types/capabilities, ReAct pattern, Tool use/function calling, Memory systems, Planning/goal decomposition, LangChain (chains/agents/tools/memory), LangGraph (state/routing/supervision), LlamaIndex (indices/routers/multi-doc), AutoGen (collaboration/consensus), CrewAI (production/delegation), Telemetry/observability, SLOs/error budgets

Sample projects

Agent architecture design • LangChain app with tools • LlamaIndex multi-doc system • AutoGen collaboration • CrewAI production crew

Tools



Module 6: Fine-Tuning, Scale & Production

(Weeks 24-28)

Tune models, harden security, deploy at scale

Key Topics

Fine-tuning strategies, LoRA/QLoRA/PEFT, Dataset prep/quality, Evaluation/benchmarking, AWS Bedrock/SageMaker, Ray Serve scaling, Load balancing, Caching/optimization, CI/CD (GitHub Actions), Blue-green/canary deploys, Prompt injection defense, Privacy/audit logging, IAM/security, Red team testing, EU AI Act compliance, Human-in-the-loop, Cost optimization

Sample projects

Fine-tuning with LoRA • Bedrock deployment • CI/CD pipeline • Security red team drill • Cost optimization report

Tools



Capstone B Production Agentic AI System

(Weeks 29-30)

Prove your mastery with a real business solution

Tracks:

- Enterprise Assistant
- Autonomous Research
- Business Operations

Deliverables

Architecture diagram | GitHub repo | AWS Bedrock deployment | RAG with governance | Monitoring dashboard | Cost report | Security review | Defense presentation | Security & cost checklists (must pass)

The Complete AI Tool Stack

Master the industry's most in-demand tools from Python to LangChain and build systems that run in real-world production.



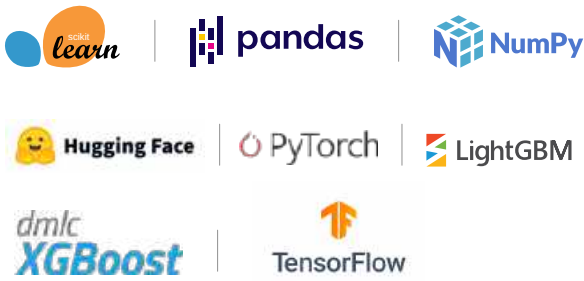
Programming & Data Foundations

Tools and environments for data manipulation, coding, and visualization



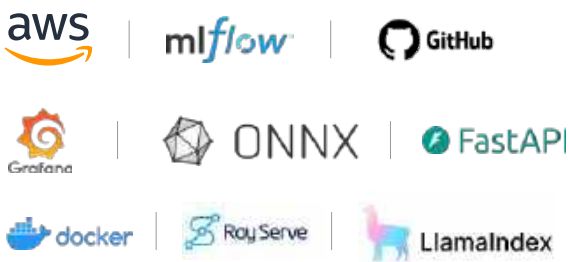
Machine Learning & Deep Learning

Core ML/DL frameworks and interpretability libraries



Cloud, MLOps & Deployment

End-to-end deployment, monitoring, and automation tools



LLMs, RAG & Vector Databases

Large Language Model tooling and RAG (Retrieval-Augmented Generation) stack



Agentic AI & Orchestration Frameworks

Tools for multi-agent design, memory, planning, and supervision



Security, Governance & Scaling

For enterprise-grade robustness, compliance, and optimization



- Privacy & Audit Logging, IAM Policies, Red Team Testing
- EU AI Act Compliance, Cost Optimization, Human-in-the-loop Systems



Visualization, Reporting & Business Tools

Tools for business storytelling, dashboards, and ROI tracking



- ROI Documentation Templates, Executive Defense Decks



Supporting Tools & Utilities

Essential for workflow, code quality, and collaboration



- Business Docs: Model Cards, Architecture Diagrams, ROI Reports

Build Real-World AI Agents. Deploy Them Across Domains

Build industry-grade AI blueprints, from adaptive tutors to compliance bots, and showcase your skills with a portfolio of deployable agents



Healthcare & Life Sciences

Autonomous Clinical Trial Recruitment Agent



BFSI

AI Compliance Officer for Financial Reporting



Retail & E-commerce

Hyper-Personalized Conversational Shopping Agent



Manufacturing & Supply Chain

Autonomous Supply Chain Disruption Response Crew



Media & Entertainment

AI-Powered Content Strategy & Generation Pipeline



Education (EdTech)

Adaptive AI Tutor or Personalized Learning Paths



Real Estate & PropTech

AI Agent for Property Valuation and Market Analysis



Energy & Utilities

Predictive Grid Management Agent Crew



Automotive

Autonomous Vehicle Sensor Data Analysis Crew



LegalTech

AI Paralegal for e-Discovery and Case Law Research



Agriculture (AgriTech)

AI-Powered Precision Farming Operations Agent



Hospitality & Travel

Autonomous Travel Concierge and Itinerary Planner

Two Capstones. Double the Impact.

From Model to Market — Deploy Twice, Learn Twice.



Applied AI Capstone

- » **Theme**
ML System in Production
- » **Goal**
Deploy a machine learning system that solves a real business problem.
- » **Focus Areas**
Data Pipelines
MLOps & FastAPI
Monitoring & ROI tracking
- » **Output**
Deployed ML API + Dashboard



Agentic AI Capstone

- » **Theme**
Autonomous Multi-Agent System
- » **Goal**
Design and deploy a production-ready multi-agent platform.
- » **Focus Areas**
LangChain, LlamaIndex
Agent Orchestration
Cost & Security Optimization
- » **Output**
Agent Platform + Monitoring Dashboard

Portfolio Presentation Deck

Combine both capstones into a professional presentation showcasing architecture, code, outcomes, and ROI.



Program Completion Certificate



Earn valuable credentials with a certificate and accredited with NAAC A+ (2021).
Join India's largest DS & AI alumni network of over 10,000 professionals.

Faculty

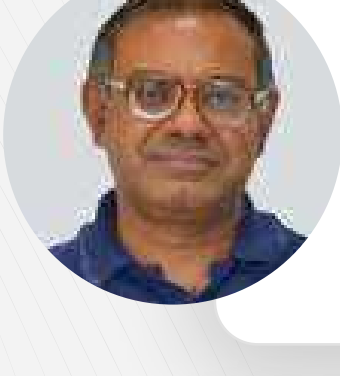


Dr. Debabrata Das

Director of IIITB



He has received his PhD from IIT-KGP. His main areas of research are IoT and Wireless Access Network.

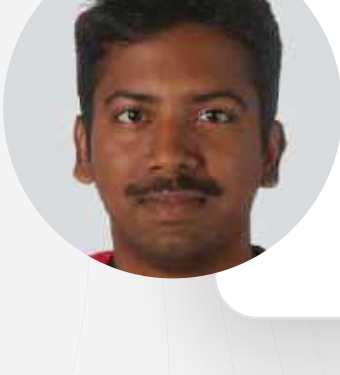


Prof. G. Srinivasaraghavan

Professor, IIITB



Prof. Srinivasaraghavan has a PhD in Computer Science from IIT-K and 18 years of experience with Infosys Technologies and several other companies.



Dr. Dinesh Babu Jayagopi

Professor, IIITB



Dr. Dinesh is currently an Associate Professor at IIIT-B where he heads the Multimodal Perception Lab. His research interests are in Audio-Visual Signal Processing, Machine Learning, and Social Computing. He obtained his doctorate from Ecole Polytechnic Federale Lausanne (EPFL), Switzerland.



Chandrashekar Ramanathan

Professor & Dean (Academics)



Prof. Chandrashekar is a faculty member at IIIT-B since 2007 serving as professor, researcher and administrator. He has been working in the field of Computing for over 25 years in various capacities across industry and academia.



Tricha Anjali

Ex-Associate Dean



Prof. Anjali has a PhD from Georgia Institute of Technology as well as an integrated MTech (EE) from IIT Bombay.



Dr. V. Sridhar

Professor



Dr. V. Sridhar, a professor at IIIT-Bangalore's CITAPP, specializes in telecommunications technology, management, and regulation. He has co-authored two books, published numerous articles in leading journals, and written about 300 articles on Telecom and IT in major business newspapers and magazines.

Industry Experts



Abhishek Vijayvargia

Senior Data Scientist



Having worked with Microsoft as a Senior Data Scientist, he is an alumnus of IIT Kharagpur with 10+ years of experience in a Data Science domain



Ex-Senior Data Scientist



Anand

CEO

Gramener

A Straive Company

CEO, Gramener A gold medallist from IIM Bangalore, an alumnus of IIT Madras and London Business School, Anand is among the top 10 data scientists in India with 20 years of experience.



Faculty



Principal



Ex-Consultant



Manish Shukla

Head of Generative AI



Leading cutting-edge GenAI platform development at NatWest Group. Expertise in OpenAI products and MLOps for optimisation of operational efficiency and seamless project delivery with high user satisfaction.



Release Manager



Release Manager



Certified Scrum Master



Deependra Singh

VP & Head of Data Science



Over 15 years of experience in leading analytics practices, data science, deep learning, and AI product development. Successfully led teams at Jungle Games, American Express Digital Business, and National Insurance Company, pioneering key projects like the analytics engine for the GOI PMJAY policy. Respected speaker at top educational institutes like IMT Hyderabad, BIT Mesra, and NMIMS.



Sajjan Kedia

Sr. Engineering manager



Senior Engineering Manager, Hotstar Sajjan has extensive experience in the field of ML, Big Data, Data Science, and AI.



Sr. Engineering manager



Machine Learning Engineer



Machine Learning Research Engineer



Mirza Rahim Baig

Startup Mentor



Analytics Lead, Zalando Mirza is a veteran professional with 10+ years of experience in application of data science, machine learning in e-commerce and healthcare.



Team Lead - Product



Marketing Analytics



Ex-Analytics Lead



Rajesh Sabapathy

Cheif Data Scientist

Sr. Data Scientist, Healthcare sector

Rajesh, an AI/ML leader, has driven innovation in healthcare using deep learning and big data. With expertise in Python, P. and cloud platforms, he has delivered significant business value and collaborated across various industries. Formerly Distinguished Engineer at Fortune 5 Healthcare Giant

Health-tech Start-up

Microsoft-Certified Advantage –

This program doesn't just prepare you for the world of Data Science & AIML—it gives you the Microsoft edge.

Learners earn industry backed certification from upGrad in association with Microsoft by completing specially designed modules integrated into the program, boosting both credibility and career readiness.



Microsoft Learn content modules Certification Modules:

- Introduction to Generative AI Concepts
- Introduction to GitHub Copilot
- Design & Manage Analytics Solutions using Power BI
- Designing & Implementing a Data Science Solution on Azure

Microsoft AI Immersion | Bangalore

Go beyond theory. In this exclusive, hands-on immersion led by Microsoft experts, you'll:

- Master advanced AI-powered collaboration & automation tools
- Build muscle memory with real-world Microsoft applications
- Gain insider strategies directly from the creators of the technology

Why It Matters:

-  Global Recognition – Microsoft badges showcase your skills worldwide
-  Career Differentiator – Employers value certified Microsoft professionals
-  Practical Edge – Learn tools you'll use daily in analytics, AI, and cloud

With industry backed certification from upGrad in association with Microsoft certification + IITB's academic rigour, you graduate with **dual credibility**: academic excellence and industry validation.



Build A Strong Portfolio



- **Commits**
Demonstrate consistency, collaboration, and coding discipline
- **Code**
Showcase well-documented repositories
- **Projects**
Host end-to-end DS/ML/AI projects that highlight real-world problem-solving

GitHub helps with

- ✓ Validating coding skills
- ✓ Showing growth and consistency
- ✓ Being interview-ready for Tech roles



- **Kernels**
Highlight data processing and EDA methodologies
- **Ranking**
Evaluate and reflect global standing among data science practitioners
- **Competitions**
Demonstrate problem-solving under tight constraints

Kaggle helps with

- ✓ Building credibility in data science circles
- ✓ Applying learning to real datasets
- ✓ Speaking confidently in Tech interviews



- **Headline**
Concise summary of goals, competencies, and professional identity
- **Summary**
Engaging overview of learnin and career journey
- **Projects**
Showcase practical experience, outcomes, and skill application

LinkedIn helps with

- ✓ Improving visibility with recruiters
- ✓ Positioning better for job openings
- ✓ Networking with peers and mentors in the field

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Rich and Dedicated Live Support

Industry Expert Sessions

Engage with industry practitioners as they help you master in-demand skills and concepts using a demonstrative hands-on approach

IIITB Faculty Sessions

Learn from some of the most accomplished academicians as they take your knowledge and understanding of data science to another level

Just-In-Time Interview Support

Participate in Technical and HR mock interviews designed to boost your confidence and prepare you to ace interviews.

Career Coaching Sessions

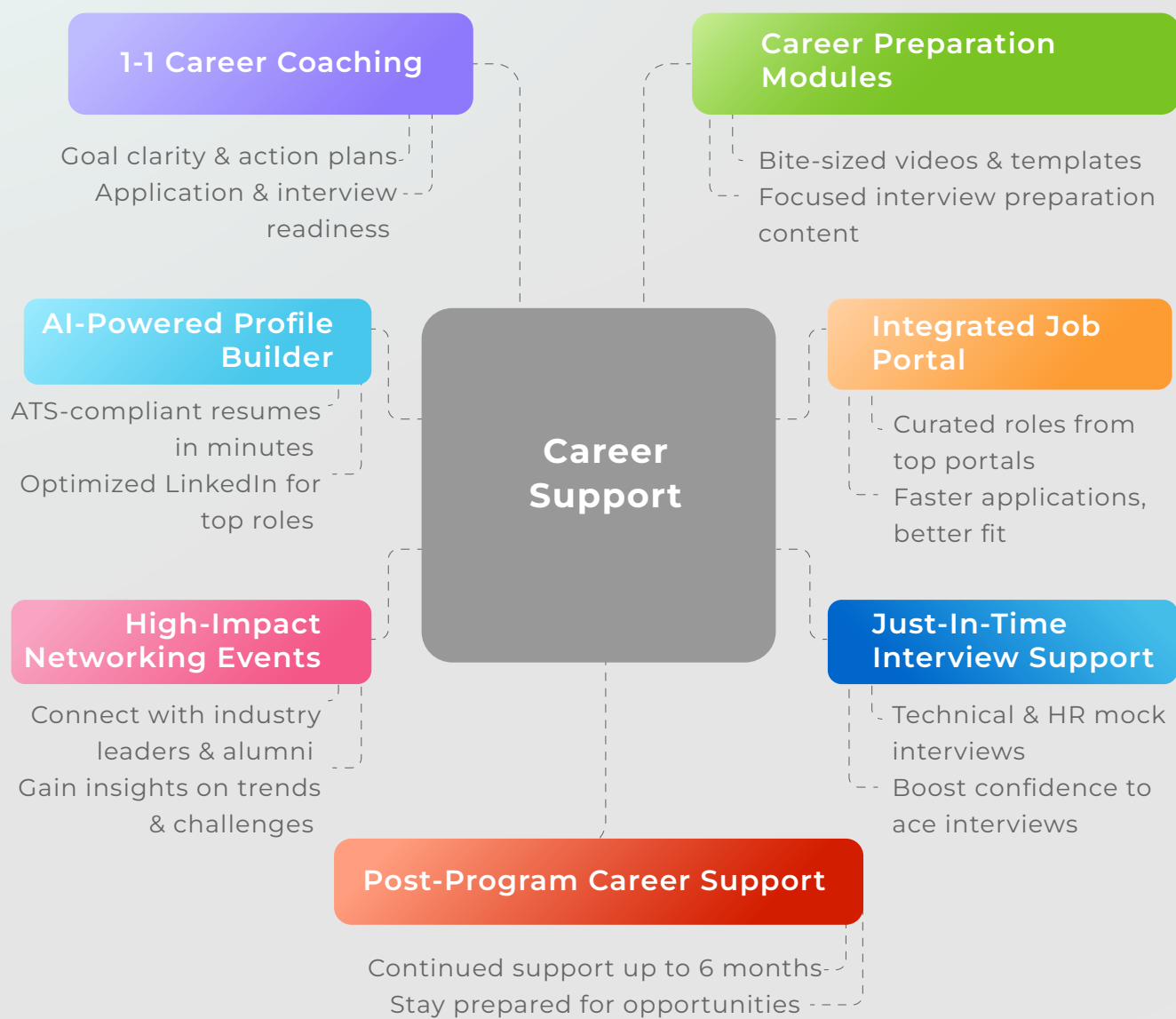
Engage in Career Coaching Sessions via Career preparation modules, High-impact networking events and Just-in-time mock interviews

Daily doubt resolution sessions

Join doubt resolution session slots, that are available daily, and have an expert available to resolve your queries for a smooth learning journey



Effective and Complete Career Support



Student Support

Telegram channel for learner communications

Cohort Telegram channel for timely program updates and announcements.

Completion Support

Personalised assistance for smooth program completion, managing backlogs, and cohort deferrals with free and paid waiver options.

Non-academic and non-technical query assistance

24 X 7 m-AI-ask bot support for non academic assistance.

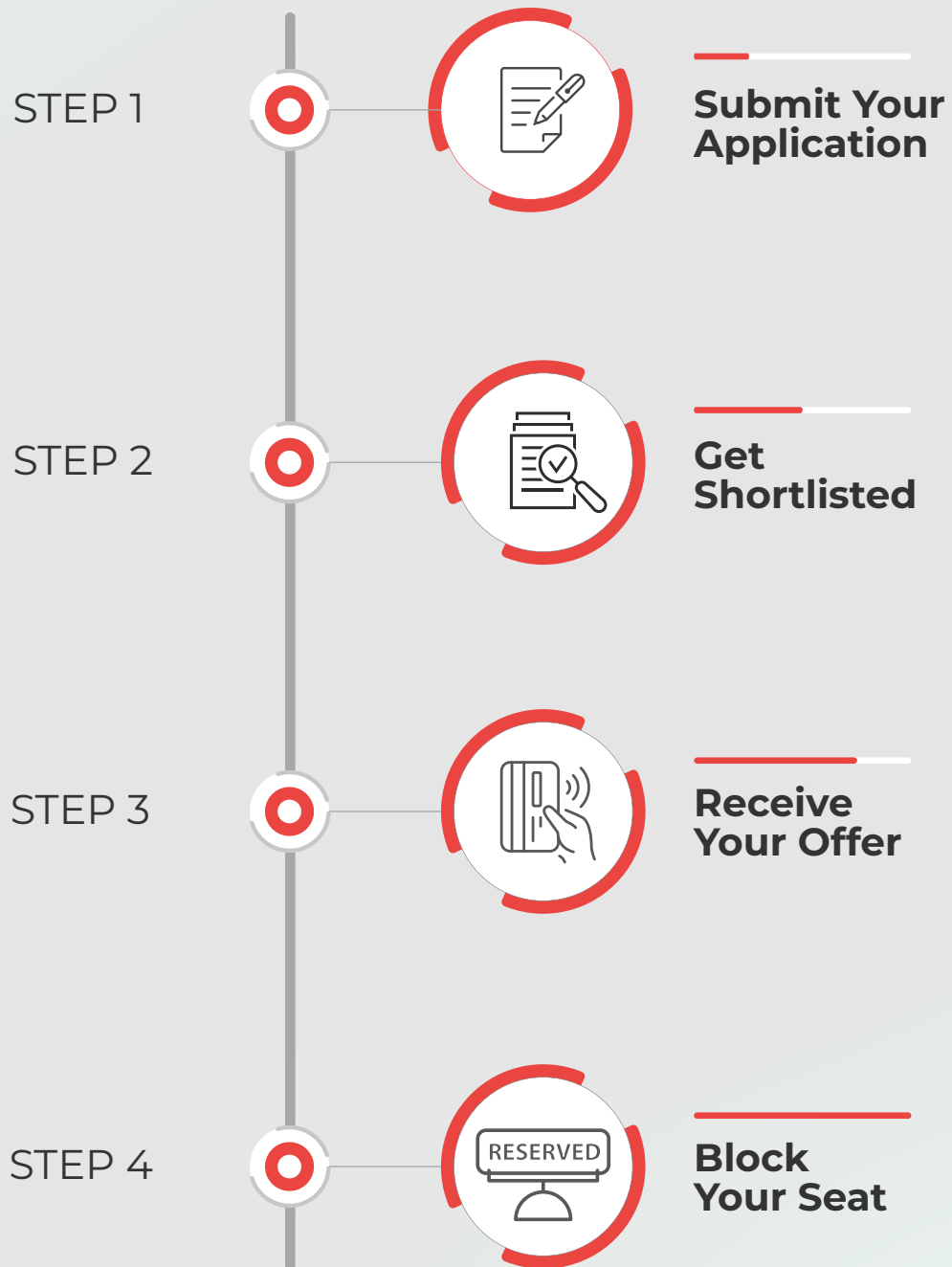
Financial benefits

Access benefits like referrals and repeats by sharing details with your program co-ordinator or student support executive.

Admissions & How To Apply

Eligibility Criteria

- Graduates & Undergraduates (preferably in the final year)





Write to us:

admissions@upgrad.com



Visit us at:

www.upgrad.com



Call: 1800-210-2020

upGrad

LET'S TALK





International
Institute of Information
Technology Bangalore



IIMU
भारतीय प्रबंध संस्थान उदयपुर
Indian Institute of Management Udaipur

C-Suite Executive Program

CTO & AI Leadership Programme

Master platform strategy, production AI, P&L ownership, and boardroom influence.

24 Weeks

Dual Certification and Double Alumni Status

Powered by

upGrad



The CTO Role Has Changed. Have You?



What CTOs Did 5 Years Ago

- ✓ Build systems and manage engineering teams
- ✓ Execute roadmaps set by product/business
- ✓ Report to CEO on technical delivery

What Boards Expect From CTOs Today

- ✓ Drive business strategy through technology bets
- ✓ Own P&L, prove ROI, speak CFO/investor language
- ✓ Scale production AI systems that create moats
- ✓ Exercise boardroom influence alongside CEO and CFO

THE GAP

Many senior tech leaders have the technical depth but lack the strategic toolkit, financial fluency, and executive presence the modern CTO role demands.

› This programme bridges that gap in 24 weeks.

From Tech Leader to CTO

Your 24-Week Transformation



Week 0

You're a technical leader ready to elevate

Months 1-3 (IIIT-B)

- › Master platform architecture
- › Cyber-governance
- › Enterprise AI
- › Production MLOps

Bridge Session

Learn To Translate Tech Into Business Value

Months 4-6 (IIMU)

- › Develop Strategic Vision
- › P&L Ownership
- › Boardroom Influence
- › Org Design

Week 24

- › Build a Board-Ready Strategic Dossier
- › Craft a concise, compelling technology strategy storyline
- › Earn Double Executive Alumni Status

THE OUTCOME

You walk away with technical depth, strategic frameworks, financial fluency, executive presence and a board-ready portfolio proving you're ready to lead at the C-suite.

8 Modules | One Board-Ready Strategic Dossier



International
Institute of Information
Technology Bangalore

Tech & AI Leadership IIIT-B

Build the deep-tech and AI engineering capabilities every modern CTO must command.

- › Cloud-native & distributed systems
- › Cyber resilience & trust engineering
- › Data foundations for GenAI & RAG
- › MLOps, LLMOps, RAG Systems & Agentic AI

Immersive AI & Strategy Workshop at IIIT-B & Microsoft

Learn to design, deploy and lead AI transformation through hands-on, real-world immersions.

- › 2-Day AI Success Planning Workshop at IIIT-B
- › 1-Day Microsoft AI Leadership Immersion
- › CTO-level AI problem framing & prototyping
- › Enterprise AI governance, ethics & operating frameworks



IIMU

भारतीय प्रबंध विज्ञान संस्थान उदयपुर
Indian Institute of Management Udaipur

Business, Strategy & P&L Leadership IIMU

Transform from engineering leader → enterprise leader

- › AI-led business & product strategy
- › P&L ownership & technology economics
- › Executive influence & boardroom communication
- › Designing high-performance, AI-ready tech orgs

Capstone Build a Board-Ready Strategic Dossier

Culminate your journey by building your Strategic CTO Dossier that proves you're ready for the C-suite.

- › Full boardroom-ready CTO deck
- › Craft a concise, compelling technology strategy storyline
- › AI & Technology Strategy Blueprint
- › Use it in CTO interviews & leadership reviews

Program Highlights

India's Only Dual-School CTO Program That Combines
Deep-Tech Mastery, Production AI, P&L Ownership
& Boardroom Leadership



Blended Learning Format

Program includes
live sessions every weekend +
recorded pre-work



Hands-On CTO Dossier Development

3–4 hrs/week of guided project
work to build your Strategic
CTO Dossier



Total Commitment

~7 hrs/week across 24 weeks
160+ total learning hours



2 Transformative Immersions

AI Success Planning Workshop at IIIT-B
Microsoft AI Leadership Immersion



Real-World Projects & Simulations

AI system blueprinting, architecture
decisions, crisis management,
ROI modeling, strategy decks



Double Executive Alumni Status

from
IIIT-B and IIMU



CXO-Led Mentorship

Guidance from CTOs, tech
leaders & industry
experts



Elite Peer Network

Directors, VPs, Engineering
Leaders & Founders
from top organizations

Why This is India's **Best CTO & AI Leadership Programme**



- **Dual Tech + Business School Programme**
Technical depth combined with strategic leadership training.
- **Production AI, Not Theory**
Learn MLOps, LLMOps, and how to industrialize AI from pilots to scale.
- **Board-Ready Strategic Dossier**
A board-ready strategy portfolio designed for CXO-level discussions.
- **P&L Ownership & FinOps Mastery**
Learn tech economics, cloud costing, AI model ROI, and M&A due diligence.
- **Product-Tech Partnership Training**
Get explicit CTO-CPO collaboration experience and frameworks.
- **3 Days of Experiential Learning**
Workshops, Microsoft immersion, and board-style presentations.
- **Dual Alumni Networks**
Join IIIT-B and IIM-U executive communities for long-term career leverage.

HOW WE COMPARE

Compared to business school programmes: deeper production AI and MLOps training

Compared to tech-only programmes: greater focus on P&L, strategy, and leadership

Compared to global executive programmes: equivalent rigor at a fraction of the cost

The Only Programme That Unites India's Premier Tech Institute With A Globally-Ranked Business School



International
Institute of Information
Technology Bangalore

India's Leading AI & Technology Institute

- ▶ A+ NAAC Accredited
- ▶ 5004 research papers, 100+ patents

What You Master

- Platform architecture
- Production AI & MLOps
- Cyber-governance & security
- 2-Day AI Success Workshop on campus



Globally Recognized Business School

- ▶ AACSB Accredited (Top 6% globally)
- ▶ Featured in FT & QS Rankings for 6 years

What You Master

- Strategic vision & product-tech partnerships
- P&L ownership & FinOps
- Boardroom influence & communication
- Organization design at scale
- Final campus capstone presentation

BRIDGE SESSION

Co-Taught by Both Schools



The Only Program With Expert Depth Across All 9 CTO Skills

A curriculum comparison based on public data from leading
CTO programs in India (2025)

Core CTO Skill	Business School	Leading IIT	IIIT-B + IIMU
Platform Architecture & DevEx			
Production AI & MLOps / LLMOps			
Cyber Security & Risk Governance			
Strategic Vision & Innovation			
P&L Ownership & FinOps			
Boardroom Influence & Crisis Comms			
CTO-CPO Product-Tech Partnership			
AI Talent & Org Design			
Board-Ready Strategic Dossier			
 Missing  Light  Adequate  Expert Depth			

Business Schools

- + Strategy, leadership, comms
- Production AI, platform eng, MLOps
- 20+ modules spread thin

Tech Institutes

- + Emerging tech, AI theory
- P&L, boardroom, org design
- Limited financial fluency

IIIT-B + IIMU

- ✓ Technical mastery (platform, AI)
- ✓ Strategic leadership (P&L, board)
- ✓ 8 focused modules, expert depth
- ✓ 40-page board-ready dossier

Based on public curriculum data from leading CTO programs in India (2025)

Learn From Academic Leaders & Industry Experts

Programme Directors (IIMU)



Prof. Kunal Kamal Kumar
Programme Director, IIMU



Teaches:
Boardroom Influence, Executive Communication, Crisis Leadership



Prof. Shankar Prakash
Programme Director, IIMU



Teaches:
Strategic Vision, P&L Ownership, Technology as Business Driver

Technology Faculty (IIIT-B)



Prof. Chandrashekar Ramanathan
Dean (Academics), IIIT-B



Teaches:
Platform Architecture, Cloud Strategy, Enterprise Systems



Prof. G. Srinivasaraghavan
Professor, IIIT-B | Former CTO



Teaches:
Platform Engineering, DevEx, SRE



Prof. V. Sridhar
Professor, IIIT-B



Teaches:
Cyber-Governance, Regulatory Foresight, Policy & Compliance

AI & Data Practitioners



Dr. Dakshinamurthy V. Kolluru

Managing Director,
TekFrameworks
Teaches the 2-day AI Success Workshop



Dr. Sridhar Pappu

Partner, TekFrameworks
Teaches: Enterprise Data Strategy, AI Readiness

Detailed Curriculum

IIIT-B (Months 1-3)

Module 1: Next-Gen Platform Engineering: Cloud, APIs & Distributed Systems

Weeks 1-3

What You'll Master

- Cloud-native architecture, modular monoliths vs microservices, multi-cloud & hybrid strategy
- Build-Buy-Partner decision frameworks; TCO, vendor lock-in, time-to-value
- API-first platforms, event-driven systems, data streaming & ecosystem governance
- Platform engineering, IDPs, DevEx, SRE, observability frameworks

Deliverable: Platform Architecture Blueprint

Module 2: Cyber Resilience for AI-Driven Enterprises: Risk, Compliance & Trust Engineering

Weeks 4-6

What You'll Master

- Board-level security governance (KIST, ISO 27001, SOC 2), material risk reporting
- Business continuity, disaster recovery, chaos engineering, resilience for cloud and AI failure modes
- Enterprise technology risk frameworks: operational, cyber, third-party & model risk
- Vendor & supply chain security, Zero-Trust principles
- AI governance: DPDP, EU AI Act, audit trails, guardrails & regulatory foresight

Deliverable: Technology Risk & Security Governance Framework

Module 3: Building AI-Ready Data Foundations: GenAI, RAG & Enterprise Architecture

Weeks 7-9

What You'll Master

- Data as strategic asset: building competitive moats, data products, real-time & analytical stacks
- AI readiness diagnostics, maturity models, change levers & opportunity identification
- Responsible AI governance: fairness, transparency, human-in-the-loop patterns
- Enterprise AI architecture: from classical ML through GenAI to Agentic AI workflows, RAG systems, and vector databases

Deliverable: Enterprise AI Opportunity Map

Module 4: Scaling AI: MLOps, LLMOps, RAG Systems & Agentic AI Workflows

Weeks 10-12

What You'll Master

- End-to-end AI lifecycle: CI/CD for ML, model registries, monitoring, drift detection, and retraining strategies
- Why most AI pilots fail to scale: identifying barriers, clarifying ROI, establishing ownership models, and building Centers of Excellence
- Integrating AI into products and workflows with clear SLAs, SLOs, and latency versus cost trade-offs
- GenAI and Agentic AI in production: RAG architectures, safety guardrails, LLMOps, prompt evaluation, and continuous improvement

Deliverable: AI Initiative Pipeline & Implementation Roadmap

Bridge Session (Co-Taught by IIIT-B + IIMU)

Strategic Value Mapping: Turning AI & Platform Ideas into Business Outcomes

- Translating technical metrics into business KPIs and strategic value narratives
- Building investment cases for platform and AI initiatives with clear ROI frameworks
- Product-tech partnership models for strategic alignment
- Preparing for business-focused module deliverables

This unique cross-school session ensures seamless transition from technical depth to strategic leadership

IIMU (Months 4-6)

Module 5: Building Enterprise Vision: Product, Innovation & Digital Strategy

Weeks 13-15

What You'll Master

- Strategic versus operational thinking: technology vision, strategic anchors, and value-driver mapping
- CTO-CPO product partnership, roadmap governance, and AI-infused product strategy
- Backcasting for long-term vision: multi-year sequencing and disruption-timing decisions
- Innovation portfolios: Core, Adjacent, and Transformational initiatives with innovation accounting and stage-gate governance
- Platform versus pipeline economics, network effects, and digital moat design

Deliverable: 18-24 Month Technology & Product Strategy Roadmap

Module 6: Financial Leadership for CTOs: Tech ROI, P&L Ownership & Financial Value Creation

Weeks 16-18

What You'll Master

- Technology as a P&L center: revenue, margin, and risk levers with disruptor versus incumbent valuation dynamics
- FinOps mastery: cloud unit economics, AI model costing, and API or data monetization models
- ROI frameworks, real-options valuation, risk-adjusted investment bets, and transformation sequencing Cross-functional budgeting across business lines with governance cycles
- M&A tech due diligence: architecture quality, data integrity, technical debt assessment, IP evaluation, and synergy capture

Deliverable: Technology Investment Case & Valuation Narrative

Module 7: CTO Executive Presence: Strategic Storytelling, Influence & Decision Leadership

Weeks 19-21

What You'll Master

- Executive narratives: crafting platform and AI transformation story arcs with narrative compression and executive briefing techniques
- Board-level investment cases: structure, sequencing, objection handling, and trade-off framing
- Communicating AI risk: fairness, privacy by default, governance frameworks, and responsible AI as a trust asset
- Crisis communication, executive presence, stakeholder navigation, and coalition-building

Deliverable: Board-Ready CTO Presentation Deck synthesizing technology strategy, AI roadmap, investment logic, and risk governance

Module 8: Building & Leading the AI-Ready Tech Organization

Weeks 22-24

What You'll Master

- Organization design: platform, product, and AI guild models with diagnostics for efficiency, ownership clarity, and decision bottlenecks
- AI talent strategy: skills architecture, performance frameworks, and talent sourcing models (buy, build, partner)
- Leading transformation: culture-behavior alignment, operating rhythms, and coalition orchestration
- Inclusive leadership, psychological safety, cultural intelligence, and human-AI collaboration patterns
- Running AI and GenAI experiments safely at scale within a learning organization

Deliverable: AI-Ready Organization & Talent Blueprint with culture, capability, and leadership enablers for 18-24 months ahead

Experiential: 1-Day Microsoft AI Immersion (Bangalore)

2-Day AI Success Planning & Strategy Workshop (IIIT-B Campus)

Integrated Capstone: CTO & AI Strategic Dossier

Throughout the 24 weeks, you'll build a comprehensive board-ready document combining all module deliverables.

- Platform Architecture Blueprint
- Technology Risk & Security Governance Framework
- Enterprise AI Opportunity Map
- AI Initiative Pipeline & Implementation Roadmap
- Technology & Product Strategy Roadmap (18-24 months)
- Technology Investment Case & valuation Narrative
- Board-Ready CTO Presentation Deck
- AI-Ready Organization & Talent Blueprint

This dossier becomes your portfolio for CTO roles, board presentations, and strategic technology leadership opportunities.

Enterprise Case Studies That Define AI-First Leadership



Maersk

AI-Optimized Global Shipping

How much can be saved when AI plans every voyage, fuel burn, and maintenance cycle across the world's largest shipping fleet?



Unilever

Demand Forecasting Weeks Before Weather Hits

How does an FMCG giant prepare production when algorithms predict demand spikes (like ice cream) weeks before heatwaves hit?



Deutsche Bank

Lifting a Live FX Trading Platform onto Distributed Cloud

What does it take to move a global trading system to distributed cloud without dropping a single trade or breaking compliance?



Walmart

Digital Twins for Every Store

What kind of tech stack keeps every shelf, aisle, and product mirrored digitally, with algorithms directing micro-level inventory moves?



Gap Inc.

Fashion Powered by a Unified AI Brain

What happens when design, demand forecasting, and omnichannel marketing are run from a single AI brain built on Google Cloud?



Marks & Spencer

Real-Time Personalisation at Scale

How do you turn a legacy loyalty card into a real-time personalization engine that adapts offers for millions of shoppers instantly?



Nike

Predicting What Sells Before It Exists

How do sneaker launches get planned when models can predict which design, colorway, and size will sell out even before production?



American Express

Real-Time Fraud Detection at Billion-Transaction Scale

How do you decide in milliseconds whether to approve or block a transaction when the cost of a wrong decision is enormous?



JPMorgan Chase

AI Watching Every Contract, Payment & Fraud Signal

What happens when every transaction, contract, and anomaly is scanned continuously by models that never sleep?

Disclaimer: This document does not present an exhaustive list of assignments and projects, and the contents are subject to change based on industry trends and institutional mandates.

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From Classroom To Boardroom

3 Days Of Immersive, Hands-On Leadership Experiences

1-DAY MICROSOFT AI IMMERSION

Microsoft Campus, Bangalore | Weeks 10-12

Real-World AI at Enterprise Scale



- **Fireside chat**
Learn from Microsoft's AI and cloud leadership about building AI-first organizations
- **Technical deep dive**
GenAI architectures, Agentic AI workflows, LLM operations at global scale
- **Reference architectures**
How Microsoft and partners deploy production AI systems reliably
- **Innovation hub tour**
See cutting-edge AI/ML implementations transforming industries
- **PoC blueprinting**
Interactive session designing proof-of-concepts for your business context

Outcome: Reference architecture framework and PoC blueprint you can immediately apply

Disclaimer: The agenda for the Microsoft Immersion is indicative and could be subject to change. The detailed agenda would be shared after the batch start.

2-DAY AI SUCCESS AND STRATEGY WORKSHOP

IIIT-B Campus | Weeks 10-12

Led by Dr. Dakshinamurthy Kolluru and Dr. Sridhar Pappu
Apply the DARWIN Framework to Your Organization



- **Define**
Identify high-impact, business-aligned AI opportunities
- **Assess**
Evaluate data readiness, technical feasibility, and ROI potential
- **Rapid prototype**
Build GenAI/Agentic AI concepts with hands-on guidance
- **Work**
Design integration into existing workflows with measurable KPIs
- **Infrastructure**
Select cloud, MLOps, and LLMOps stack for experimentation through production
- **Navigate**
Engineer responsible AI with fairness, transparency, and bias mitigation

Outcome: Prioritized AI initiative pipeline and implementation roadmap validated by expert practitioners

Your Board-Ready CTO Portfolio:

Across every module, you build a real CTO strategy for your own organization.



What's Inside Your Strategic Dossier



Platform Architecture Blueprint
(Module 1)



18-24 Month Technology & Product Strategy
(Module 5)



Technology Risk & Security Governance Framework
(Module 2)



Investment Case with Financial Modeling & ROI
(Module 6)



Enterprise AI Opportunity Map
(Module 3)



Board-Ready Executive Presentation Deck
(Module 7)



AI Implementation Roadmap with KPIs
(Module 4)

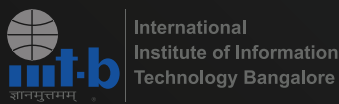


AI-Ready Organization & Talent Design
(Module 8)

This Isn't An Academic Exercise

- Use it to present to your CEO and board
- Show it in CTO job interviews as proof of strategic thinking
- Apply it immediately to transform your organization
- Faculty-validated, board-tested through final boardroom presentation

Join Two Elite Executive Alumni Networks



IIIT-B Executive Alumni

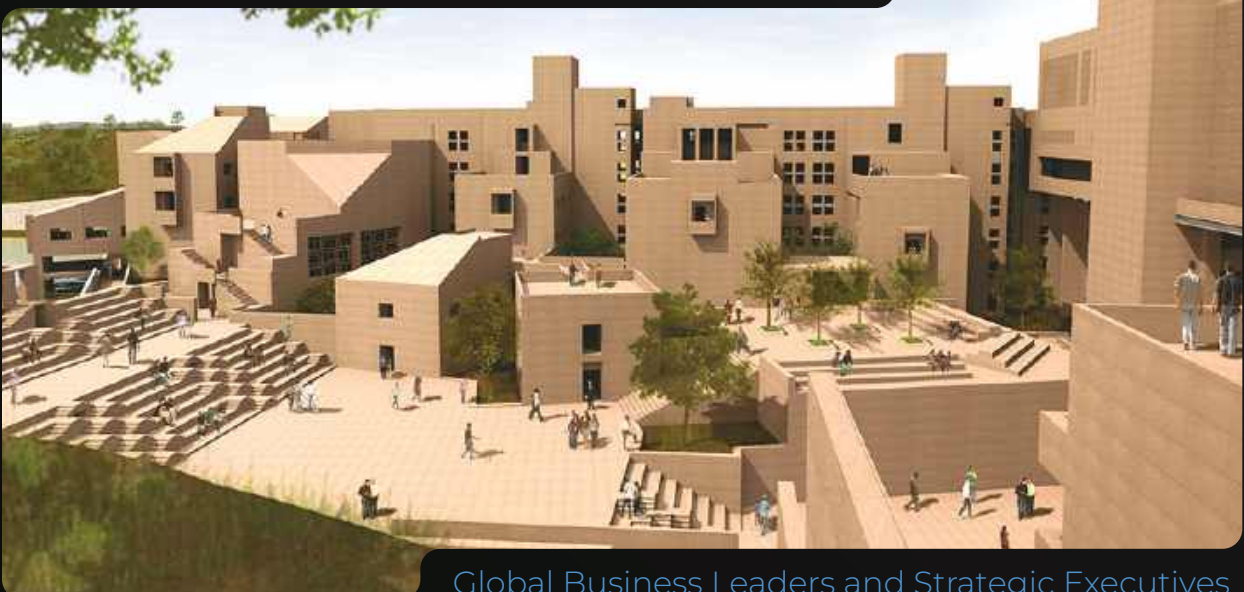


India's Leading AI and Tech Professionals

- Access to annual alumni events, tech talks, research seminars
- Networking opportunities with AI researchers, tech entrepreneurs, and engineering leaders



IIMU Executive Alumni



Global Business Leaders and Strategic Executives

- A permanent lifetime mail id with iimu.ac.in domain will be provided
- Invite to Alumni events across cities for networking with overall institute alums
- Lifetime ID card
- IIMU alum will be provided with a health card with unique benefits
- Any newsletters or articles released by IIMU will be shared/circulated with them
- Any program (non-degree) under ExEd will attract a 10% discount for the Alumni of IIMU
- As a stakeholder, ExEd Alumni will have the privilege to represent IIMU in participating in the ranking process of FT. A questionnaire will be sent to select Alumni for filling the data
- Select Alumni in senior positions may be invited for mentoring under L&D for the full-time students
- Will have access to yearly magazines part of IIMU releases

Disclaimer: Executive Alumni status of IIIT-B is subject to the paying of an annual executive alumni membership fee

For The First Time, **Two Academic Leaders** Unite To Certify The Future Of Tech Leadership.



International
Institute of Information
Technology Bangalore



IIMU
भारतीय प्रबंध संस्थान उदयपुर
Indian Institute of Management Udaipur



Is This Programme Right For You?

Designed for Senior Tech Leaders Ready to Lead at the C-Suite



**Technology Director/VP
Engineering**

You lead teams but want to shape business strategy

You need P&L thinking, boardroom skills, cross-functional leadership



**Principal/Senior
Engineering Manager**

You're on the path to VP/CTO but need to build executive presence

You want to learn platform arch, AI strategy, financial modeling



**Early-Stage or
First-Time CTO**

You have the title but need to elevate strategic impact

You want frameworks for board comm, org design



**Tech Entrepreneur/
Founder**

You're building an AI-first product or platform company

You need to professionalize tech strategy and architecture

Eligibility Requirements

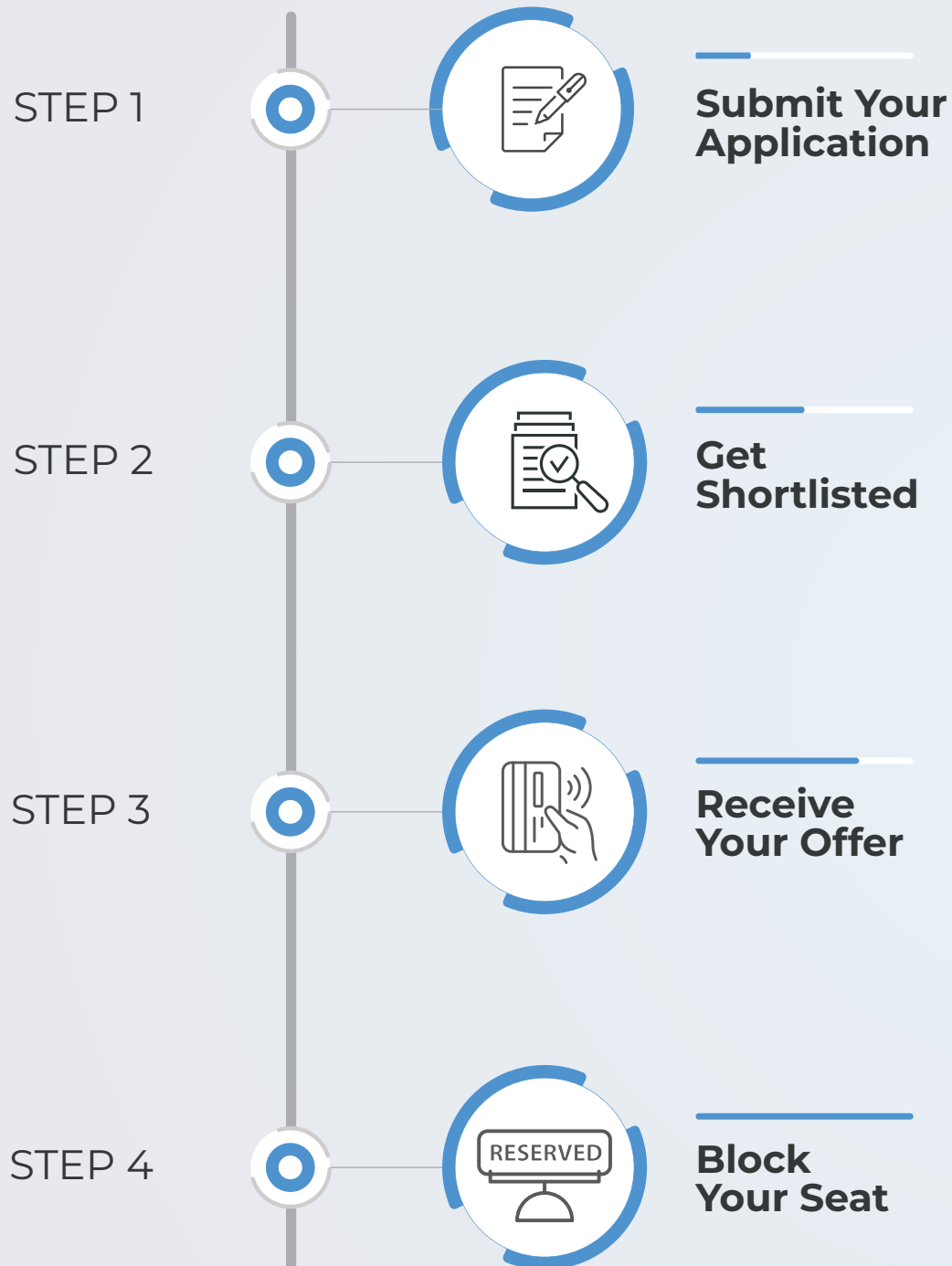
- Bachelor's degree (any discipline)
- Minimum 8 years of professional experience

Prior AI/ML experience is NOT required; we'll build that expertise together.

Admissions & How To Apply

Eligibility Criteria

Bachelor's Degree with a minimum of 8 years of work experience





Write to us:

admissions@upgrad.com



Visit us at:

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LET'S TALK

